LIFTING MOD p GALOIS REPRESENTATIONS OVER GLOBAL FIELDS TO CHARACTERISTIC p^2

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Abstract. Let k be a finite field of characteristic p. We study the Khare–Serre problem of lifting continuous representations $\bar{\rho} : G_K \to \mathrm{GL}_n(k)$ of the absolute Galois group of a global field K to the ring $W_2(k)$ of Witt vectors of length two. We prove that this problem has an affirmative answer for all global fields K and all dimensions $n \geq 1$ if and only if $p = 2$. For every global field K and every finite field k, we prove that all representations of dimension at most four lift to $W_2(k)$. A five-dimensional counterexample for $p = 5$ shows that this bound is sharp. For odd p and any global field K of characteristic different from p with $\mu_p \subset K$ and $\mu_{p^2} \not\subset K$, we construct nonliftable representations in every dimension $n \geq 6$ for $p = 3$ and in every dimension $n \geq p$ for $p \geq 5$.

Furthermore, we prove that every continuous semisimple representation $\bar{\rho} : G_K \to \mathrm{GL}_n(k)$ admits a $W_2(k)$-lift. Over $\mathbf{Q}$ for $p = 5$ and over global function fields for every $p \geq 5$, we construct two-dimensional absolutely irreducible representations that admit no $W_2(k)$-lift without enlarging the set of ramified primes.

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2020 *Mathematics Subject Classification.* 11F80, 11R34, 14H30, 20C20.

Key words and phrases. Galois representations, Witt vectors, lifting obstructions, prescribed ramification, Hurwitz spaces.

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1. Introduction

Let k be a finite field of characteristic p, and write $W_r(k) = W(k)/p^r W(k)$ for the ring of Witt vectors of length r. Khare and Serre posed a lifting problem for Galois representations (see [22, Question 1.1] and [25, §1.2]). We consider the case of global fields.

Question 1 (Serre–Khare)**.** Let K be a global field with absolute Galois group G_K, and let $\bar{\rho} : G_K \to \mathrm{GL}_n(k)$ be a continuous representation with $n \geq 1$. Does $\bar{\rho}$ admit a continuous lift to $W_2(k)$, i.e., a continuous representation $\widetilde{\rho} : G_K \longrightarrow \mathrm{GL}_n(W_2(k))$ whose reduction modulo p is $\bar{\rho}$?

In this paper we study this lifting problem over global fields and give a complete answer. We prove that every such representation lifts when $p = 2$ and construct counterexamples for every finite coefficient field of odd characteristic.

1.1. **Previous work.** In dimension two, Khare proved liftability over number fields for every finite coefficient field [21]. Serre observed that the Kummer theory argument applies over every field; see also [22, Proposition 3.3]. Khare and Larsen asked whether lifting a given two-dimensional representation over a global field can force new ramification [22, Remark 3.4]. De Clercq and Florence extended the unrestricted two-dimensional lifting result to arbitrary coefficient fields and proved liftability in dimensions at most four over $\mathbf{F}_2$ over arbitrary ground fields [9, Theorem 6.1]. Böckle proved that for every local field K and every $n \geq 1$, any continuous representation $\bar{\rho} : G_K \to \mathrm{GL}_n(k)$ admits a continuous lift to $W_2(k)$ [2, Theorem 1.1]. He also gave a cohomological criterion for lifting over global fields [2, Theorem 1.5]. The author proved that for every local field K, every continuous representation $G_K \to \mathrm{GL}_n(k)$ admits a continuous lift to $W(k)$ [23, Theorem 1.1]. Conti, Demarche, and Florence developed stronger flag-preserving local lifting results under roots-of-unity hypotheses [6, Theorem 7.21 and Corollary 7.25]. Khare and Larsen proved three-dimensional $\mathbf{F}_p$-liftability for odd p over nonarchimedean local and global fields containing μ_{p^2} [22, Theorem B]. Florence observed that over a number field containing μ_{p^2}, the three-dimensional lifting result for odd p extends to arbitrary coefficient fields of characteristic p [16, Remark 6.3].

In equal characteristic, Böckle proved that for every connected affine noetherian scheme X over $\mathbf{F}_p$ and every finite field k of characteristic p, each continuous representation $\pi_1^{\mathrm{et}}(X) \to \mathrm{GL}_n(k)$ lifts to $\mathrm{GL}_n(W(k))$ [2, Theorem 1.7]. This includes representations of G_K in characteristic p and of $G_{K,S}$ for global function fields of characteristic p with S nonempty and finite.

Merkurjev and Scavia constructed nonliftable representations in every dimension at least three for odd p over suitable rational function fields [24], and subsequently solved the universal lifting problem over arbitrary fields [25, Theorem 1.1]. Their quantification over arbitrary fields is distinct from restricting the ground field itself to be global. In particular, a rational function field in several variables over a finite field is not a global field.

1.2. **Main results.**

1.2.1. *Arbitrary representations.*

Theorem 1.1. *Let K be a global field and let k be a finite field of characteristic 2. Then every continuous representation $G_K \to \mathrm{GL}_n(k)$ has a continuous lift to $\mathrm{GL}_n(W_2(k))$.*

For a number field, the lift can preserve any finite ramification set containing the archimedean primes and the primes above 2. For a function field of odd characteristic it can preserve any finite ramification set, including the empty set. For a function field of characteristic two, Böckle's theorem gives a $W(k)$-lift on $G_{K,S}$ whenever $S \neq \varnothing$ [2, Theorem 1.7].

Theorem 1.2. *Let K be a global field and k a finite field. If $n \leq 4$, then every continuous $G_K \to \mathrm{GL}_n(k)$ lifts to $\mathrm{GL}_n(W_2(k))$.*

Theorem 1.3. *Let $p \geq 5$ and let K be a global field of characteristic different from p containing μ_p but not μ_{p^2}. Put*

$$N_p = \sum_{i=1}^{p-1} E_{i,i+1}, \qquad P_p = \langle I + N_p, I + E_{23} \rangle \subset \mathrm{GL}_p(\mathbf{F}_p).$$

Then there exists a continuous surjection $G_K \twoheadrightarrow P_p$ having no lift to $\mathrm{GL}_p(\mathbf{Z}/p^2\mathbf{Z})$. For every finite field k of characteristic p and every $n \geq p$, there is consequently a nonliftable representation $G_K \to \mathrm{GL}_n(k)$.

The group P_p has order p^p and exponent p. Its defining representation is not semisimple.

Theorem 1.4. *Let K be a global field of characteristic different from 3 containing μ_3 but not μ_9, and put*

$$x = I + E_{16} + E_{23} + E_{24} + E_{25} - E_{45},$$
$$y = I + E_{14} + E_{36}, \qquad P_3 = \langle x, y \rangle \subset \mathrm{GL}_6(\mathbf{F}_3).$$

Then there exists a continuous surjection $G_K \twoheadrightarrow P_3$ having no lift to $\mathrm{GL}_6(\mathbf{Z}/9\mathbf{Z})$. Nonliftable representations exist in every dimension $n \geq 6$ over every finite coefficient field of characteristic three.

The group P_3 is nonabelian of order 27 and exponent 3. For a global function field $K = \mathbf{F}_q(C)$ with complete constant field $\mathbf{F}_q$, the root-of-unity condition in these two theorems is exactly $v_p(q-1) = 1$. Example 5.5 gives explicit cyclic quotient extensions over $\mathbf{F}_{11}(t)$ for $p = 5$ and over $\mathbf{F}_7(t)$ for $p = 3$.

Corollary 1.5. *(1) For a finite field k of characteristic p, all continuous representations $G_K \to \mathrm{GL}_n(k)$ lift to $W_2(k)$, for all global fields K and all n, if and only if $p = 2$.*
(2) The same equivalence holds when K ranges only over global function fields, with no fixed restriction on their characteristic.
(3) The largest integer d for which every representation of dimension at most d lifts, uniformly in all global fields and finite coefficient fields, is 4. It is also 4 when the ground fields range only over global function fields.

The equivalence in the first part of Corollary 1.5 also holds with $K = \mathbf{Q}$.

Corollary 1.6. *Let k be a finite field of characteristic p. Then all continuous representations $G_{\mathbf{Q}} \to \mathrm{GL}_n(k)$, as n ranges over the positive integers, lift to $\mathrm{GL}_n(W_2(k))$ if and only if $p = 2$. For odd p, a nonliftable representation of $G_{\mathbf{Q}}$ exists in every dimension $n \geq 12$ if $p = 3$ and in every dimension $n \geq p(p-1)$ if $p \geq 5$.*

1.2.2. *Semisimple representations and prescribed ramification.*

Theorem 1.7. *Let K be a global field and k a finite field. Then every continuous semisimple representation $G_K \to \mathrm{GL}_n(k)$ has a continuous lift to $\mathrm{GL}_n(W_2(k))$.*

The coefficient ring and dimension remain fixed, but the finite set of ramified primes may grow. The following results show that this freedom is necessary over both number fields and function fields.

Theorem 1.8. *Put $S = \{5, 1951, \infty\}$. There exists an even, absolutely irreducible continuous representation*

$$\bar{\rho} : G_{\mathbf{Q},S} \twoheadrightarrow \mathrm{SL}_2(\mathbf{F}_5) \subset \mathrm{GL}_2(\mathbf{F}_5)$$

unramified at every finite prime other than 1951, which has no $\mathbf{Z}/25\mathbf{Z}$-lift on $G_{\mathbf{Q},S}$. Its scalar extension to every finite extension $k'/\mathbf{F}_5$ has no $W_2(k')$-lift on that same group. It does have a finite-image lift over $\mathbf{Z}_5[\sqrt{5}]$ on the same $G_{\mathbf{Q},S}$.

The number-field example is proved using an exact arithmetic certificate. For function fields we obtain a family of counterexamples by a different method, in every coefficient characteristic $p \geq 5$.

Theorem 1.9. *For every prime $p \geq 5$, there is a finite set Σ_p of rational primes, containing p, such that for every $\ell \notin \Sigma_p$ there are a power $q = \ell^a$, an element $\lambda \in \mathbf{F}_q \setminus \{0, 1, 2\}$, and a continuous representation*

$$\bar{\rho} : G_{\mathbf{F}_q(x),S} \longrightarrow \mathrm{SL}_2(\mathbf{F}_p), \qquad S = \{0, \lambda, 1, \infty\},$$

with geometric image $\mathrm{SL}_2(\mathbf{F}_p)$ and no lift to $\mathrm{GL}_2(\mathbf{Z}/p^2\mathbf{Z})$ on the same group. For every finite extension $k/\mathbf{F}_p$, its scalar extension has no $W_2(k)$-lift on that group.

Theorem 10.1 proves this assertion without restricting the lifted determinant or inertia orders. The dimension is minimal. These representations lift on $G_{\mathbf{F}_q(x)}$, and every such lift introduces ramification outside S. The construction gives q and λ by a Frobenius existence argument.

Corollary 1.10. *For each prime $p \geq 5$ and all but finitely many primes ℓ, there exist a finite field $\mathbf{F}_q$ of characteristic ℓ, a smooth projective geometrically connected curve $C/\mathbf{F}_q$ of genus 11, and a continuous representation*

$$\bar{\rho}_C : \pi_1(C) \longrightarrow \mathrm{SL}_2(\mathbf{F}_p)$$

with geometric image $\mathrm{SL}_2(\mathbf{F}_p)$. *For every finite extension* $k/\mathbf{F}_p$, *the representation* $\bar{\rho}_C \otimes_{\mathbf{F}_p} k$ *has no* $W_2(k)$*-lift on* $\pi_1(C) = G_{\mathbf{F}_q(C),\varnothing}$.

The proof in Section 10 pulls back the four-point example along a cover of degree 12. For both curves, de Jong's rank-two theorem gives a characteristic-zero lift with the same allowed ramification after a finite extension of the coefficient field [10]. Every such coefficient field must be ramified over $\mathbf{Q}_p$, as explained in Section 10.

1.3. **Method and the function-field additions.** The congruence extension gives an obstruction in $H^2(G, \mathrm{End}_k(k^n))$. In cross characteristic, Böckle's theorem gives local liftability, and global vanishing is detected by arithmetic duality.

The characteristic-two argument identifies the congruence extension with a theta automorphism extension. Explicit cochain primitives and local homotopies annihilate its Poitou–Tate pairing. A separate proper-curve duality argument treats empty ramification over a finite field of odd characteristic. The low-dimensional argument reduces to unipotent images and computes the remaining duality contractions, including the characteristic-three reciprocity case.

For function fields, the detector lemma includes the degree component of the unramified class group, hence constant extensions. The common arithmetic construction uses a ray divisor identity modulo p, rather than an impossible unrestricted prescription of the degree of a single prime divisor. The residual finite groups, matrix cocycles, and local contractions are unchanged. The prescribed contractions give a nonzero sum of local invariants, contradicting global Brauer reciprocity if any lift exists. Semisimple liftability follows instead from trace detection on the full endomorphism module and prime-to-p cyclotomic descent.

For the fixed-ramification function-field examples, we use a four-point representation with geometric image $\mathrm{SL}_2(\mathbf{F}_p)$. The pth power of a separating Dehn twist fixes its residual class, but has no fixed point on the set of all $\mathbf{Z}/p^2\mathbf{Z}$-lift classes. We encode these classes by finite étale Hurwitz spaces, including lifts whose image is a proper subgroup of the congruence preimage. After good reduction, a Frobenius realization argument produces a residual arithmetic representation for which any lift would give a forbidden fixed point.

Section 2 supplies these arithmetic tools. Sections 3 and 4 prove the positive assertions. Section 5 gives the common global construction; Sections 6 and 7 supply its two matrix realizations. Section 8 treats semisimple representations. Sections 9 and 10 give the number-field and function-field counterexamples with prescribed ramification. Section 11 records the remaining questions. The appendices separate exact arithmetic certificates and supplementary finite matrix checks from the general proofs.

1.4. **Conventions.** For any field F, write $G_F = \mathrm{Gal}(F^{\mathrm{sep}}/F)$. All cohomology is continuous. Finite matrix groups carry the discrete topology; $W(k)$ carries its p-adic topology. We use left actions and

$$dX(g) = gX - X, \qquad df(g,h) = gf(h) - f(gh) + f(g).$$

Put $\Delta_g X = gXg^{-1} - X$. Matrix units are E_{ij}. The first-superdiagonal coordinates in dimension four are (a, b, c). A finite additive subgroup of a k-space need only be $\mathbf{F}_p$-linear; statements about its k-span do not make the subgroup itself a k-subspace. In a function field, $\deg(v)$ is the residue degree over its complete constant field. All Frobenius elements used with local reciprocity are arithmetic Frobenius elements.

2. Obstructions and arithmetic detection

2.1. **The actual Witt obstruction.** Reduction gives an exact sequence of finite groups

$$1 \longrightarrow M_n(k) \longrightarrow \mathrm{GL}_n(W_2(k)) \longrightarrow \mathrm{GL}_n(k) \longrightarrow 1, \qquad X \longmapsto 1 + p\widetilde{X}. \tag{2.1}$$

Here $\widetilde{X}$ is any lift. Its choice is immaterial, and the kernel has its additive group structure and the usual conjugation action. With Teichmüller notation, $p[x] = (0, x^p)$ in Witt coordinates; our identification is $x \mapsto p[x]$, not the second Witt coordinate. Thus there is no Frobenius twist in the module in (2.1).

Proposition 2.1 ([29, Theorem 1.2.4])**.** *Let G be a profinite group and $\bar{\rho} : G \to \mathrm{GL}_n(k)$ continuous. Write $M = \mathrm{End}_k(k^n)$ with its residual adjoint action. The pullback of* (2.1) *defines*

$$\beta_k(\bar{\rho}) \in H^2(G, M),$$

which vanishes if and only if $\bar{\rho}$ admits a continuous $W_2(k)$-lift. For any continuous matrix section R reducing to $\bar{\rho}$, the actual factor

$$R_g R_h R_{gh}^{-1} = 1 + pe(g, h)$$

represents this class. A correction $(1 + pu_g)R_g$ is multiplicative exactly when $e = -du$.

Lemma 2.2. *If k'/k is a finite extension of finite fields and $\bar{\rho} \otimes_k k'$ lifts over $W_2(k')$, then $\bar{\rho}$ lifts over $W_2(k)$.*

Proof. The natural map on obstruction modules is $M \to M \otimes_k k'$. Choose a k-linear map $r : k' \to k$ with $r(1) = 1$. Since the residual action is defined over k, $1 \otimes r$ is an equivariant retraction. Hence the map on H^2 is injective. Functoriality of (2.1) identifies the image of the obstruction with the scalar-extended obstruction, which vanishes. No ring retraction $W_2(k') \to W_2(k)$ is required. □

Lemma 2.3. *Liftability may be checked after any finite separable extension of K of degree prime to p. For arbitrary $\bar{\rho} : G_K \to \mathrm{GL}_n(k)$, it therefore suffices to treat a p-group image, in an upper unitriangular basis, over a global field containing μ_p.*

The prime-to-p descent assertion is [9, Lemmas 3.2 and 3.4]. Passing to the fixed field of a Sylow p-subgroup gives an upper unitriangular representation by [25, Lemma 2.6]. Adjoining μ_p has degree dividing $p - 1$.

Lemma 2.4 ([9, Lemma 3.4])**.** *If $V \oplus W$ has a $W_2(k)$-lift, then V and W each have such a lift.*

2.2. **Affine vanishing in characteristic p.**

Lemma 2.5 ([9, Proposition 4.10])**.** *Let X be a connected affine noetherian scheme of characteristic p, let $\Gamma = \pi_1^{\mathrm{et}}(X, \bar{x})$, and let M be a finite continuous $\mathbf{F}_p[\Gamma]$-module. Then $H^2(\Gamma, M) = 0$.*

By Böckle's lifting theorem [2, Theorem 1.7], every continuous representation of $\pi_1^{\mathrm{et}}(X)$ over k lifts to $W(k)$ for X as above. Taking $X = \operatorname{Spec} K$ gives a $W(k)$-lift for every such representation of G_K when $\operatorname{char} K = p$. If K is a global function field of characteristic p with smooth projective model C and S is a nonempty finite set of primes, then $C \setminus S$ is affine and $\pi_1^{\mathrm{et}}(C \setminus S) = G_{K,S}$. Hence a representation of $G_{K,S}$ admits a $W(k)$-lift unramified outside the same S.

2.3. **Kummer lifts.** Suppose $\operatorname{char} K \neq p$ and $\mu_p \subset K$, and let $\chi : G_K \to (\mathbf{Z}/p^2)^\times$ be the cyclotomic character. Write $\chi = 1 + p\kappa$ and view it in $W_2(k)^\times$ through $\mathbf{Z}/p^2 \subset W_2(k)$.

Lemma 2.6. *Every additive continuous character $a : G_K \to k$ has a continuous lift $A : G_K \to W_2(k)$ satisfying*

$$A(gh) = A(g) + \chi(g)A(h).$$

For the untwisted connecting homomorphism $\beta_0 : H^1(K, k) \to H^2(K, k)$ associated with multiplication by p on $W_2(k)$, one has

$$\beta_0(a) = -\kappa \cup a.$$

The lifting assertion is the Kummer surjectivity $H^1(K, W_2(k)(1)) \twoheadrightarrow H^1(K, k)$ in [25, proof of Theorem 3.1]. The Kummer class of a chosen primitive pth root is κ, since $\chi = 1 + p\kappa$. The Bockstein formula of [15, proof of Proposition 3.2], applied in each $\mathbf{F}_p$-coordinate of k, gives $\beta_0(a) = a \cup \kappa = -\kappa \cup a$. Both statements apply over the completions of K as well.

Lemma 2.7. *Every representation with matrices $\left(\begin{smallmatrix} I_r & C_g \\ 0 & I_s \end{smallmatrix}\right)$ over k lifts over $W_2(k)$. In particular, if $(\bar{\rho}(g) - I)(\bar{\rho}(h) - I) = 0$ for all g, h, then the representation lifts over $W_2(k)$.*

The submodule k^r and the quotient k^s are trivial permutation modules, so the lifting criterion of De Clercq and Florence applies [9, Corollary 4.8, Propositions 4.9–4.10, and Theorem 5.2]. For the last assertion, put $U = \sum_g \mathrm{im}(\bar{\rho}(g) - I)$. The product condition makes U and V/U trivial G_K-modules.

2.4. **A finite set of cyclic detectors.** For a finite group H and finite $\mathbf{F}_p[H]$-module N, put

$$H^1_{\mathrm{cyc}}(H,N) = \ker\left(H^1(H,N) \longrightarrow \prod_{\substack{C\le H\\ C \text{ cyclic}}} H^1(C,N)\right).$$

In the remainder of this section assume $\operatorname{char} K \neq p$. A finite set T of primes is called *admissible* if, in the number-field case, it contains all archimedean primes and all primes above p, and, in the function-field case, it is nonempty. Primes ramified in any module under consideration are also included when that module is used. Write $G_{K,T}$ for the Galois group of the maximal extension unramified outside T. For a finite module unramified outside T, set

$$Ш^i_T(K,N) = \ker\left(H^i(G_{K,T},N) \longrightarrow \bigoplus_{v\in T} H^i(K_v,N)\right), \qquad i=1,2.$$

At real primes use Tate-modified local complexes, which agree with ordinary group cohomology in positive degrees. A function field has no archimedean primes. The nonempty hypothesis is part of the finite-set duality used below; it will not be silently imposed on the separate proper-curve argument.

Böckle's fixed-module detection statement is [2, Lemma 4.2]. Here one set of primes is required to work for all finite $\mathbf{F}_p[H]$-modules.

Lemma 2.8. *Let E/K be finite Galois with group H, and let T_0 be admissible and contain the ramification of E/K. There is a finite $\Sigma \supseteq T_0$ such that, for every finite $T \supseteq \Sigma$ and every finite $\mathbf{F}_p[H]$-module N, each class of $Ш^1_T(K,N)$ inflates uniquely from a class of $H^1_{\mathrm{cyc}}(H,N)$.*

Proof. Let U/E be the maximal abelian extension of exponent p unramified at all finite primes. In the number-field case its group is $\mathrm{Cl}^+(E)/p\mathrm{Cl}^+(E)$; the narrow class group permits ramification at real primes when $p=2$. In the function-field case, let C_E be the smooth projective model over the complete constant field of E. Unramified class field theory gives

$$\mathrm{Gal}(U/E) \simeq \widehat{\mathrm{Pic}(C_E)}/p \simeq \mathrm{Pic}(C_E)/p\,\mathrm{Pic}(C_E).$$

The degree-zero subgroup is finite, and the image of the degree map is a nonzero subgroup of $\mathbf{Z}$. Hence this quotient is finite. Its degree part must be retained: it detects the constant extension of degree p. These class field statements are the unramified cases of [27, Chapter I, Appendix A] and [38, §1] in the function-field case, and [28, Chapter V] in the number-field case. The defining properties are invariant under $\mathrm{Gal}(E/K)$, so U/K is finite Galois.

Choose a basis of $\mathrm{Gal}(U/E)$ over $\mathbf{F}_p$. By Chebotarev, choose primes of K outside T_0 split completely in E, and choose primes above them in U, whose Frobenius elements are precisely the selected basis vectors. The choice of a prime in U above a given prime of K selects the desired representative of its Frobenius conjugacy class. In a function field the action of that representative on the constants specifies a degree congruence; Chebotarev supplies primes of arbitrarily large degree in that allowed congruence class; see [39, Theorem 4.2]. These primes may have degree greater than one. Also choose primes whose Frobenius in H generates each cyclic subgroup up to conjugacy. Let Σ be the union of these primes and T_0.

For $[c] \in Ш^1_T(K,N)$, inflate a representative to G_K. Its restriction to G_E is a homomorphism, since G_E acts trivially on N. It is unramified outside T. At a prime above T, local triviality makes that homomorphism zero: any local coboundary restricts to zero on G_E. It therefore factors through $\mathrm{Gal}(U/E)$ and vanishes on the selected Frobenius basis. Inflation–restriction gives a unique class in $H^1(H,N)$. At each cyclic detector, inflation from the local residual image is injective in degree one, so the resulting class has zero restriction to the corresponding cyclic subgroup. This proves the claim. The choice of Σ is independent of N and of any further enlargement T: each newly allowed ramification prime is also required to be a prime of local triviality. □

2.5. **The duality criterion used below.** In this subsection $\operatorname{char} K \neq p$. For a finite p-torsion module M, let $M^D = \operatorname{Hom}_{\mathbf{F}_p}(M, \mathbf{F}_p)(1)$. Finite-set Poitou–Tate duality gives a perfect pairing

$$Ш^2_T(K, M) \times Ш^1_T(K, M^D) \longrightarrow \mathbf{Q}/\mathbf{Z} \tag{2.2}$$

when T is admissible in the sense above and contains the ramification of the modules. We use the local Tate modifications at real primes. These are the finite-module statements of [27, Chapter I, Theorem 4.10(a)].

Corollary 2.9. *Suppose the actions on M and M^D factor through a finite Galois group H, and $H^1_{\mathrm{cyc}}(H, M^D) = 0$. After a finite enlargement Σ of any admissible initial set, for every finite $T \supseteq \Sigma$ the localization map*

$$H^2(G_{K,T}, M) \longrightarrow \bigoplus_{v \in T} H^2(K_v, M)$$

is injective.

Proof. Lemma 2.8 kills $Ш^1_T(K, M^D)$; perfectness of (2.2) kills $Ш^2_T(K, M)$. □

For the remaining criterion assume $\mu_p \subset K$ and $M = M_n(k)$ with adjoint action. The perfect invariant pairing

$$(X, Y) \longmapsto \operatorname{Tr}_{k/\mathbf{F}_p} \operatorname{tr}(XY)$$

identifies M with M^D after choosing $\mathbf{F}_p \simeq \mu_p$. In particular, it uses all matrices, not the trace-zero submodule.

Proposition 2.10. *Let $e \in Z^2(G_{K,T}, M)$ represent a locally trivial class, and suppose T contains the detector set of Lemma 2.8. For every $\eta \in Ш^1_T(K, M)$, choose an inflated finite-group representative and common local primitives Q_v with $\eta|_{G_{K_v}} = dQ_v$. Suppose these choices can be made so that*

$$\operatorname{Tr}_{k/\mathbf{F}_p} \operatorname{tr}(e \cup \eta) = 0 \tag{2.3}$$

as a cochain and

$$\sum_{v \in T} \operatorname{inv}_v\big[\operatorname{Tr}_{k/\mathbf{F}_p} \operatorname{tr}(e|_v \cup Q_v)\big] = 0. \tag{2.4}$$

Then $[e] = 0$ in $H^2(G_{K,T}, M)$. In particular it suffices that every local class in (2.4) vanish.

Proof. All cup products include the action; in particular

$$(e \cup Q_v)(g, h) = e(g, h)\ \operatorname{Ad}(\bar\rho(gh))Q_v.$$

The Leibniz identity and (2.3) make its scalar trace a local 2-cocycle. In the shifted mapping cone with differential $d(c, b) = (dc, \operatorname{res}(c) - db)$, the dual class has compactly supported representative $(\eta, (Q_v)_v)$. Its product with the degree-two class e has components $\operatorname{Tr}_{k/\mathbf{F}_p} \operatorname{tr}(e \cup \eta)$ and $\operatorname{Tr}_{k/\mathbf{F}_p} \operatorname{tr}(e|_v \cup Q_v)$. Thus the hypotheses say that (2.2) vanishes against every dual class, and perfectness gives $[e] = 0$. This is the explicit pairing of [27, Chapter I, §4, p. 65], written with the degree-two factor first. Changing Q_v by an invariant matrix does not change the local class, since $[e|_v] = 0$. At real primes the complete Tate complex agrees with ordinary cochains in the nonnegative degrees used here. □

Corollary 2.11. *Assume $k = \mathbf{F}_p$ and $\mu_p \subset K$. Let e be the actual factor of a continuous matrix section of $\bar\rho$, and let $\eta \in Z^1(K, M)$ be locally inner, with $\eta|_v = dQ_v$. If $\operatorname{tr}(e \cup \eta) = 0$ as a cochain and*

$$\sum_{v} \operatorname{inv}_v\big[\operatorname{tr}(e|_v \cup Q_v)\big] \neq 0,$$

then $\bar\rho$ has no $W_2(k)$-lift, with no restriction on the ramification of a hypothetical lift.

Proof. A lift would give $e = -du$ by Proposition 2.1. Then $\xi = \operatorname{tr}(u \cup \eta)$ is a global scalar 2-cocycle and

$$d \operatorname{tr}(u \cup Q_v) = -\operatorname{tr}(e|_v \cup Q_v) - \xi|_v.$$

The displayed nonzero sum contradicts global Brauer reciprocity for $[\xi]$. Only finitely many terms can be nonzero: outside a finite set the actions and cochains are unramified, and scalar unramified H^2 vanishes. □

The order of choices in applications of Proposition 2.10 is as follows. First fix the residual basis and the matrix section, including its Kummer lifts and coefficient cochains. These continuous cochains have finite values and factor through finite quotients. Choose an admissible T containing their ramification and the detector set of Lemma 2.8. Only then vary the dual class in $Ш^1_T$. We normalize it on the fixed finite residual group by subtracting the coboundary of a constant matrix. This introduces no new ramification. Thus the same T works for every dual class. At every completion, local triviality of the Witt obstruction follows from [2, Theorem 1.1]. No injectivity from restricted-ramification H^2 to absolute H^2 is assumed.

3. Representations in characteristic two

We first prove a lifting theorem with a fixed ramification set. The absolute Galois statement of Theorem 1.1 will follow as Corollary 3.12.

Theorem 3.1. *Let K be a global field of characteristic different from 2. Let S be a finite set of primes of K, containing the archimedean primes and the primes above 2 if K is a number field. For a function field, S may be empty. Let k be a finite field of characteristic two. For every $n \geq 1$, each continuous representation*

$$\bar{\rho} : G_{K,S} \longrightarrow \mathrm{GL}_n(k)$$

admits a continuous lift

$$\widetilde{\rho} : G_{K,S} \longrightarrow \mathrm{GL}_n(W_2(k)).$$

Over a number field the conclusion also holds for an infinite set S containing the same required primes. The finite extension cut out by $\bar{\rho}$ ramifies at only finitely many primes. Choose a finite $S_0 \subset S$ containing these primes, the archimedean primes, and the primes above 2. The lift given by Theorem 3.1 on G_{K,S_0} pulls back along $G_{K,S} \twoheadrightarrow G_{K,S_0}$.

The proof first establishes an arithmetic vanishing theorem for theta automorphisms on $G_{K,S}$, then identifies the resulting extension with the matrix congruence extension in odd rank. The passage to even rank and to arbitrary finite coefficient fields also works on the same group $G_{K,S}$. No roots-of-unity hypothesis beyond $\mu_2 \subset K$ is needed.

For a finite $\mathbf{F}_2[G_{K,S}]$-module N, write

$$N^* = \operatorname{Hom}_{\mathbf{F}_2}(N, \mathbf{F}_2), \qquad Ш^i_S(K, N) = \ker\left(H^i(G_{K,S}, N) \longrightarrow \bigoplus_{v \in S} H^i(K_v, N) \right), \quad i = 1, 2,$$

as in Section 2. We identify $\mathbf{F}_2$ with μ_2 when evaluating scalar cochains in the Brauer group. At real primes, duality uses Tate-modified cohomology; in positive degrees it agrees with ordinary cohomology. All scalar cochain calculations in this section are carried out in $\mathbf{F}_2$, where the two signs agree. We write $[x, y] = xyx^{-1}y^{-1}$.

3.1. **The automorphism extension.** Let M be a finite-dimensional $\mathbf{F}_2$-vector space and let $q : M \to \mathbf{F}_2$ be a quadratic form with nondegenerate polar form

$$b(x, y) = q(x + y) + q(x) + q(y).$$

Choose a bilinear form $f : M \times M \to \mathbf{F}_2$ such that $f(x, x) = q(x)$. Then $f(x, y) + f(y, x) = b(x, y)$. Define

$$\Theta_q = \mathbf{F}_2 \times M, \qquad (z, x)(w, y) = (z + w + f(x, y), x + y). \tag{3.1}$$

Its center is $\mathbf{F}_2 \times \{0\}$; its square map is q and its commutator pairing is b. For $\ell \in M^*$ put

$$\beta_\ell(z, x) = (z + \ell(x), x).$$

Every automorphism acting trivially on the center and on M has this form. Conjugation by any lift of $C \in M$ is

$$\mathrm{Inn}(C) = \beta_{b(C,-)}. \tag{3.2}$$

Nondegeneracy of b identifies these inner automorphisms with M^*.

For $M \neq 0$, the square map q and its nondegenerate commutator pairing b make Θ_q an extraspecial 2-group. Its classical automorphism sequence [18, §1, p. 403] is

$$1 \longrightarrow M^* \xrightarrow{\beta} \mathrm{Aut}(\Theta_q) \longrightarrow \mathrm{O}(M,q) \longrightarrow 1. \tag{3.3}$$

The sequence also holds for $M = 0$, when all three groups are trivial. The isometry group acts on the kernel by $(s\ell)(y) = \ell(s^{-1}y)$. Different choices of f give isomorphic central extensions, since their difference is an alternating bilinear form and hence a quadratic coboundary.

Theorem 3.4 proves that its extension class vanishes on $G_{K,S}$. Proposition 3.6 identifies its pullback under the adjoint action in odd rank with the congruence extension.

Let Γ be a profinite group acting continuously on (M,q). Choose normalized functions a_g, depending only on the finite action image, with

$$a_g(x+y) + a_g(x) + a_g(y) = f(gx, gy) + f(x,y). \tag{3.4}$$

Set $a_1 = 0$ and

$$\alpha_g(z,x) = (z + a_g(x), gx).$$

Define

$$k(g,h)(x) = a_g(hx) + a_h(x) + a_{gh}(x), \tag{3.5}$$

$$e(g,h)(y) = k(g,h)((gh)^{-1}y). \tag{3.6}$$

The polarization of $k(g,h)$ vanishes, so it is linear. Moreover,

$$\alpha_g\alpha_h = \beta_{e(g,h)}\alpha_{gh}. \tag{3.7}$$

Associativity gives $de = 0$ for the stated action on M^*. Thus e represents the pullback of (3.3). Its class vanishes if and only if the action of Γ on M lifts to a continuous action on Θ_q.

3.2. **Two cochain identities.** The following identities will compute the global duality pairing. The transport in a degree-$(2,0)$ cup product is essential:

$$(e \cup Q)(g,h) = e(g,h)(ghQ), \qquad (e \cup \eta)(g,h,l) = e(g,h)(gh\eta_l). \tag{3.8}$$

Lemma 3.2. *Let $\eta \in Z^1(\Gamma, M)$ and write $\eta_g = \eta(g)$. Put*

$$F_\eta(g,h) = f(\eta_g, g\eta_h) + a_g(\eta_h). \tag{3.9}$$

Then $dF_\eta = e \cup \eta$. If

$$H_\eta(g,h) = e(g,h)(\eta_{gh}), \qquad \gamma_\eta = F_\eta + H_\eta,$$

then

$$dH_\eta = \eta \cup e + e \cup \eta, \qquad d\gamma_\eta = \eta \cup e. \tag{3.10}$$

Proof. Use $\eta_{gh} = \eta_g + g\eta_h$. Bilinearity of f reduces the expansion of $dF_\eta(g,h,l)$ to

$$\begin{aligned} &f(\eta_h, h\eta_l) + f(g\eta_h, gh\eta_l) \\ &\quad + a_h(\eta_l) + a_{gh}(\eta_l) + a_g(\eta_h + h\eta_l) + a_g(\eta_h). \end{aligned}$$

By (3.4), the last two terms equal

$$a_g(h\eta_l) + f(g\eta_h, gh\eta_l) + f(\eta_h, h\eta_l).$$

The bilinear terms cancel. The remaining expression is $k(g,h)(\eta_l) = e(g,h)(gh\eta_l)$, proving the first assertion.

For the second assertion, the cocycle identity for e gives

$$\begin{aligned} dH_\eta(g,h,l) &= e(h,l)(\eta_{hl} + g^{-1}\eta_{ghl}) + e(g,h)(\eta_{ghl} + \eta_{gh}) \\ &= e(h,l)(g^{-1}\eta_g) + e(g,h)(gh\eta_l). \end{aligned}$$

The two terms are respectively $(\eta \cup e)(g,h,l)$ and $(e \cup \eta)(g,h,l)$. This proves (3.10). □

Lemma 3.3. *Suppose that, on a subgroup $D \subseteq \Gamma$, one has $\eta_g = gQ + Q$ for some $Q \in M$. Define*

$$t_Q(g) = q(Q) + a_g(Q) + f(Q, gQ). \tag{3.11}$$

Then, on D, there are identities of scalar cochains

$$F_\eta + dt_Q = e \cup Q, \qquad Q \cup e - \gamma_\eta = dt_Q. \tag{3.12}$$

Proof. Substitute $\eta_g = Q + gQ$ and $g\eta_h = gQ + ghQ$ in (3.9). Using (3.4) and the invariance of q, one obtains

$$\begin{aligned} F_\eta(g,h) &= q(Q) + f(Q,gQ) + f(Q,ghQ) + f(Q,hQ) \\ &\quad + a_g(Q) + a_g(hQ), \\ dt_Q(g,h) &= q(Q) + f(Q,gQ) + f(Q,hQ) + f(Q,ghQ) \\ &\quad + a_g(Q) + a_h(Q) + a_{gh}(Q). \end{aligned}$$

Their sum is $k(g,h)(Q) = e(g,h)(ghQ)$, as required. Furthermore, $\eta_{gh} = ghQ + Q$, so

$$H_\eta = e \cup Q + Q \cup e.$$

Combining this with the first identity proves the second. □

3.3. **Theta vanishing with fixed ramification.**

Theorem 3.4. *Let K and S satisfy the cross-characteristic hypotheses of Theorem 3.1. Let (M,q) be a finite quadratic $\mathbf{F}_2[G_{K,S}]$-module with nondegenerate polar form. The pullback of (3.3) to $G_{K,S}$ has zero class in $H^2(G_{K,S}, M^*)$. Equivalently, the prescribed action of $G_{K,S}$ on M lifts to a continuous action of the same group on Θ_q.*

Proof. First assume $S \neq \varnothing$. Put $\Gamma = G_{K,S}$. Choose the functions a_g on the finite action image, as above, and let $e \in Z^2(\Gamma, M^*)$ be the cocycle (3.6). First fix $v \in S$ and an invariant $Q \in M^{G_{K_v}}$. For $g, h \in G_{K_v}$ one has $hQ = ghQ = Q$, and hence

$$e(g,h)(Q) = a_g(Q) + a_h(Q) + a_{gh}(Q).$$

Thus evaluation at Q sends $[e_v]$ to zero in $H^2(K_v, \mu_2)$. The Cartier dual of M^* is M, since $\mu_2 = \mathbf{F}_2$ over K. Local Tate duality therefore gives $[e_v] = 0$ at every finite prime in S [27, Corollary I.2.3]. At a real prime the dual degree-zero group is the quotient $\widehat{H}^0(K_v, M)$ of the invariant vectors; vanishing against all invariant vectors is sufficient [27, Theorem I.2.13(a)]. At a complex prime $H^2 = 0$. Consequently

$$[e] \in Ш^2_S(K, M^*).$$

Since S is nonempty and admissible, finite-set Poitou–Tate duality gives the perfect pairing

$$Ш^1_S(K, M) \times Ш^2_S(K, M^*) \longrightarrow \mathbf{Q}/\mathbf{Z}; \tag{3.13}$$

see (2.2) and [27, Theorem I.4.10(a)]. Let $[\eta] \in Ш^1_S(K, M)$ and choose a continuous representative $\eta \in Z^1(\Gamma, M)$. Lemma 3.2 constructs

$$\gamma_\eta \in C^2(\Gamma, \mathbf{F}_2), \qquad d\gamma_\eta = \eta \cup e.$$

The functions a_g and the cocycle η are defined on this same Γ, so the formula for γ_η is a continuous cochain on Γ itself. Through $\mathbf{F}_2 = \mu_2$, whose elements are S-units, it is an admissible global primitive for the finite-set pairing. Thus S remains fixed while $[\eta]$ varies.

For each $v \in S$, choose local cochains

$$\eta_v = dQ_v, \qquad e_v = d\lambda_v, \quad Q_v \in M, \quad \lambda_v \in C^1(K_v, M^*). \tag{3.14}$$

Under the explicit construction of (3.13), a local representative is

$$c_v = \eta_v \cup \lambda_v - (\gamma_\eta)_v \in Z^2(K_v, \mu_2). \tag{3.15}$$

The value of the pairing is $\sum_{v\in S}\operatorname{inv}_v[c_v]$, where these cochains are included in $(K_v^{\mathrm{sep}})^\times$ to take their Brauer invariants. Formula (3.15) is the representative with the degree-one class first; see [27, Chapter I, §4, p. 65].

The local primitive λ_v is accounted for explicitly. The cochain Leibniz rule and Lemma 3.3 give

$$\begin{aligned} d(Q_v\cup\lambda_v) &= \eta_v\cup\lambda_v + Q_v\cup e_v,\\ c_v &= d(Q_v\cup\lambda_v) + Q_v\cup e_v - (\gamma_\eta)_v\\ &= d(Q_v\cup\lambda_v + t_{Q_v}). \end{aligned} \tag{3.16}$$

Thus the Brauer class of c_v is zero for every $v\in S$. At real primes the positive Tate degrees agree with ordinary cohomology, so (3.14) and (3.16) remain valid.

It follows that $[e]$ pairs trivially with every class in $Ш^1_S(K,M)$. Perfectness of (3.13) gives $[e]=0$ in $H^2(\Gamma,M^*)$. By the definition of continuous cohomology, there is a continuous $u\in C^1(\Gamma,M^*)$ with $e=du$. The corrected maps $\beta_{u_g}\alpha_g$ have defect $e+du=0$ in characteristic two, by (3.7). They therefore give a continuous homomorphism $\Gamma\to\operatorname{Aut}(\Theta_q)$ lifting the prescribed action.

It remains to treat $S=\varnothing$, which can occur here only for a function field $K=\mathbf{F}_q(C)$ of odd characteristic. In this case $\Gamma=\pi_1^{\mathrm{et}}(C)$ and $\mu_2=\mathbf{F}_2$ is a constant etale sheaf. Use the same cocycle e and, for every $\eta\in Z^1(\Gamma,M)$, the identity $e\cup\eta=dF_\eta$ of Lemma 3.2. The low-degree comparison for finite locally constant sheaves [36, §2, p. 1501] gives

$$H^1(\Gamma,M)\simeq H^1_{\mathrm{et}}(C,\mathcal{M}),\qquad H^2(\Gamma,M^*)\hookrightarrow H^2_{\mathrm{et}}(C,\mathcal{M}^*).$$

Proper-curve duality over a finite field gives a perfect pairing

$$H^2_{\mathrm{et}}(C,\mathcal{M}^*)\times H^1_{\mathrm{et}}(C,\mathcal{M})\longrightarrow H^3_{\mathrm{et}}(C,\mu_2)\simeq \tfrac{1}{2}\mathbf{Z}/\mathbf{Z};$$

see [27, Chapter II, Corollary 3.3]. Compatibility of the comparison maps with cup products and the displayed cochain primitive show that the image of $[e]$ pairs to zero with every class in degree one. Perfectness and then injectivity give $[e]=0$ in $H^2(\Gamma,M^*)$. The same continuous cochain correction as above now lifts the action on this unchanged Γ. This argument does not apply finite-set Poitou–Tate duality with an empty set.

□

Remark 3.5. The global primitive in Lemma 3.2 is explicit and is defined on $G_{K,S}$. The proof therefore requires neither the vanishing of $H^3(G_{K,S},\mathbf{F}_2)$ nor an enlargement of S by detector primes. It proves vanishing in $H^2(G_{K,S},M^*)$ before inflation to G_K. The conclusion concerns the particular automorphism class (3.3); it does not assert injectivity of localization, or of inflation to $H^2(K,M^*)$, for arbitrary classes.

3.4. **The congruence extension in odd rank.** Fix an odd integer $n\geq 3$ and put

$$G=\mathrm{GL}_n(\mathbf{F}_2),\qquad M=\mathfrak{sl}_n(\mathbf{F}_2).$$

Reduction gives an exact sequence

$$1\longrightarrow M\longrightarrow \mathrm{SL}_n(\mathbf{Z}/4)\longrightarrow G\longrightarrow 1,\qquad A\longmapsto I+2\widehat{A}. \tag{3.17}$$

The choice of the lift $\widehat{A}$ is immaterial. Indeed, $\det(I+2\widehat{A})=1+2\operatorname{tr}(A)$, and the reduction map is onto: take an invertible lift over $\mathbf{Z}/4$ and multiply one row by the inverse of its determinant. This does not change its reduction. The induced action on the kernel is the adjoint action.

Define

$$q(A)=e_2(A)=\sum_{a<b}(A_{aa}A_{bb}+A_{ab}A_{ba}). \tag{3.18}$$

It is invariant under conjugation, and its polarization on all matrices is $\operatorname{tr}(A)\operatorname{tr}(B)+\operatorname{tr}(AB)$. On M this is $b(A,B)=\operatorname{tr}(AB)$. The trace pairing on all matrices is nondegenerate, and the orthogonal complement of M is the scalar line. Since n is odd, that line meets M trivially. Thus b is nondegenerate on M. It is alternating there because $\operatorname{tr}(A^2)=\operatorname{tr}(A)^2=0$.

Via $A \mapsto b(A,-)$, identify M with M^*. Pulling (3.3) back along $\mathrm{Ad} : G \to \mathrm{O}(M,q)$ gives an extension of G by M.

Proposition 3.6. *Under this identification, the class of the pulled-back theta automorphism extension equals the class of* (3.17) *in* $H^2(G,M)$.

We prove the proposition by constructing the comparison over the Sylow subgroup $U_n(\mathbf{F}_2)$.

3.4.1. *Explicit root automorphisms.* Eliminate the last diagonal coordinate by $A_{nn} = \sum_{a<n} A_{aa}$ and choose

$$f(A,B) = \sum_{a<b} A_{ab}B_{ba} + \sum_{a<n} A_{aa}B_{aa} + \sum_{a<b<n} A_{aa}B_{bb}. \tag{3.19}$$

Then $f(A,A) = q(A)$ and $f(A,B) + f(B,A) = \mathrm{tr}(AB)$ on M. For $i<j$ put

$$\begin{aligned} t_{ij} &= I + E_{ij}, & R_{ij}(A) &= t_{ij}At_{ij}, \\ D_{ij}(A) &= \sum_{i\le p<j} A_{pp}, & r_{ij}(A) &= A_{ji}D_{ij}(A) + \sum_{i<p<j} A_{pi}A_{jp}. \end{aligned} \tag{3.20}$$

Lemma 3.7. *The map*

$$T_{ij}(z,A) = (z + r_{ij}(A), R_{ij}(A))$$

is an automorphism of Θ_q. *Moreover,*

$$T_{ij}^2 = \mathrm{Inn}(E_{ij}), \qquad T_{ij}^4 = 1. \tag{3.21}$$

Proof. Write $f = F + D$, separating the first sum in (3.19) from the two diagonal sums. Set $a = A_{ji}$, $c = B_{ji}$, and $S(A) = D_{ij}(A)$. The entry formula

$$(R_{ij}A)_{uv} = A_{uv} + \delta_{ui}A_{jv} + \delta_{vj}A_{ui} + \delta_{ui}\delta_{vj}A_{ji} \tag{3.22}$$

gives

$$\begin{aligned} F(R_{ij}A, R_{ij}B) + F(A,B) &= (A_{ii} + A_{jj})c + ac \\ &\quad + \sum_{i<p<j} (A_{jp}B_{pi} + A_{pi}B_{jp}). \end{aligned} \tag{3.23}$$

Here is an explicit account of the cancellation. Among the upper entries of A, only row i and column j change; among the paired lower entries of B, only row i and column j change. Before cancelling, their contributions are

$$\begin{aligned} &\sum_{v>i,\ v\neq j} A_{jv}B_{vi} + \sum_{u<j,\ u\neq i} A_{ui}B_{ju} + (A_{ii} + A_{jj} + A_{ji})B_{ji} \\ &\qquad + \sum_{u<i} A_{ui}B_{ju} + \sum_{v>j} A_{jv}B_{vi}. \end{aligned}$$

The terms with $u<i$ and with $v>j$ occur twice. The remaining terms are exactly (3.23).

On the independent diagonal coordinates, let δ have entries i,j equal to one if $j<n$, and only entry i equal to one if $j=n$. Writing $d_A = (A_{11}, \dots, A_{n-1,n-1})$, one has

$$D(\delta, d_B) = S(B), \quad D(d_A, \delta) = S(A) + A_{ii} + A_{jj}, \quad D(\delta,\delta) = 1.$$

For clarity, if $d = (d_1, \dots, d_{n-1})$, then

$$D(\delta,d) = \sum_s \Big(\sum_{r\le s} \delta_r\Big) d_s, \qquad D(d,\delta) = \sum_r d_r \Big(\sum_{s\ge r} \delta_s\Big).$$

The first coefficient is the indicator of $i \le s < j$. When $j<n$, the second is the indicator of $i < r \le j$. When $j=n$, it is the indicator of $r \le i$, which equals the claimed expression after using $A_{nn} = \sum_{r<n} A_{rr}$. Finally $D(\delta,\delta) = 1$: there is either one nonzero diagonal coordinate or two of them and exactly one cross-term. Consequently

$$D(R_{ij}A, R_{ij}B) + D(A,B) = aS(B) + c(S(A) + A_{ii} + A_{jj}) + ac. \tag{3.24}$$

Adding (3.23) and (3.24) gives exactly $r_{ij}(A+B) + r_{ij}(A) + r_{ij}(B)$. This is the condition that T_{ij} preserve (3.1).

Under R_{ij} the entry A_{ji} and all intermediate entries A_{pi}, A_{jp} remain fixed, while D_{ij} changes by A_{ji}. Thus $r_{ij}(R_{ij}A) + r_{ij}(A) = A_{ji}^2 = A_{ji}$. Together with $R_{ij}^2 = 1$ and (3.2), this proves (3.21). □

3.4.2. *Root relations over* $\mathbf{Z}/4$.

Lemma 3.8. *The automorphisms T_{ij} satisfy*

$$[T_{ij}, T_{kl}] = 1 \qquad \text{if } j \neq k \text{ and } i \neq l, \tag{3.25}$$

$$T_{ij}T_{jk} = T_{jk}T_{ik}T_{ij} \qquad \text{if } i < j < k. \tag{3.26}$$

In particular T_{ik} commutes with T_{ij} and T_{jk}, and $[T_{ij}^a, T_{jk}^b] = T_{ik}^{ab}$ for all $a, b \in \mathbf{Z}/4$.

Proof. The phase calculations hold on arbitrary matrices over $\mathbf{F}_2$. Write $\Delta_{kl}r(A) = r(R_{kl}A) + r(A)$. If $j \neq k$ and $i \neq l$, then the entry formula gives

$$\Delta_{kl}r_{ij}(A) = \epsilon A_{ji}A_{lk} + \epsilon' A_{li}A_{jk}, \tag{3.27}$$

where, with indicator values taken in $\mathbf{F}_2$,

$$\epsilon = \mathbf{1}_{[i,j)}(k) + \mathbf{1}_{[i,j)}(l), \qquad \epsilon' = \mathbf{1}_{(i,j)}(k) + \mathbf{1}_{(i,j)}(l).$$

Indeed, A_{ji} is unchanged because $j \neq k$ and $i \neq l$. The diagonal sum changes by ϵA_{lk}. For $i < p < j$, the only changes in the remaining factors are

$$A_{pi} \longmapsto A_{pi} + \mathbf{1}_{p=k}A_{li}, \qquad A_{jp} \longmapsto A_{jp} + \mathbf{1}_{p=l}A_{jk}.$$

There is no product of the two new terms because $k \neq l$. Their sum is the second term of (3.27).

The expression (3.27) is unchanged when the two roots are exchanged. For four distinct endpoints, the two indicators have the same sum, and this sum is one precisely when the intervals cross. If the roots share their initial endpoint, then the two monomials coincide and their combined coefficient is one in both orders. If they share their terminal endpoint, then the combined coefficient is zero in both orders. Identical roots give the same conclusion directly. These are all possibilities under the hypotheses. Thus $\Delta_{kl}r_{ij} = \Delta_{ij}r_{kl}$, proving (3.25).

For a chain $i < j < k$, put $u = A_{ji}$, $v = A_{ki}$, and $w = A_{kj}$. Under R_{jk}, the factors in r_{ij} change by $u \mapsto u + v$, $D_{ij} \mapsto D_{ij}$, and $A_{jp} \mapsto A_{jp} + A_{kp}$ for $i < p < j$. Under R_{ij}, the factors in r_{jk} change by $w \mapsto w + v$, $D_{jk} \mapsto D_{jk} + u$, and $A_{pj} \mapsto A_{pj} + A_{pi}$ for $j < p < k$. Thus

$$\Delta_{jk}r_{ij} = vD_{ij} + \sum_{i<p<j} A_{pi}A_{kp},$$

$$\Delta_{ij}r_{jk} = vD_{jk} + wu + vu + \sum_{j<p<k} A_{pi}A_{kp}.$$

Since $D_{ij} + D_{jk} = D_{ik}$, it follows that

$$\Delta_{jk}r_{ij} + \Delta_{ij}r_{jk} = r_{ik}(A) + uv = r_{ik}(R_{ij}A). \tag{3.28}$$

The term indexed by $p = j$ in r_{ik} is uw; under R_{ij} only its factor w changes, giving the extra uv in (3.28). Also $\Delta_{ik}r_{jk} = 0$ because none of its displayed factors changes. The scalar phase of the left side of (3.26) is $r_{jk}(A) + r_{ij}(R_{jk}A)$; that of the right side is

$$r_{ij}(A) + r_{ik}(R_{ij}A) + r_{jk}(R_{ik}R_{ij}A).$$

Their equality follows from (3.28) and $\Delta_{ik}r_{jk} = 0$. Their matrix actions agree by elementary matrix multiplication. Finally (3.25) gives the asserted centrality of T_{ik}. The class-two commutator identity now gives the relations for all powers, including the parameters 2 and 3. □

Lemma 3.9. *There is a homomorphism*

$$T : U_n(\mathbf{Z}/4) \longrightarrow \mathrm{Aut}(\Theta_q), \qquad I + aE_{ij} \longmapsto T_{ij}^a.$$

It induces the adjoint action after reduction, and for every strictly upper triangular matrix B over $\mathbf{F}_2$ it satisfies

$$T(I + 2\widehat{B}) = \mathrm{Inn}(B). \tag{3.29}$$

Proof. Use the presentation of $U_n(\mathbf{Z}/4)$ on the adjacent transvections $s_i = I + E_{i,i+1}$ in [1, Theorem 1]. Send s_i to $T_{i,i+1}$. Their fourth powers are trivial by (3.21), and nonadjacent generators commute by (3.25). The adjacent commutator maps to $T_{i,i+2}$, which commutes with both factors and with $T_{i+1,i+3}$ whenever the latter is defined. These are all the defining relations, including the last relation for even modulus. Iterating (3.26) gives $T(I + E_{ij}) = T_{ij}$ for every $i < j$. Finally, $I + 2\widehat{B}$ is the product of the roots $I + 2B_{ij}E_{ij}$ in any order, since all cross-products vanish modulo four. Their images are $\mathrm{Inn}(E_{ij})^{B_{ij}}$ by (3.21), and their product is $\mathrm{Inn}(B)$. □

3.4.3. *Comparison of the full extensions.*

Proof of Proposition 3.6. Put $U = U_n(\mathbf{F}_2)$, and let E be its inverse image in $\mathrm{SL}_n(\mathbf{Z}/4)$. Write

$$C = \{I + 2\widehat{A} : A \in M\}, \qquad R = U_n(\mathbf{Z}/4).$$

Then $E = CR$: for any element of E, choose an element of R with the same reduction, and divide by it. Moreover,

$$C \cap R = \{I + 2\widehat{B} : B \text{ strictly upper triangular}\}.$$

Define

$$\Psi\big((I + 2\widehat{A})r\big) = \mathrm{Inn}(A)T(r). \tag{3.30}$$

The maps on C and R agree on $C \cap R$ by (3.29). Explicitly, if $cr = c'r'$, then $k = (c')^{-1}c = r'r^{-1} \in C \cap R$, so $c = c'k$ and $r' = kr$; these equalities show that (3.30) is well-defined.

Every automorphism of Θ_q transports inner automorphisms by its action on M. Hence

$$T(r)\,\mathrm{Inn}(A)T(r)^{-1} = \mathrm{Inn}(\mathrm{Ad}(\bar{r})A),$$

which is exactly the conjugation action of R on C in E. It follows that Ψ is a homomorphism. Pairing it with the residual quotient map gives a morphism

$$E \longrightarrow U \times_{\mathrm{O}(M,q)} \mathrm{Aut}(\Theta_q)$$

which is the identity on U and on the entire kernel M, under (3.2). It is injective because an element in its kernel first has trivial image in U and then is zero in M. It is surjective because its image surjects onto U and contains the whole kernel M. Thus it is an isomorphism of the full extensions, not only of their upper triangular subgroups.

The subgroup U is a Sylow 2-subgroup of G and

$$[G : U] = \prod_{j=1}^{n} (2^j - 1)$$

is odd. The composite $\mathrm{cor} \circ \mathrm{res}$ on $H^2(G, M)$ is multiplication by this index, hence the identity. Restriction to U is consequently injective. The two extension classes agree over U, so they agree over G. □

3.5. **Completion in all ranks and coefficient fields.**

Proposition 3.10. *Let K and S satisfy the hypotheses of Theorem 3.1. For every $n \geq 1$, each continuous representation $\bar{\rho} : G_{K,S} \to \mathrm{GL}_n(\mathbf{F}_2)$ admits a continuous lift on $G_{K,S}$ to $\mathrm{GL}_n(\mathbf{Z}/4)$. When n is odd, it has such a lift to $\mathrm{SL}_n(\mathbf{Z}/4)$.*

Proof. Put $\Gamma = G_{K,S}$. In rank one the representation is trivial and its trivial lift has determinant one. Suppose $n \geq 3$ is odd. Equip $M = \mathfrak{sl}_n(\mathbf{F}_2)$ with the action induced by $\bar{\rho}$ and the quadratic form (3.18). By Theorem 3.4, the corresponding theta class vanishes in $H^2(\Gamma, M)$, using the equivariant identification $M \simeq M^*$ furnished by the polar form. Proposition 3.6 is an equality of finite-group extension classes, so pulling it back along $\bar{\rho}$ identifies this zero class with the obstruction for (3.17) on Γ. A continuous cochain correcting a matrix section therefore gives an $\mathrm{SL}_n(\mathbf{Z}/4)$ lift on Γ. Inclusion into $\mathrm{GL}_n(\mathbf{Z}/4)$ is an actual lift of $\bar{\rho}$.

If n is even, then $\bar{\rho} \oplus \mathbf{1}$ has odd dimension and lifts on Γ by the preceding paragraph. Lemma 2.4 gives a lift of $\bar{\rho}$ on Γ. □

3.5.1. *Finite coefficient fields.*

Lemma 3.11. *Let Γ be a profinite group, let k be a finite field of characteristic two, and let V be a finite-dimensional continuous $k[\Gamma]$-module. If its underlying $\mathbf{F}_2[\Gamma]$-module lifts to $\mathbf{Z}/4$, then V lifts to $W_2(k)$.*

Proof. Write $f = [k : \mathbf{F}_2]$. The Teichmüller lifts of an $\mathbf{F}_2$-basis of k form a $\mathbf{Z}/4$-basis of $W_2(k)$ by [8, (2.33) and its proof]. The regular representation of $W_2(k)$ gives a commutative diagram

$$\begin{array}{ccc} \mathrm{GL}_n(W_2(k)) & \longrightarrow & \mathrm{GL}_{nf}(\mathbf{Z}/4) \\ \downarrow & & \downarrow \\ \mathrm{GL}_n(k) & \longrightarrow & \mathrm{GL}_{nf}(\mathbf{F}_2). \end{array} \tag{3.31}$$

On congruence kernels its map is the usual inclusion

$$j : \mathrm{End}_k(V) \hookrightarrow \mathrm{End}_{\mathbf{F}_2}(V).$$

Indeed, the regular image of $I + 2\widehat{A}$ is $I + 2$ times the regular image of $\widehat{A}$; reducing that coefficient gives the underlying $\mathbf{F}_2$-linear operator A. Thus the map on kernels is exactly j in the coordinates of (2.1).

For $a \in k^\times$ let m_a denote scalar multiplication on V. Define

$$\Pi(S) = \sum_{a \in k^\times} m_a S m_a^{-1}, \qquad S \in \mathrm{End}_{\mathbf{F}_2}(V).$$

Since $|k^\times|$ is odd, this is the averaging projection onto the centralizer of $k^\times$, which is exactly $\mathrm{End}_k(V)$. Moreover, $\Pi j = 1$, and Π is Γ-equivariant because the action of Γ is k-linear. Consequently the induced map

$$H^2(\Gamma, \mathrm{End}_k(V)) \longrightarrow H^2(\Gamma, \mathrm{End}_{\mathbf{F}_2}(V))$$

is injective. By naturality in (3.31), it sends $\beta_k(V)$ to the obstruction of the underlying $\mathbf{F}_2$-module. The latter vanishes by hypothesis, and hence $\beta_k(V) = 0$. □

Proof of Theorem 3.1. Put $\Gamma = G_{K,S}$ and $f = [k : \mathbf{F}_2]$. The underlying $\mathbf{F}_2$-representation of $\bar\rho$ has dimension nf and is a continuous representation of this same Γ. Proposition 3.10 gives it a $\mathbf{Z}/4$-lift on Γ. Lemma 3.11, applied to Γ, then gives a continuous $W_2(k)$-lift of $\bar\rho$ on $G_{K,S}$. □

Corollary 3.12. *Let K be a global field, let k be a finite field of characteristic two, and let $n \geq 1$. Every continuous representation $\bar\rho : G_K \to \mathrm{GL}_n(k)$ admits a continuous lift to $\mathrm{GL}_n(W_2(k))$.*

Proof. If $\mathrm{char}\, K = 2$, then [2, Theorem 1.7] gives a $W(k)$-lift, whose reduction modulo 4 is the desired lift. Otherwise the finite image of $\bar\rho$ cuts out a finite Galois extension E/K. Choose a finite set S containing its ramification; in the number-field case include the archimedean primes and the primes above 2. Then $\bar\rho$ factors through $G_{K,S}$. Theorem 3.1 supplies a continuous $W_2(k)$-lift on $G_{K,S}$; its composition with the quotient map $G_K \to G_{K,S}$ is the required lift. This proves Theorem 1.1. □

4. Representations of dimension at most four

Let K be a global field and let k be a finite field of characteristic p. A lift means an unrestricted continuous lift, with no prescribed flag, determinant, or ramification set. We prove the following uniform low-dimensional result.

Theorem 4.1. *Let K be a global field and let k be any finite field. Every continuous representation $G_K \longrightarrow \mathrm{GL}_n(k)$ of dimension $n \leq 4$ admits a continuous lift to $\mathrm{GL}_n(W_2(k))$.*

If $\mathrm{char}\, K = p$, then a $W(k)$-lift exists by [2, Theorem 1.7]. We may therefore assume $\mathrm{char}\, K \neq p$. The characteristic-two coefficient case then follows from Corollary 3.12. For the remainder of this section, assume that p is odd and $\mathrm{char}\, K \neq p$, and put $R = W_2(k)$. By Lemma 2.3, we may work with a residual image in $U_4(k)$. Write a, b, c for its first-superdiagonal coordinates as in (4.1). If a coordinate vanishes identically, then the three-layer cases of Propositions 4.6 and 4.5 apply. Otherwise let d be the dimension of the k-span of the image of (a, b, c). Propositions 4.8

and 4.9 treat $d = 1$ and $d = 3$, respectively. For $d = 2$, Proposition 4.10 treats independent outer coordinates a, c. For proportional outer coordinates we use Proposition 4.12 if $p \geq 5$ and Proposition 4.15 if $p = 3$.

4.1. **Common matrix and duality calculations.** We use $\Delta_g X = \mathrm{Ad}(g)X - X$ and call a cocycle on a finite group *cyclically trivial* if its restriction to each cyclic subgroup is a coboundary. Thus $z(g) = \Delta_g X_g$ for some X_g depending on g. We use the following form of the scalar-multiple criterion from [7, §2.5, proof of Theorem 2.8], applied to the additive subgroup $D(H)$ and the induced map $D(h) \mapsto F(h)$.

Lemma 4.2. *Let $D, F : H \longrightarrow k^m$ be additive maps. Suppose $F(h) \in kD(h)$ for every h, with $F(h) = 0$ when $D(h) = 0$, and suppose the k-span of $D(H)$ has dimension at least two. Then $F = tD$ for one $t \in k$.*

Let $\mathfrak{n}$ be the algebra of strictly upper triangular 4×4 matrices over k. Write

$$g = \begin{pmatrix} 1 & a_g & d_g & f_g \\ 0 & 1 & b_g & e_g \\ 0 & 0 & 1 & c_g \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad Z = E_{14}, \qquad \mathcal{C} = \mathfrak{n}^2 = \langle E_{13}, E_{24}, Z \rangle_k. \tag{4.1}$$

For $u = X_{31}$, $v = X_{42}$, $w = X_{41}$, direct multiplication gives

$$\begin{aligned} &(\Delta_g X)_{31} = c_g w, && (\Delta_g X)_{42} = -a_g w, \\ &(\Delta_g X)_{21} = b_g u + e_g w, && (\Delta_g X)_{43} = -b_g v + (a_g b_g - d_g) w, \\ &(\Delta_g X)_{32} = c_g v - a_g u - a_g c_g w. \end{aligned} \tag{4.2}$$

These formulas also specify every lower entry needed below. For clarity, we record the upper-triangular identities that will supply the subsequent cocycle calculations. If Y is upper triangular with diagonal (h_1, h_2, h_3, h_4), then direct multiplication gives

$$\begin{aligned} (\Delta_g Y)_{12} &= a_g(h_2 - h_1), \\ (\Delta_g Y)_{23} &= b_g(h_3 - h_2), \\ (\Delta_g Y)_{34} &= c_g(h_4 - h_3), \\ (\Delta_g Y)_{13} &= a_g Y_{23} - b_g Y_{12} + d_g(h_3 - h_1) + a_g b_g(h_1 - h_2), \\ (\Delta_g Y)_{24} &= b_g Y_{34} - c_g Y_{23} + e_g(h_4 - h_2) + b_g c_g(h_2 - h_3). \end{aligned} \tag{4.3}$$

Indeed

$$g^{-1} = \begin{pmatrix} 1 & -a_g & a_g b_g - d_g & -f_g + a_g e_g + d_g c_g - a_g b_g c_g \\ 0 & 1 & -b_g & b_g c_g - e_g \\ 0 & 0 & 1 & -c_g \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

and substituting this inverse in $gYg^{-1} - Y$ gives every displayed entry. In particular the diagonal of an upper triangular cocycle is additive. If $v_g = z(g)_{23}$ and $\delta = h_3 - h_2$, then its cocycle identity is

$$v_{gh} = v_g + v_h + b_g \delta_h, \qquad v_{[g,h]} = b_g \delta_h - b_h \delta_g,$$

where $[g, h] = ghg^{-1}h^{-1}$. For the second formula, the first formula gives $v_{g^{-1}} = -v_g + b_g \delta_g$; applying it successively to $ghg^{-1}h^{-1}$ gives the result. Thus no commutator coordinate below uses an unstated linearity assumption on the residual subgroup.

Assume $\mu_p \subset K$ and write $\chi = 1 + p\kappa$. Choose Kummer lifts A, B, L of a, b, c, and coefficient lifts D, E, F of d, e, f. Use the section

$$R_g = \begin{pmatrix} \chi_g^3 & \chi_g^2 A_g & \chi_g D_g & F_g \\ 0 & \chi_g^2 & \chi_g B_g & E_g \\ 0 & 0 & \chi_g & L_g \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{4.4}$$

Its adjacent multiplication equations hold exactly, so its actual defect e, defined by $R_gR_hR_{gh}^{-1} = 1 + pe(g,h)$, lies in $\mathcal{C}$. Put

$$\begin{aligned} N_D &= D_g + \chi_g A_g B_h + \chi_g^2 D_h - D_{gh}, \\ N_E &= E_g + \chi_g B_g L_h + \chi_g^2 E_h - E_{gh}, \\ N_F &= F_g + \chi_g^2 A_g E_h + \chi_g D_g L_h + \chi_g^3 F_h - F_{gh}. \end{aligned}$$

Then, after reduction,

$$e_{13} = N_D/p, \qquad e_{24} = N_E/p, \qquad e_{14} = N_F/p - e_{13}c_{gh}. \tag{4.5}$$

The last term is the inverse-section correction and will not be omitted. More explicitly, the raw difference $R_gR_h - R_{gh}$ has only the entries (13), (24), (14), which are respectively $\chi_{gh}N_D, N_E, N_F$. The (33) and (34) entries of R_{gh}^{-1} are χ_{gh}^{-1} and $-L_{gh}\chi_{gh}^{-1}$. Consequently the corresponding entries of $(R_gR_h - R_{gh})R_{gh}^{-1}$ are exactly $N_D, N_E, N_F - N_DL_{gh}$. Division by p and reduction gives (4.5).

Lemma 4.3. *Suppose a dual cocycle $z : P \to M_4(k)$ is upper triangular and satisfies*

$$z_{11} = z_{44}, \qquad z_{11} + z_{22} = 0. \tag{4.6}$$

At every prime where $z = \Delta X$ for one common primitive X, the local trace-cup class with the defect of (4.4) *is zero. Moreover that actual defect is locally trivial in $H^2(K_v, M_4(k))$ for every residual representation into $U_4(k)$.*

Proof. Use u, v, w as in (4.2). If some local a or c is nonzero, then $w = 0$. If also b is nonzero locally, then $u = v = 0$, so the pairing vanishes pointwise. If instead $b = 0$, then write $B = p\beta$ locally. The residual d, e are additive. Comparing D, E with local Kummer lifts gives, in the trivial quotient $\mathcal{C}/kZ$,

$$[e_{13}] = [\kappa \cup d + a \cup \beta], \qquad [e_{24}] = [\kappa \cup e + \beta \cup c]. \tag{4.7}$$

The 32 entry gives $ua = vc$, whereas $(\Delta X)_{11} + (\Delta X)_{22} = du + ev = 0$. Thus the weighted sum of (4.7) is $\kappa \cup (du + ev) + (ua) \cup \beta + \beta \cup (vc) = 0$ by graded commutativity.

If $a = c = 0$ locally, then write $A = p\alpha$, $L = p\gamma$. The module $\mathcal{C}$ is locally fixed; d, e, f are additive, and the inverse correction in (4.5) vanishes. The actual classes are

$$\begin{aligned} [e_{13}] &= [\kappa \cup d + \alpha \cup b], \\ [e_{24}] &= [\kappa \cup e + b \cup \gamma], \\ [e_{14}] &= [2\kappa \cup f + \alpha \cup e + d \cup \gamma]. \end{aligned} \tag{4.8}$$

Their weighted trace is

$$\kappa \cup (du + ev + 2fw) + \alpha \cup (bu + ew) + (bv + dw) \cup \gamma.$$

The last two characters vanish by the $21, 43$ entries; the first is $(\Delta X)_{11} - (\Delta X)_{44} = 0$.

These are the actual cup products: for a 0-cochain X the local trace contains $\mathrm{Ad}(\rho(gh))X$. When $u = v = w = 0$ this makes no difference; when $w = 0$ only the fixed quotient $\mathcal{C}/kZ$ contributes; and when $a = c = 0$ the entire module $\mathcal{C}$ is fixed. The computation therefore proves the first assertion without discarding an action term.

For the second assertion, Böckle's local lifting theorem [2, Theorem 1.1] gives a $W_2(k)$-lift at every prime. Hence the restriction of the obstruction class is zero. □

For later use, the calculation also shows that when z is upper triangular with only $z_{11} = z_{44}$ imposed, its sole potentially nonzero local term occurs at $b = 0$ with some a or c nonzero, and is

$$[\kappa \cup (du + ev)] = [\kappa \cup (z_{11} + z_{22})]. \tag{4.9}$$

For upper triangular z, the global trace $\mathrm{tr}(e \cup z)$ is identically zero because e is strictly upper triangular. Consequently Lemma 2.8 and Proposition 2.10 turn the finite-group normalizations below into global lifting results.

The preceding local-vanishing assertion concerns the obstruction class, so it applies to every section used below. We therefore need not check local vanishing again in the individual cases. After the prime-to-p reduction of Lemma 2.3, assume $\mu_p \subset K$. For each representation fix the basis and section first, and then choose a finite set T containing all ramification of the representation, χ, and the chosen Kummer and coefficient cochains. In the number-field case also include all archimedean primes and all primes above p; in the function-field case require $T \neq \varnothing$. Since $\operatorname{char} K \neq p$, there are no primes above p in the latter case. Enlarge it by Lemma 2.8. Every dual class is represented by an inflated cyclically trivial cocycle on P. All subsequent coboundary normalizations factor through P and introduce no ramification. Proposition 2.10 reduces the proof to the global trace-cup identity and the sum of the local contractions. When that sum is a reciprocity sum, the global scalar classes used below are unramified outside T, where their local second cohomology classes vanish. This convention applies throughout the section.

Lemma 4.4. *Suppose a, b are nonzero on $P \subset U_4(k)$ and $\zeta_0 = 1 + t_0 Z \in P$ with $t_0 \neq 0$. A cyclically trivial cocycle z with $z(\zeta_0) = 0$ is upper triangular and satisfies $z_{11} = z_{44}$. For every pure-central $\zeta \in P$,*

$$z(\zeta) = r_\zeta E_{13} + s_\zeta E_{24} + t_\zeta Z, \qquad a s_\zeta = c r_\zeta.$$

In particular, if a, c are independent, then

$$z(\zeta) \in kZ. \tag{4.10}$$

If $c = a$, then $r_\zeta = s_\zeta$.

Proof. The commuting cocycle identity puts $z(g)$ in $\operatorname{Cent}(Z)$. Thus its entries $21, 31, 41, 42, 43$ vanish, its outer diagonal entries coincide, and its remaining lower entry z_{32} is additive. For $b_g \neq 0$, cyclic innerness and (4.2) give $z_{32}(g) = 0$. If a_g or c_g is nonzero, then $w = 0$, which in turn gives $u = v = 0$. If both vanish, then the 32 formula itself is zero. Translation by an element with $b \neq 0$ also kills this entry on $\ker b$.

For a nonidentity pure-central $\zeta = 1 + tZ$, the commuting identity shows that $z(\zeta)$ is P-invariant. A cyclic primitive X gives possible diagonal $(tX_{41}, 0, 0, -tX_{41})$, zero 23 entry, and possible upper entries $12 = tX_{42}$, $13 = tX_{43}$, $24 = -tX_{21}$, $34 = -tX_{31}$, and 14. Commutation with P in position 12 kills X_{41} because $a \neq 0$; positions $13, 24$ then kill entries $12, 34$ because $b \neq 0$. The remaining 14 equation is $a s_\zeta = c r_\zeta$. □

4.2. **Three-layer representations.**

Proposition 4.5. *Every representation with image in the three-layer block group of type $(1, 2, 1)$ lifts.*

Proof. Write

$$g = \begin{pmatrix} 1 & A_g & f_g \\ 0 & I_2 & B_g \\ 0 & 0 & 1 \end{pmatrix}, \qquad Z = E_{14}.$$

Then $(g - I)(h - I) = (A_g B_h)Z$ and $[g, h] = I + (A_g B_h - A_h B_g)Z$. If all $A_g B_h$ vanish, then Lemma 2.7 applies. Otherwise we show that every cyclically trivial $c : P \to M_4(k)$ can be normalized so that

$$[c(g), Z] = 0 \quad (g \in P), \qquad c(I + tZ)_{11} = 0 \quad (I + tZ \in P). \tag{4.11}$$

If a nontrivial pure-central z_0 exists, then normalize $c(z_0) = 0$. Centrality gives the first assertion. For another pure-central z, $c(z)$ is invariant under P and belongs to $\operatorname{im} \Delta_z$. Its diagonal blocks have shape $(\alpha, 0, -\alpha)$. Commutation with P gives $\alpha A_g = \alpha B_g = 0$. Some A_g or B_g is nonzero, so $\alpha = 0$.

If no nontrivial pure center exists, then P is abelian and the nonzero bilinear form $A_g B_h$ is symmetric. Since p is odd, some x has $A_x B_x \neq 0$: otherwise evaluate at gh for a pair with $A_g B_h \neq 0$. Normalize $c(x) = 0$. Then $c(g)$ commutes with $(x - I)^2 = (A_x B_x)Z$. The second assertion of (4.11) is vacuous in the absence of pure center.

For a common primitive X on a subgroup D, put $w = X_{41}$. The lower blocks of $\Delta_g X$ are $B_g w$ and $-wA_g$. Equation (4.11) makes these zero. Hence $w = 0$ if some A_g or B_g is nonzero on D; otherwise a nontrivial pure-central element gives $0 = c(I + tZ)_{11} = tw$. If $D = 1$, then take $X = 0$.

Assume $\mu_p \subset K$ and choose Kummer lifts $\widetilde{A}, \widetilde{B}$. The section

$$R_g = \begin{pmatrix} \chi_g^2 & \chi_g \widetilde{A}_g & \widetilde{f}_g \\ 0 & \chi_g I_2 & \widetilde{B}_g \\ 0 & 0 & 1 \end{pmatrix}$$

has actual defect in the fixed line kZ, with numerator $\widetilde{f}_g + \chi_g^2 \widetilde{f}_h + \chi_g \widetilde{A}_g \widetilde{B}_h - \widetilde{f}_{gh}$. Both the normalized cocycle and its common local primitives annihilate kZ. Finite-set duality proves global vanishing, and prime-to-p descent completes the proof. □

Proposition 4.6. *Every representation with image contained in the three-layer block group of type* $(1,1,2)$ *or* $(2,1,1)$ *lifts to* $\mathrm{GL}_4(R)$.

Proof. It suffices to treat $(1,1,2)$; inverse transpose treats the other type. By Lemma 2.3, we may assume $\mu_p \subset K$. Write the image as

$$g = (a_g, B_g, C_g) = \begin{pmatrix} 1 & a_g & C_g \\ 0 & 1 & B_g \\ 0 & 0 & I_2 \end{pmatrix}, \qquad B_g, C_g \in k^{1\times 2}.$$

Then a, B are additive and $C_{gh} = C_g + C_h + a_g B_h$. If $B(P)$ spans at most one line, then choose the last two basis vectors so that $B_g = (0, b_g)$. Relative to blocks $(1,2,1)$ the resulting matrices belong to the group of Proposition 4.5, which applies. If $a = 0$, then Lemma 2.7 applies. We may therefore assume $a \neq 0$ and that $B(P)$ spans k^2. The image is then nonabelian: commutativity would give $B_g = (a_g/a_x)B_x$ for any x with $a_x \neq 0$. Its pure-central rows consequently span a nonzero subspace.

A pure-central element is $(0,0,Z)$. For an arbitrary matrix X, use block notation $T = X_{31}$, $s = X_{21}$, $r = X_{32}$. Multiplication gives

$$(\Delta_g X)_{31} = 0, \qquad (\Delta_g X)_{21} = B_g T, \qquad (\Delta_g X)_{32} = -a_g T, \tag{4.12}$$

$$(\Delta_g X)_{11} = a_g s + C_g T, \qquad (\Delta_g X)_{33} = -TC_g + a_g T B_g - r B_g. \tag{4.13}$$

For $z = (0,0,Z)$ the diagonal blocks of $\Delta_z X$ are $(ZT, 0, -TZ)$, and its 12 and 23 blocks are Zr and $-sZ$.

If $c : P \to M_4(k)$ is cyclically trivial, then its bottom-left block is zero. Consequently

$$s_g = c(g)_{21}, \qquad F_g = -c(g)_{32}$$

are additive, and pointwise $s_g = B_g T_g$, $F_g = a_g T_g$.

For a nonzero pure-central $z = (0,0,Z)$, the 12 and 23 blocks of the commuting cocycle identity give

$$ZF_g = a_g Z T_z, \qquad s_g = B_g T_z. \tag{4.14}$$

If the central rows span k^2, then two independent such rows yield a fixed T_0 with $F_g = a_g T_0$. For $a_g \neq 0$, pointwise innerness gives $s_g = B_g T_0$; additivity extends this to all g. Subtract a coboundary with bottom-left block T_0. We now have $s = F = 0$, so c is block upper triangular. Equation (4.14) then forces $T_z = 0$ at every nonzero pure-central element. Put

$$c(z)_{12} = u_z, \quad c(z)_{23} = W_z, \quad c(g)_{11} = h_g, \quad c(g)_{33} = D_g.$$

The remaining central identity is

$$Z(D_g - h_g I_2) = a_g W_z - u_z B_g. \tag{4.15}$$

Choose g, h with $R_0 = a_h B_g - a_g B_h \neq 0$, which is possible because the image is nonabelian. Multiply (4.15) for g by a_h and subtract its instance for h multiplied by a_g. This gives $ZL = -u_z R_0$, where

$$L = a_h(D_g - h_g I_2) - a_g(D_h - h_h I_2).$$

Choose a nonzero coordinate $(R_0)_j$ and put $v = -L_{\bullet j}/(R_0)_j$. Then $u_z = Zv$ for every actual central row Z. Subtracting the coboundary with 32 block v kills every u_z. Choosing g_0 with $a_{g_0} \neq 0$ now gives $W_z = ZH$ and $D_g - h_g I_2 = a_g H$ for a fixed H. Pointwise innerness at z says $ZH \in kZ$. Choose independent actual central rows Z_1, Z_2. Their sum is again an actual central row. If $Z_i H = \lambda_i Z_i$ and $(Z_1 + Z_2)H = \lambda_{12}(Z_1 + Z_2)$, then independence gives $\lambda_1 = \lambda_2 = \lambda_{12}$, hence $H = \lambda I_2$. Only closure under addition is used; the set of central rows need not be a k-vector space. Subtract the coboundary with 21 block $-\lambda$. Then $W = 0$ and $D_g = h_g I_2$. For $a_g \neq 0$, a cyclic primitive has $T_g = 0$, so $D_g = -r_g B_g$ has rank at most one. Hence $h_g = 0$. Additivity gives $h = D = 0$ everywhere, and trace zero kills the remaining diagonal block.

If the central rows span only one line, then choose it to be $k(1, 0)$. The commutator formula implies $B_g = (b_g, \lambda a_g)$ with $\lambda \neq 0$. Subtracting a coboundary with bottom-left block T_z for one nonzero central element gives $s = F_1 = 0$ by (4.14). For $a_g \neq 0$, pointwise innerness gives $(T_g)_1 = 0$ and $0 = s_g = \lambda F_{g,2}$; for $a_g = 0$ we already have $F_g = 0$. Thus $F = 0$. The same central identities then give $T_z = 0$ for all nonzero central z, and (4.15) applies again. The preceding elimination makes $u_z = 0$. Cyclic innerness makes W_z a multiple of Z, and a fixed g_0 with $a_{g_0} \neq 0$ shows $W_z = wZ$ for one scalar w. Subtract the coboundary with 21 block $-w$. Now $W = 0$, and the first row of D_g is $(h_g, 0)$. For $a_g \neq 0$, $T_g = 0$ and $D_g = -r_g(b_g, \lambda a_g)$. Its first row then forces $(r_g)_1 = 0$ and $h_g = 0$. Additivity extends this to all g. Both central-span cases have therefore produced

$$c_{31} = c_{21} = c_{32} = c_{11} = 0 \quad \text{in block notation.} \tag{4.16}$$

Choose Kummer lifts $A, \widetilde{B}$ of a, B and any lift $\widetilde{C}$. The section

$$R_g = \begin{pmatrix} \chi_g^2 & \chi_g A_g & \widetilde{C}_g \\ 0 & \chi_g & \widetilde{B}_g \\ 0 & 0 & I_2 \end{pmatrix}$$

has actual defect in the fixed module $\mathcal{C} = kE_{13} + kE_{14}$. If $a = 0$ locally, then write $A = p\alpha$. Comparing $\widetilde{C}$ with a local Kummer lift of the now additive row C gives

$$[e_v] = [\kappa \cup C + \alpha \cup B] \quad \text{in } H^2(K_v, \mathcal{C}). \tag{4.17}$$

Indeed the top-right raw defect is $\widetilde{C}_g + \chi_g^2 \widetilde{C}_h + \chi_g A_g \widetilde{B}_h - \widetilde{C}_{gh}$; replacement by a Kummer lift leaves $(\chi_g^2 - \chi_g)C_h$ and $p\alpha_g B_h$ modulo p^2. For a common primitive of a normalized dual cocycle, (4.16) gives the same alternatives $T = 0$, or $BT = CT = 0$. The global trace cup is identically zero, and all local trace-cup classes vanish. Since $\mathcal{C}$ is fixed, this calculation includes the cup action. Apply Lemma 2.8 and Proposition 2.10.

□

4.3. **A regular element and a one-dimensional first projection.**

Proposition 4.7. *An abelian image in $U_4(k)$ containing a regular unipotent element lifts.*

Proof. Choose a basis in which a regular element has centralizer $k[N]$, with $N = E_{12} + E_{23} + E_{34}$. Then every image element is $g = I + a_g N + b_g N^2 + c_g N^3$. The polynomial section $S_g = I + [a_g]N + [b_g]N^2 + [c_g]N^3$ has defect in $Nk[N]$. If a local image lies in $1 + N^r k[N]$, then its defect lies in $N^r k[N]$; this follows coefficientwise from the free polynomial basis over R.

We prove the trace assertion

$$[f(N), X] \in k[N], \quad f(N) - 1 = t_r N^r + \cdots,\ t_r \neq 0 \quad \Longrightarrow \quad \operatorname{tr}(N^j X) = 0\ (j \geq r). \tag{4.18}$$

For $r = 1$, take $H = \operatorname{diag}(0, 1, 2, 3)$. Since $[f, H] = Nf'(N)$ is N times a unit, each N^j is $[f, h_j(N)H]$. Writing $q(N) = [f, X]$, trace cyclicity gives $\operatorname{tr}(N^j X) = -h_j(0)q(0)\operatorname{tr}(H)$. Since p is odd, $4q(0) = \operatorname{tr}(q) = 0$ gives $q(0) = 0$. For $r = 2$ the same argument works because $Nf'(N) = N^2(2t_2 + 3t_3 N)$ and $2t_2$ is nonzero in odd characteristic. Thus each N^j, $j \geq 2$, is again a commutator $[f, h_j(N)H]$. For $r = 3$, $f = 1 + t_3 E_{14}$, and the $22, 33$ entries of $[f, X]$ are zero. Its polynomial shape forces its 11 entry $t_3 X_{41}$ to be zero. This proves (4.18).

Normalize a cyclically trivial dual cocycle z to vanish at a regular element. Commutativity gives $z(g) \in k[N]$. At a nontrivial local image choose f of least order r in N. For one common

local primitive X_v, $[f, X_v] = z(f)f \in k[N]$, so (4.18) annihilates the local defect in $N^r k[N]$. At a trivial image the polynomial section is the identity. Globally $\mathrm{tr}(e \cup z) = 0$ because $e \in Nk[N]$. The polynomial algebra is fixed by P, so the cup action introduces no additional term. Finite-set duality completes the proof. □

Proposition 4.8. *Suppose the first-superdiagonal projection of $P \subset U_4(k)$ spans one k-line containing a regular vector. Every representation with image P lifts.*

Proof. The abelian case is Proposition 4.7. Otherwise a diagonal change makes $a = b = c$, and the commutator subgroup is a nontrivial subgroup of $1 + kZ$. Normalize a cyclically trivial cocycle at a nonzero element of that subgroup. Lemma 4.4 makes it upper triangular with equal outer diagonal entries. The only possible local contribution in (4.9) requires $b = 0$ and some outer character nonzero, which cannot occur when $a = b = c$. Thus every local contraction vanishes, and duality applies. □

4.4. **Three independent first-superdiagonal coordinates.**

Proposition 4.9. *If the k-span of the first-superdiagonal image of $P \subset U_4(k)$ is k^3, then every representation with image P lifts.*

Proof. Write $Q = [P, P]$. Its second-superdiagonal coordinates span k^2, since commutators have coordinates

$$(a_g b_h - a_h b_g,\ b_g c_h - b_h c_g).$$

The induced map $\bigwedge^2 k^3 \to k^2$ sends $e_1 \wedge e_2$ to $(1, 0)$ and $e_2 \wedge e_3$ to $(0, 1)$. Pairs of vectors from the actual first-superdiagonal image span $\bigwedge^2 k^3$, so their commutator coordinates span k^2. Also P contains a nonzero pure-central element: the triple commutator coordinate

$$a_l(b_g c_h - b_h c_g) - (a_g b_h - a_h b_g)c_l$$

cannot vanish on all triples from a spanning set, because its value on $(l, g, h) = (e_1, e_2, e_3)$ is one. These multilinear arguments do not require the standard basis vectors themselves to occur in P.

Normalize a cyclically trivial z at one such pure-central element. Lemma 4.4 makes it upper triangular with $h_1 = h_4$ and gives (4.10). Its diagonal entries h_i are additive.

For $q \in Q$, all diagonal entries of $z(q)$ vanish. Its first superdiagonal coordinates are additive on Q. Cyclic innerness gives

$$(z(q)_{12}, -z(q)_{34}) \in k(d_q, e_q).$$

Indeed, for a cyclic primitive, $w = 0$ when $(d_q, e_q) \neq 0$, and $d_q u = e_q v = 0$ from the diagonal. The displayed pair equals $(d_q X_{32} + f_q v, e_q X_{32} + f_q u)$, which has the asserted proportionality also when exactly one coordinate is zero. If both are zero, then (4.10) applies. Lemma 4.2 and subtraction of $\Delta(tE_{32})$ now give $z(q)_{12} = z(q)_{34} = 0$ for all $q \in Q$. This preserves all preceding conditions. Put $v_q = z(q)_{23}$. The identity $gqg^{-1} = q\zeta$ with ζ pure central gives, in entries $13, 24$,

$$a_g v_q = d_q(h_3 - h_1)(g), \qquad c_g v_q = e_q(h_2 - h_1)(g). \tag{4.19}$$

Choosing g_0 with $a_{g_0} \neq 0$ yields $v_q = \lambda d_q$ and $h_3 - h_1 = \lambda a$. Independence of (d_q, e_q) in the second equation gives $\lambda c_g = 0$ and $h_2 - h_1 = 0$ for every g. Hence $\lambda = 0$ and all h_i coincide. Trace zero gives $4h_1 = 0$, hence $h_1 = 0$. The normalized cocycle satisfies (4.6). Lemma 4.3 and finite-set duality finish the proof. □

4.5. **A plane with independent outer coordinates.**

Proposition 4.10. *Suppose the first-superdiagonal image spans a k-plane, a and c are k-linearly independent, and $b \neq 0$. Assume that P contains a regular element. Every representation with image P lifts.*

Proof. Write $b = \alpha a + \beta c$, where $(\alpha, \beta) \neq (0,0)$. For a commutator, its second-superdiagonal projection is $s(\beta, \alpha)$ with $s = a_g c_h - a_h c_g$. Some such s is nonzero. Independence of a, c gives a nonzero triple commutator, hence a nontrivial pure center. Lemma 4.4 normalizes every cyclically trivial cocycle to be upper triangular with $h_1 = h_4$ and gives (4.10). Set $\tau = h_1 + h_2$. For $q \in Q = [P,P]$, write $u_q = z(q)_{12}$, $v_q = z(q)_{23}$, $w_q = z(q)_{34}$. Comparing the $13, 24$ coordinates of $gq = qg\zeta$ gives

$$d_q(h_3 - h_1) = av_q - bu_q, \qquad e_q(h_1 - h_2) = bw_q - cv_q. \tag{4.20}$$

At $d_q e_q \neq 0$, cyclic innerness gives $u_q/d_q = -w_q/e_q$. At $d_q = 0, e_q \neq 0$ it gives $u_q = 0$; at $e_q = 0, d_q \neq 0$ it gives $w_q = 0$. These follow from the same primitive calculation used in Proposition 4.9.

Trace zero and $h_1 = h_4$ give $h_3 - h_2 = -2\tau$. The 23 cocycle identity implies, exactly,

$$v_{[g,h]} = -2(b_g \tau_h - b_h \tau_g). \tag{4.21}$$

If $\alpha\beta \neq 0$, then choose $q = [g,h]$ with $s \neq 0$ and put $d = s\beta$, $e = s\alpha$, $U = u_q/d$, $v_0 = v_q$. Solving (4.20) and the trace identity gives

$$h_1 = -\frac{v_0}{4}\left(\frac{a}{d} + \frac{c}{e}\right) = -\frac{v_0 b}{4s\alpha\beta}, \qquad \tau = Ub - \frac{v_0(\alpha a - \beta c)}{2s\alpha\beta}.$$

Substituting the latter expression into (4.21) for this commutator yields $v_0 = -2v_0$, hence $3v_0 = 0$. For $p \neq 3$, $v_0 = 0$ and $\tau = Ub$. For $p = 3$, choose a regular x. Since $x^3 = 1 + a_x b_x c_x Z$, (4.10) gives zero $13, 24$ entries of $z(x^3)$. But $z(x^3) = (\mathrm{Ad}(x) - 1)^2 z(x)$, whose two entries are

$$a_x b_x (h_1 - 2h_2 + h_3)(x), \qquad b_x c_x (h_2 - 2h_3 + h_4)(x).$$

Each parenthesis equals $-h_1(x)$ in characteristic three. Thus $h_1(x) = 0$, and the displayed formula for h_1 forces $v_0 = 0$. Again $\tau \in kb$. If $\alpha = 0$ and $\beta \neq 0$, then $e_q = 0$ for every $q \in Q$. For $d_q \neq 0$, cyclic innerness gives $w_q = 0$, so the second equation of (4.20) gives $v_q = 0$. For $d_q = 0$ use (4.10). Equation (4.21) therefore gives $b \wedge \tau = 0$, or $\tau \in kb$. The case $\beta = 0$ is symmetric.

In every case subtract $\Delta(tE_{32})$ where $\tau = tb$. This preserves upper triangularity and $h_1 = h_4$, and kills $h_1 + h_2$. Lemma 4.3 and duality apply. □

4.6. **Proportional outer coordinates in odd characteristic.**

Lemma 4.11. *Suppose p is odd, $c = a$, and $(a,b)(P)$ spans k^2. Let z be an upper triangular cyclically trivial cocycle, write $h_i = z_{ii}$, and suppose $h_1 = h_4$ and*

$$z(\zeta) \in k(E_{13} + E_{24}) + kZ \qquad (\zeta \in P \cap (1 + kZ)). \tag{4.22}$$

After subtracting $\Delta(rE_{32})$ for some $r \in k$, it satisfies $h_3 - h_2 = 2sa$ for a scalar $s \in k$ with $3s = 0$.

Proof. For $q \in Q = [P,P]$, the second-superdiagonal coordinates are $(d_q, -d_q)$. Its cocycle value has zero diagonal. Cyclic innerness gives $z(q)_{12} = z(q)_{34}$: when $d_q \neq 0$, the zero $21, 43$ entries first give $w = 0$, the zero diagonal gives $u = v = 0$, and the two entries both equal $d_q X_{32}$; when $d_q = 0$, use (4.22). Write $F_q = z(q)_{12}$ and $V_q = z(q)_{23}$. These are additive on Q and vanish on its pure-central subgroup.

Write $S(\zeta)$ for the common $13, 24$ entry in (4.22); no k-linearity of S is needed. Since $[g,q] = 1 - 2a_g d_q Z$, the $13, 24$ entries of the conjugation cocycle identity give

$$S([g,q]) = a_g V_q - b_g F_q - d_q(h_3 - h_1)(g),$$
$$S([g,q]) = b_g F_q - a_g V_q + d_q(h_1 - h_2)(g).$$

Subtracting yields

$$2a_g V_q - 2b_g F_q = d_q(h_3 - h_2)(g). \tag{4.23}$$

Some commutator q_0 has $d_{q_0} \neq 0$, because $(a,b)(P)$ spans k^2. Put $r = F_{q_0}/d_{q_0}$ and $s = V_{q_0}/d_{q_0}$. Equation (4.23) for q_0 gives $h_3 - h_2 = 2sa - 2rb$. Substituting this identity back into the same equation for arbitrary q yields

$$a_g(V_q - sd_q) = b_g(F_q - rd_q) \qquad (g \in P).$$

Two independent actual values of (a_g, b_g) then give $F_q = rd_q$ and $V_q = sd_q$. Subtracting $\Delta(rE_{32})$ makes $F = 0$ and $h_3 - h_2 = 2sa$, while preserving upper triangularity and the outer diagonal. The 23 cocycle identity at $q = [g,h]$ now gives

$$sd_q = z(q)_{23} = 2s(b_g a_h - b_h a_g) = -2sd_q.$$

Hence $3s = 0$. □

Proposition 4.12. *Suppose $p \geq 5$, the first-superdiagonal image spans a plane containing a regular vector, and its outer coordinates are proportional. Every representation with image P lifts.*

Proof. Normalize $c = a$. A commutator has coordinates $(d, -d)$ with $d \neq 0$, and $[g,q] = 1 - 2a_g dZ$ supplies a nontrivial pure center. Lemmas 4.4 and 4.11 normalize a cyclically trivial cocycle so that $h_1 = h_4$ and $h_2 = h_3$, because 3 is invertible. Trace zero then gives $h_1 + h_2 = 0$. Apply Lemma 4.3 and duality. □

4.7. **Proportional outer coordinates in characteristic three.** We now treat the remaining plane over an arbitrary finite field of characteristic three. The argument uses a global cup product whose local restrictions account for the primes omitted from the restricted sum in the obstruction pairing. In particular, no cyclicity or abelianness condition is imposed on the local residual images.

Lemma 4.13. *Let k be a finite field of characteristic three, and let $P \subset U_4(k)$ have a regular element. Suppose its first-superdiagonal k-span has dimension two and its outer coordinates are proportional. After a change of basis one may write these coordinates as (a,b,a) and choose $x = 1 + N \in P$, where $N = E_{12} + E_{23} + E_{34}$. Every cyclically trivial cocycle $\eta : P \to M_4(k)$ is cohomologous to an upper triangular cocycle with*

$$\operatorname{diag} \eta(g) = (0, -sa_g, sa_g, 0) \tag{4.24}$$

for one scalar $s \in k$.

Proof. First remove the nonzero outer proportionality constant by diagonal conjugation. Choose a regular element x_0 with adjacent coordinates (a_0, b_0, a_0) and put $T_0 = x_0 - I$. If v_4 is the last standard basis vector, then the matrix

$$S = (T_0^3 v_4,\ T_0^2 v_4,\ T_0 v_4,\ v_4)$$

is upper triangular with diagonal $(a_0^2 b_0, a_0 b_0, a_0, 1)$ and satisfies $T_0 S = SN$. Thus $S^{-1} x_0 S = 1 + N$. The change $g \mapsto S^{-1} g S$ divides both outer adjacent coordinates by a_0 and the middle one by b_0. It preserves their outer equality and the two-dimensional span, as required.

Use cyclic triviality to subtract a coboundary and arrange $\eta(x) = 0$. Since $x^3 = 1 + Z$, also $\eta(1 + Z) = 0$. Lemma 4.4 makes η upper triangular. Its additive diagonal entries satisfy

$$h_1 = h_4, \qquad 2h_1 + h_2 + h_3 = 0. \tag{4.25}$$

For every pure-central $z_t = 1 + tZ \in P$, commutativity with x gives $\eta(z_t) \in k[N]$. Cyclic innerness at z_t gives zero $22, 33, 23$ entries, so

$$\eta(z_t) = F(t)N^2 + G(t)Z. \tag{4.26}$$

Both functions are additive on the actual subgroup of parameters $\{t \in k : 1 + tZ \in P\}$. We do not extend this subgroup to a k-vector space, and no k-linearity of F or G is asserted. Since $\eta(x^3) = 0$, we have $F(1) = 0$.

Let $q = [x,g]$. Its second-superdiagonal coordinates are $(d_q, -d_q)$, where $d_q = b_g - a_g$, and $[x,q] = z_{d_q}$. The diagonal of $\eta(q)$ is zero because the h_i are characters. Cyclic innerness at q gives $\eta(q)_{12} = \eta(q)_{34}$. To verify this when $d_q \neq 0$, its zero $21, 43$ entries force $w = 0$ for a

cyclic primitive, its zero diagonal then forces $u = v = 0$, and its $12, 34$ entries both equal $d_q X_{32}$. When $d_q = 0$, use (4.26). Since $\eta(x) = 0$ and $\eta(q)$ commutes with Z,

$$\eta([x, q]) = \Delta_x \eta(q).$$

The $13, 24$ entries on the right are opposites, whereas on the left both equal $F(d_q)$. Thus

$$F(b_g - a_g) = 0 \qquad (g \in P). \tag{4.27}$$

The exact matrix identity $g^3 = 1 + a_g^2 b_g Z$ gives $\eta(g^3) = (\mathrm{Ad}(g) - 1)^2 \eta(g)$. The $13, 24$ entries of the latter expression are

$$a_g b_g (h_1 - 2h_2 + h_3)(g), \qquad a_g b_g (h_2 - 2h_3 + h_4)(g).$$

In characteristic three, (4.25) makes each parenthesis equal to $-h_1(g)$. Hence

$$F(a_g^2 b_g) = -a_g b_g h_1(g). \tag{4.28}$$

Apply this equation to $x^i g$, for $i = 0, 1, -1$. All three elements have the same value of h_1 because $\eta(x) = 0$. In characteristic three,

$$\sum_{i=0,1,-1} (a + i)^2 (b + i) = -(b - a), \qquad \sum_{i=0,1,-1} (a + i)(b + i) = 2.$$

Using additivity of F and (4.27), the sum of the left sides of (4.28) is zero; the sum of its right sides is $-2h_1(g) = h_1(g)$. Thus $h_1 = h_4 = 0$. Every argument of F used in this calculation is the parameter of an actual central element of P.

Finally apply Lemma 4.11. Its hypotheses follow from (4.26); its coboundary preserves $h_1 = h_4 = 0$ and gives $h_3 - h_2 = 2sa$. Trace zero proves (4.24). □

Lemma 4.14. *Let H be a profinite group and let $a : H \to k$ be an additive continuous character, where k is a finite field of characteristic three. Write β_0 for the connecting map associated with the constant coefficient sequence*

$$0 \longrightarrow k \xrightarrow{z \mapsto 3[z]} W_2(k) \longrightarrow k \longrightarrow 0.$$

Then the continuous 2-cochain

$$B_3(a) = a \cup a^2 + a^2 \cup a$$

is a cocycle, and

$$[B_3(a)] = -\beta_0(a^3). \tag{4.29}$$

Here powers denote pointwise powers in k; a^3 is additive, whereas a^2 need not be.

Suppose moreover that a representation $H \to U_4(k)$ has coordinates (a, b, a, d, e, f) as in (4.1), and that $b = \beta a$ for a fixed $\beta \in k^*$. Then $t = e - d$ is additive and

$$\beta\, \beta_0(a^3) = -[a \cup t]. \tag{4.30}$$

For $H = G_{K_v}$ with $\mu_3 \subset K_v$, this becomes

$$[\kappa \cup a^3] = \beta^{-1} [a \cup t]. \tag{4.31}$$

Proof. Let $V : k \to W_2(k)$ denote Verschiebung. The length-two Witt addition formulas in characteristic three are

$$[x] + [y] - [x + y] = V(-x^2 y - x y^2), \qquad 3[z] = V(z^3).$$

Apply the first formula with $x = a(g)^3$ and $y = a(h)^3$. The resulting difference is

$$V\big(-(a(g)^2 a(h) + a(g) a(h)^2)^3\big) = 3\big[-a(g)^2 a(h) - a(g) a(h)^2\big].$$

Thus the Teichmüller lift $g \mapsto [a(g)^3]$ represents $\beta_0(a^3)$ by the cochain $-B_3(a)$. This proves both cocyclehood and (4.29). In particular, the cube cannot in general be omitted when $k \neq \mathbf{F}_3$.

Write d for the scalar cochain differential, to distinguish it from the matrix coordinate d. Matrix multiplication gives

$$\mathrm{d}d = -a \cup b, \qquad \mathrm{d}e = -b \cup a, \qquad \mathrm{d}f = -a \cup e - d \cup a.$$

Since $\mathrm{d}(a^2) = -2a\cup a = a\cup a$, the functions $r = d+\beta a^2$ and $t = e-d$ are additive. Substitution gives

$$\mathrm{d}f = \beta B_3(a) - a\cup t - a\cup r - r\cup a.$$

The last two cup products cancel in cohomology. Thus $\beta[B_3(a)] = [a\cup t]$, proving (4.30). For a local absolute Galois group, Lemma 2.6 gives $\beta_0(a^3) = -[\kappa\cup a^3]$, and (4.31) follows. □

Proposition 4.15. *Let K be a global field of characteristic different from three, and k any finite field of characteristic three. Suppose $\rho : G_K \to U_4(k)$ has image P containing a regular element, and its first-superdiagonal k-span is a plane with proportional outer coordinates. Then ρ admits a continuous lift to $\mathrm{GL}_4(W_2(k))$.*

Proof. Assume $\mu_3 \subset K$ by Lemma 2.3. Fix the basis and element x of Lemma 4.13, and use the same Kummer lift for the two outer characters in (4.4). Choose T by the convention following Lemma 4.3; in particular it is fixed before the dual class. Normalize a representative η of any dual class using Lemma 4.13. Its diagonal is (4.24) for some $s \in k$, and its global trace cup with the actual defect vanishes because it is upper triangular.

For each $v \in T$ choose one matrix X_v which is a primitive for the restriction of η to the whole local group. The calculation in Lemma 4.3 and (4.9) gives the local trace-cup class

$$[\mathrm{tr}(e|_v \cup X_v)] = \begin{cases} -[s\kappa\cup a]_v, & b|_{G_{K_v}} = 0, \\ 0, & b|_{G_{K_v}} \neq 0. \end{cases} \tag{4.32}$$

This uses the actual defect, including its inverse-section correction and cup action. If $s = 0$, then all terms vanish and duality applies. Assume $s \neq 0$.

We construct a global scalar class to evaluate their total invariant. The upper triangular cocycle identities and (4.24) give

$$\mathrm{d}\eta_{12} = s(a\cup a), \qquad \mathrm{d}\eta_{34} = s(a\cup a).$$

It follows that

$$J = \eta_{12} + \eta_{34} + sa^2 \tag{4.33}$$

is an additive global character: its differential is $3s(a\cup a) = 0$. In particular $[a\cup J]$ is a global k-valued cohomology class.

First consider a prime at which a_v, b_v are both nonzero. For its common primitive X put

$$\begin{gathered} u = X_{31}, \qquad v' = X_{42}, \qquad w = X_{41}, \\ P_0 = X_{21}, \qquad Q_0 = X_{43}, \qquad m = X_{32}, \\ R_0 = X_{22} - X_{11} + X_{44} - X_{33}. \end{gathered}$$

The zero lower entries give $u = v' = w = 0$. The first and last diagonal entries then give $P_0 = Q_0 = 0$, and the second diagonal entry gives

$$bm = -sa. \tag{4.34}$$

Since $s \neq 0$ and $a_v \neq 0$, $m \neq 0$. Thus $b = \beta a$ locally for the nonzero scalar $\beta = -s/m$. Direct multiplication of $\rho(g)X\rho(g)^{-1} - X$ in positions $12, 34$ yields

$$\eta_{12} + \eta_{34} = aR_0 - m(e-d) + abm = aR_0 - m(e-d) - sa^2.$$

Writing $t = e - d$, equation (4.33) therefore becomes

$$J = aR_0 + (s/\beta)t.$$

Since $[a\cup a] = 0$ in odd characteristic, Lemma 4.14 gives

$$[a\cup J]_v = (s/\beta)[a\cup t]_v = s[\kappa\cup a^3]_v. \tag{4.35}$$

No assertion about the local group's commutators is used.

Next suppose $b_v = 0$ and $a_v \neq 0$. For the same notation, the lower equations give $w = 0$ and $u = v'$. The first and last diagonal equations, and the sum of the first two, are

$$aP_0 + du = 0, \qquad -eu - aQ_0 = 0, \qquad (d+e)u = -sa. \tag{4.36}$$

Thus $u \neq 0$, both d, e are constant scalar multiples of a, and $P_0 + Q_0 = s$. The exact sum of the outer entries of $\Delta_g X$ is now

$$\eta_{12} + \eta_{34} = aR_0 + (d-e)m - a^2(P_0 + Q_0).$$

The terms involving f cancel because $u = v'$. Adding sa^2 shows $J = aR_0 + (d-e)m$, a scalar multiple of a on this local group. Consequently

$$[a \cup J]_v = 0 \qquad \text{if } b_v = 0. \tag{4.37}$$

The same conclusion is automatic when $a_v = 0$, regardless of b_v.

Define the global class

$$\Theta = \mathrm{Tr}_{k/\mathbf{F}_3}(s^2[a \cup J]) \in H^2(K, \mathbf{F}_3). \tag{4.38}$$

At $v \in T$ with $b_v \neq 0$, equation (4.35) gives

$$\Theta_v = \mathrm{Tr}_{k/\mathbf{F}_3}(s^3[\kappa \cup a^3])_v = \mathrm{Tr}_{k/\mathbf{F}_3}(s[\kappa \cup a])_v.$$

The second equality is Frobenius invariance of the finite-field trace: $s^3\kappa(g)a(h)^3 = (s\kappa(g)a(h))^3$ because κ is $\mathbf{F}_3$-valued. Equation (4.37) handles $b_v = 0$. Both global classes are unramified outside T and have zero local class there. Thus, over all primes,

$$\Theta_v = \begin{cases} 0, & b_v = 0, \\ \mathrm{Tr}_{k/\mathbf{F}_3}(s[\kappa \cup a])_v, & b_v \neq 0. \end{cases} \tag{4.39}$$

Brauer reciprocity for these two global classes gives

$$\sum_{v : b_v = 0} \mathrm{inv}_v \, \mathrm{Tr}_{k/\mathbf{F}_3}(s[\kappa \cup a]) = 0. \tag{4.40}$$

This is the negative of the invariant sum in (4.32). Hence every dual class pairs to zero with the locally trivial obstruction, whose global trace cup also vanishes. Proposition 2.10 and then Proposition 2.1 give a lift. Prime-to-three descent finishes the proof. □

4.8. **Exhaustion of cases.**

Proof of Theorem 4.1. If $\mathrm{char}\, K = p$, then [2, Theorem 1.7] gives a $W(k)$-lift. If $p = 2$, then Corollary 3.12 applies. Henceforth assume p odd and $\mathrm{char}\, K \neq p$. By Lemma 2.3, pass to a prime-to-p extension so that the residual image is a p-group, and triangularize it. We first consider dimension four, with coordinates (4.1). If one of a, b, c is identically zero, then the image lies in a three-layer group of type $(2,1,1)$, $(1,2,1)$, or $(1,1,2)$ respectively, and Propositions 4.6 and 4.5 apply. Hence assume all three characters are nonzero.

Let $D \subset k^3$ be the first-superdiagonal image and let $d = \dim_k \langle D \rangle_k$. If $d = 1$, then every nonzero projected vector has all three coordinates nonzero, so Proposition 4.8 applies. If $d = 3$, then Proposition 4.9 applies. Suppose $d = 2$. The proper coordinate kernels of a, b, c cannot cover the $\mathbf{F}_p$-space D, since a cover by proper subspaces requires at least $p+1$ members [20, Theorem 1.2]. Thus D contains a regular vector. If a, c are independent, then Proposition 4.10 applies. If they are dependent, then $c = \lambda a$ with $\lambda \neq 0$. Use Proposition 4.12 when $p \geq 5$ and Proposition 4.15 when $p = 3$. This exhausts all four-dimensional residual images over every finite coefficient field of odd characteristic.

For $n = 1$, the Teichmüller character lifts ρ. The case $n = 2$ follows from [25, Theorem 3.1]. For $n = 3$, apply the four-dimensional result to $\rho \oplus 1$ and use Lemma 2.4. Finally descend vanishing of the actual obstruction along every prime-to-p extension using Lemma 2.3. □

5. A common arithmetic construction

The counterexamples in Sections 6 and 7 use the same family of finite groups and the same arithmetic realization, but different matrix representations. Throughout this section p is odd, $k = \mathbf{F}_p$, and

$$\mathcal{R} = k[C_p], \qquad C_p = \langle \sigma \rangle, \qquad \nu = 1 + \sigma + \cdots + \sigma^{p-1}.$$

Put $V = \mathcal{R}/k\nu$, let $q : \mathcal{R} \to V$ be the quotient, and write $y = q(1)$ additively. Set

$$P = V \rtimes \langle x \rangle, \qquad |x| = p, \qquad xvx^{-1} = \sigma v.$$

Thus $V \simeq k[D]/D^{p-1}$ for $D = \sigma - 1$, and the norm operator on V is zero. In particular P has order p^p and exponent p. Augmentation induces $\ell : V \to k$ with $\ell(y) = 1$ and kernel DV. The defining module sequence is

$$0 \longrightarrow k \xrightarrow{\iota} \mathcal{R} \xrightarrow{q} V \longrightarrow 0, \qquad \iota(1) = \nu. \tag{5.1}$$

5.1. **Controlled Kummer approximation.**

Lemma 5.1. *Let p be odd and let L/K be a finite Galois extension of global fields of characteristic different from p, with $\mu_p \subset L$. Let S be a finite set of finite primes of K, containing all primes above p when K is a number field. Given a class $t_w \in L_w^\times/L_w^{\times p}$ for every $w \mid S$, there exist $t \in L^\times$ and a prime $\mathfrak{q}$ of L outside S such that t represents all the prescribed classes and*

$$v_w(t) \equiv 0 \pmod{p} \quad (w \nmid S,\ w \neq \mathfrak{q}), \qquad v_{\mathfrak{q}}(t) \equiv 1 \pmod{p}.$$

The prime below $\mathfrak{q}$ splits completely in L/K, and it may be required to avoid any additional finite set of primes. In the number-field case one may strengthen these congruences to $v_w(t) = 0$ and $v_{\mathfrak{q}}(t) = 1$.

Proof. Put $S_L = \{w : w \mid S\}$. The subgroup $L_w^{\times p}$ is open in $L_w^\times$. Weak approximation therefore gives $t_0 \in L^\times$ with all the prescribed local classes. Choose a ray modulus $\mathfrak{m}$ with support exactly S_L (the zero modulus if $S_L = \varnothing$) and exponents large enough that

$$d \equiv 1 \pmod{\mathfrak{m}} \quad \Longrightarrow \quad d \in L_w^{\times p} \quad (w \in S_L).$$

Here a ray congruence includes $v_w(d) = 0$ at its support. There are no real primes in the number-field case because p is odd and $\mu_p \subset L$. Write the divisor of t_0 as $D_S + I$, where D_S is supported on S_L and I is supported outside S_L; for a number field use the corresponding additive notation for fractional ideals.

First consider a function field. Its full ray divisor class group $\mathrm{Cl}_{\mathfrak{m}}(L)$ has finite degree-zero subgroup and a rank-one degree quotient. Indeed, the kernel of its map to the ordinary divisor class group is a quotient of the finite residue-unit group of $\mathfrak{m}$, and Pic^0 over a finite field is finite. Consequently

$$A = \mathrm{Cl}_{\mathfrak{m}}(L)/p\mathrm{Cl}_{\mathfrak{m}}(L)$$

is finite. Global class field theory realizes this quotient by a finite abelian extension $H_{\mathfrak{m},p}/L$; importantly, this extension includes the appropriate constant-field quotient. Choose a finite Galois extension F/K containing $H_{\mathfrak{m},p}$. The Artin element corresponding to $[I] \in A$ extends to an element $\widetilde{\tau} \in \mathrm{Gal}(F/L)$. Chebotarev supplies a prime of K outside all excluded sets and a prime of F above it with Frobenius exactly $\widetilde{\tau}$. The restriction $\mathfrak{q}$ to L lies above a prime split completely in L/K. Since its relative residue degree is one, its Artin element in A is $[I]$. The constant action of $\widetilde{\tau}$ specifies the allowed degrees of the base prime; there are arbitrarily large degrees in this congruence class. Thus constant extensions do not obstruct this choice. For finite-quotient class field existence, see [38, §1]; for Chebotarev with this constant-field degree congruence, see [39, Theorem 4.2]. Reciprocity is also reviewed in [27, Chapter I, Appendix A].

The equality $[I] = [\mathfrak{q}]$ in A means that

$$I - \mathfrak{q} = pD + \mathrm{div}(d)$$

for some divisor D supported outside S_L and some $d \equiv 1 \pmod{\mathfrak{m}}$. Set $t = t_0/d$. Then

$$\mathrm{div}(t) = D_S + \mathfrak{q} + pD. \tag{5.2}$$

The local classes at S_L are unchanged, and the claimed valuation congruences follow.

For a number field, the full ray class group is already finite. Apply the same construction to its ray class field, without quotienting by p. Then $[I] = [\mathfrak{q}]$ in the full ray class group gives $I - \mathfrak{q} = \mathrm{div}(d)$ with the same ray congruence. This proves the stronger exact valuation statement; see [28, Chapter V, Theorems 3.6 and 3.23]. □

Remark 5.2. The term pD in (5.2) is essential for a function field. An exact formula $\operatorname{div}(t) = D_S + \mathfrak{q}$ would force $\deg \mathfrak{q} = -\deg D_S$ and cannot in general be obtained by Chebotarev outside arbitrary finite sets. The congruences suffice because the Kummer ramification criterion [32, Proposition 10.3] shows that pD contributes no ramification.

5.2. **A single realization for all odd primes.** Fix $\mu_p \subset K$, identify $k \simeq \mu_p$, and write the cyclotomic character as $\chi_{p^2} = 1 + p\kappa$. With arithmetic Frobenius as reciprocity convention, an unramified local character θ satisfies

$$\operatorname{inv}_v(\theta \cup \operatorname{Kum}(u)) = \frac{\theta(\operatorname{Frob}_v)v(u)}{p}. \tag{5.3}$$

The order of factors and sign are those of [28, Chapter III, §4, Step 2 and Proposition 4.3].

Proposition 5.3. *Let K be a global field of characteristic different from p with $\mu_p \subset K$ and $\mu_{p^2} \not\subset K$. Put*

$$(r_1, r_2, r_3) = (1, 1, -2), \qquad (\lambda_1, \lambda_2, \lambda_3) = (-1, 0, -1/2).$$

There are distinct finite primes v_1, v_2, v_3 away from p and a continuous surjection $\rho : G_K \twoheadrightarrow P$ with quotient character $a : G_K \to C_p$ such that

$$\operatorname{inv}_{v_i}(\kappa \cup a) = r_i/p, \qquad \operatorname{inv}_v(\kappa \cup a) = 0 \quad (v \notin \{v_1, v_2, v_3\}). \tag{5.4}$$

At v_i the local image is a V-conjugate of $\langle y^{\lambda_i} x\rangle$. At every other prime the local image is cyclic or contained in V. The construction can retain any prescribed Kummer character a satisfying the full invariant prescription (5.4)*, provided κ is unramified and nonzero at v_1, v_2, v_3.*

Proof. Choose v_1, v_2, v_3 away from p with $\kappa(\operatorname{Frob}_{v_i}) \neq 0$. By the Albert–Brauer–Hasse–Noether exact sequence [19, Theorem 4], there is a class α with invariants r_i/p at v_i and zero elsewhere, since $\sum r_i = 0$. Put $E = K(\mu_{p^2})$. The restriction of α to E is zero because the local degree is p at each prime in its support. For the cyclic extension E/K with character κ, the isomorphism $K^\times/\operatorname{N}_{E/K}(E^\times) \simeq \operatorname{Br}(E/K)$ sends the class of t to $\kappa \cup \operatorname{Kum}(t)$ [28, Chapter III, §4, Steps 1–2, pp. 111–112]. Thus for some $t \in K^\times$ we have

$$\alpha = \kappa \cup a, \qquad a = \operatorname{Kum}(t).$$

Set $L = K(t^{1/p})$. All r_i are nonzero for odd p, so (5.3) implies $p \nmid v_i(t)$. Thus L/K is cyclic of degree p and totally ramified at each v_i. If a was prescribed, then use the corresponding extension L.

Inflate (5.1) through a. Choose $\gamma \in \operatorname{Br}(K)[p]$ with the following complete invariant list:

	v_1	v_2	v_3	$v \notin \{v_1, v_2, v_3\}$
$\operatorname{inv}_v(\alpha)$	$1/p$	$1/p$	$-2/p$	0
$\operatorname{inv}_v(\gamma)$	$1/p$	0	$-1/p$	$0.$

(5.5)

Its restriction to L is zero, because restriction multiplies local invariants by the local degree. Under Shapiro, the map $H^2(K, k) \to H^2(K, \mathcal{R})$ induced by ι is restriction to L: the norm vector is the constant function 1, and Shapiro restricts cochains and evaluates at the identity coset. Exactness gives

$$c_0 \in H^1(K, V), \qquad \delta(c_0) = \gamma.$$

For $\lambda \in k$, the cocycle

$$c_\lambda(\sigma^j) = \lambda q(1 + \sigma + \cdots + \sigma^{j-1}), \qquad 0 \le j < p,$$

has graph $\langle y^\lambda x\rangle$. Lifting the displayed sum to $\mathcal{R}$ gives differential $\lambda\lfloor (i+j)/p\rfloor \nu$. Lemma 2.6 therefore fixes the connecting sign:

$$\delta(c_\lambda \circ a) = \lambda\beta_0(a) = -\lambda(\kappa \cup a). \tag{5.6}$$

Since $(-\lambda_i r_i) = (1, 0, -1)$, the local classes $c_{\lambda_i} \circ a$ have connecting images $\gamma|_{v_i}$.

Choose a continuous representative of c_0, and only then choose a finite set S containing the three distinguished primes and all ramification of (c_0, a); in the number-field case include all primes above p. Prescribe $c_{\lambda_i} \circ a$ at v_i and zero in $H^1(K_v, V)$ at every other $v \in S$. Every

prescription has connecting image $\gamma|_v$. Local exactness and Shapiro lift its difference from $c_0|_v$ to Kummer classes in $L_w^\times/L_w^{\times p}$ at all w above S. Lemma 5.1 realizes these by

$$h \in H^1(L,k) \simeq H^1(K,\mathcal{R}), \qquad c = c_0 + q_* h, \qquad \rho = (c,a) : G_K \to P.$$

Equality of local cohomology classes means conjugacy by V, so the prescribed images occur up to these conjugacies.

At the one new prime below $\mathfrak{q}$, the entire decomposition group has $a = 0$, because the prime splits completely in L/K. Its image therefore lies in V, even when it is noncyclic. Elsewhere outside S, the valuation of the Kummer representative of h is divisible by p. Its local class is therefore represented by a unit and is unramified by the étale Kummer sequence [35, Tag 03PL]. The extension L/K and (c_0, a) are also unramified. Shapiro preserves this property: the restrictions to inertia are the restrictions at the conjugate primes of L, all zero. Thus ρ is unramified there and its local image is cyclic. The prescribed images at S are cyclic. In the number-field case all infinite primes are complex; function fields have no archimedean primes. This proves the assertion about every local image.

Finally the image contains elements $w_i x$ with $\ell(w_i) = \lambda_i$, for $i = 1, 2$. Their quotient belongs to V and has nonzero augmentation $\lambda_1 - \lambda_2$. It therefore generates $V \simeq k[D]/D^{p-1}$ as a $k[C_p]$-module. Conjugation by either $w_i x$ acts as σ, so the image contains all of V and surjects onto P/V. Hence it is P. □

5.3. **Constant fields and explicit rational-function inputs.**

Proposition 5.4. *Let $K = \mathbf{F}_q(C)$ be a global function field with complete constant field $\mathbf{F}_q$, and let $p \neq \operatorname{char} K$ be odd. The condition $\mu_p \subset K$, $\mu_{p^2} \not\subset K$ is equivalent to $v_p(q-1) = 1$. Under this condition, with $c = (q-1)/p \bmod p$, the character $\chi_{p^2} = 1 + p\kappa$ is unramified everywhere and*

$$\kappa(\mathrm{Frob}_v) = c \deg(v) \pmod p.$$

In particular Proposition 5.3 applies to every such K.

Proof. The prime-to-char K roots of unity in K belong to $\mathbf{F}_q^\times$, which is cyclic of order $q-1$. In a finite constant extension, every prime is unramified and arithmetic Frobenius at v acts on constants by the $q^{\deg(v)}$-power map [32, Propositions 8.5 and 9.10 and the definition preceding the latter]. Under the stated condition, $q^{\deg(v)} \equiv 1 + pc\deg(v) \pmod{p^2}$, as claimed. □

Example 5.5. For explicit cyclic quotients in the two smallest constructions, put $h(t) = t(t-1)/(t-2)^2$. For $K = \mathbf{F}_{11}(t)$ and $p = 5$, take

$$a = \mathrm{Kum}(h^3), \qquad L = K\big((h^3)^{1/5}\big).$$

Here $c = 2$, and the valuations of h^3 at $0, 1, 2$ are $3, 3, -6$. The invariant formula (5.3) gives $1/5, 1/5, -2/5$ in $\mathbf{Q}/\mathbf{Z}$. For $K = \mathbf{F}_7(t)$ and $p = 3$, take

$$a = \mathrm{Kum}(h^2), \qquad L = K\big((h^2)^{1/3}\big).$$

Again $c = 2$, and valuations $2, 2, -4$ give $1/3, 1/3, -2/3$. In either example the valuation at infinity is zero, as are those at all other primes. Since κ is unramified, every other invariant is zero. Each L/K has degree p and is totally ramified at the three specified points. Proposition 5.3 completes these explicit cyclic quotients to the required P-extension; the formulas above alone are not defining equations for the full nonabelian extension.

6. Counterexamples for every prime $p \geq 5$

We construct, for every prime $p \geq 5$, a p-dimensional representation over $\mathbf{F}_p$ whose obstruction to lifting modulo p^2 is nonzero. The construction works over every global field of characteristic different from p containing μ_p but not μ_{p^2}, and permits arbitrary ramification in a hypothetical lift.

Theorem 6.1. *Let $p \geq 5$ be prime and let K be a global field of characteristic different from p such that $\mu_p \subset K$ and $\mu_{p^2} \not\subset K$. Put*

$$N_p = \sum_{i=1}^{p-1} E_{i,i+1}, \qquad P = \langle I + N_p, I + E_{23} \rangle \subset \mathrm{GL}_p(\mathbf{F}_p).$$

There exists a continuous surjection $\rho : G_K \twoheadrightarrow P$ that has no continuous lift to $\mathrm{GL}_p(\mathbf{Z}/p^2\mathbf{Z})$.

The arithmetic realization is supplied by Section 5; in the function-field case it uses the ray divisor class group modulo p. All matrix identities below are over finite coefficient rings, and the only field-theoretic input in the local contraction calculation is Kummer theory in characteristic different from p. We determine the finite matrix cocycle and its actual local contractions.

6.1. **The finite group and a locally inner cocycle.** Throughout this section put $k = \mathbf{F}_p$, $M = M_p(k)$, $N = N_p$, $x = I + N$, and $y = I + E_{23}$. The action on M is conjugation. Let $\sigma = \mathrm{Ad}(x)$, $D = \sigma - 1$, and $f_0 = E_{23}$. If $(e_1, \ldots, e_p)$ is the standard basis with dual basis $(e_1^*, \ldots, e_p^*)$, then

$$\sigma^t f_0 = (e_2 + te_1) \sum_{j=0}^{p-3} \binom{-t}{j} e_{3+j}^* \qquad (t \in \mathbf{F}_p). \tag{6.1}$$

Indeed $x^t e_2 = e_2 + te_1$, and the displayed row is $e_3^* x^{-t}$. All binomial polynomials in (6.1) have denominators prime to p. The right side has degree $p-2$ in t, with leading coefficient $E_{1p}/(p-3)!$. Taking successive forward differences therefore gives

$$D^{p-2} f_0 = -2E_{1p} \neq 0, \qquad D^{p-1} f_0 = 0.$$

It follows that $f_0, Df_0, \ldots, D^{p-2} f_0$ are linearly independent: apply successive powers of D to a relation, beginning with its least nonzero index. Let $\mathcal{V}$ be their span. It has dimension $p-1$ and is supported in rows $1, 2$ and columns $3, \ldots, p$. Hence $\mathcal{V}^2 = 0$. Identify its additive group with $V = I + \mathcal{V}$ by $F \mapsto I + F$.

The conjugates of y by powers of x generate V, since their span agrees with that of the successive differences. Moreover, x has order p, normalizes V, and has $(1,2)$-entry j in its jth power. Consequently

$$P = V \rtimes \langle x \rangle, \qquad |P| = p^p. \tag{6.2}$$

The $k[C_p]$-module V, where $C_p = \langle \sigma \rangle$, is a single unipotent block of length $p-1$. Its norm operator is zero, because

$$\sum_{i=0}^{p-1} \sigma^i = (\sigma - 1)^{p-1} = 0 \quad \text{on } V.$$

For $j \neq 0$, the powers of σ^j give the same norm operator. The semidirect product law thus shows that every element of P has order dividing p.

The cyclic module V is the module of Section 5, under $q(1) \mapsto f_0$. Its augmentation ℓ is the $(2,3)$-coordinate, and $\ker \ell = DV$.

The first-superdiagonal coordinates define characters

$$a(g) = g_{12} = g_{34} = \cdots = g_{p-1,p}, \qquad b(g) = g_{23}. \tag{6.3}$$

If $g = vx^j$, then $a(g) = j$ and $b(g) = j + \ell(v)$.

Lemma 6.2. *There is a cocycle $\eta \in Z^1(P, M)$ with*

$$\eta(x) = 0, \qquad \eta(y) = E_{33} - E_{22} + E_{23}.$$

It takes upper triangular, trace-zero values and satisfies

$$\operatorname{diag} \eta(g) = (0, -(b(g) - a(g)), b(g) - a(g), 0, \ldots, 0). \tag{6.4}$$

Its restriction to every cyclic subgroup is a coboundary. Its restriction to V has a common primitive Q, given, with $c = -1/2 \in k$, by

$$Q_{i+1,i} = ic \ (1 \leq i < p), \qquad Q_{11} = 2c, \quad Q_{22} = 3c, \quad Q_{ii} = ic \ (3 \leq i \leq p), \tag{6.5}$$

all other entries being zero.

Proof. The successive subdiagonal entries of Q give the constant diagonal cI in $[N, Q]$, including its last entry because $pc = 0$. Its successive diagonal differences give $cN - cE_{23}$. Thus $[N, Q] = cI + cN - cE_{23}$, and direct multiplication by y^{-1} gives

$$\Delta_x Q = cI - cE_{23}x^{-1}, \qquad \Delta_y Q = E_{33} - E_{22} + E_{23}. \tag{6.6}$$

The nonscalar part of $\Delta_x Q$ is supported in row 2 and columns $3, \dots, p$, so it commutes with V. It follows that $\Delta_{xvx^{-1}} Q = \mathrm{Ad}(x)\Delta_v Q$ for $v \in V$. The same identity holds for every power of x. Therefore

$$\eta(vx^j) = \Delta_v Q$$

is a cocycle on (6.2): the equivariance just proved and the coboundary identity on V give

$$\eta(vx^j wx^l) = \Delta_v Q + \mathrm{Ad}(vx^j)\Delta_w Q.$$

For $v = I + F$ with $F \in \mathcal{V}$, one has $\eta(v) = FQ - QF - FQF$. Since Q vanishes below the first subdiagonal and F has the stated rectangular support, this matrix is upper triangular. Its only possible nonzero diagonal entries are $-\ell(F)$ in position 2 and $\ell(F)$ in position 3. This proves (6.4) and the trace assertion.

Let $g \in P$. If $a(g) = 0$, then $g \in V$ and Q is a primitive. If $a(g)b(g) \neq 0$, then every first-superdiagonal entry of $g - I$ is nonzero, so $g - I$ is regular nilpotent and its centralizer is $k[g - I]$. Under the nondegenerate trace pairing on M, the image of $\mathrm{Ad}(g) - 1$ is the orthogonal complement of this centralizer. The upper triangular, trace-zero matrix $\eta(g)$ is orthogonal to I and to every positive power of $g - I$. Hence $\eta(g) \in (\mathrm{Ad}(g) - 1)M$.

Suppose now that $j = a(g) \neq 0$ and $b(g) = 0$. Conjugation by V in the coset Vx^j changes its V-coordinate by $(1 - \sigma^j)V$. The quotient $V/(\sigma^j - 1)V$ is one-dimensional and is identified with k by ℓ: indeed $\sigma^j - 1$ is D times an invertible polynomial in D. The condition $b(g) = 0$ fixes $\ell(v) = -j$. Consequently g is V-conjugate to g_0^j, where

$$g_0 = xy^{-1} = I + H_0, \qquad H_0 = E_{12} - E_{13} + \sum_{i=3}^{p-1} E_{i,i+1}. \tag{6.7}$$

Set $Q_0 = E_{22} - E_{21} - E_{31} - E_{42}$. Writing $z = \eta(y)$, direct multiplication yields

$$[g_0, Q_0] = E_{12} + E_{22} - E_{23} - E_{33} = -g_0 z.$$

Since $\eta(g_0) = -\mathrm{Ad}(g_0)z$, this proves $\Delta_{g_0} Q_0 = \eta(g_0)$. A cocycle and a coboundary that agree on a generator agree on its cyclic subgroup. Finally, if Q' is a primitive on a subgroup C, then $\mathrm{Ad}(v)Q' - \eta(v)$ is a primitive on vCv^{-1}. This proves cyclic local triviality in the remaining case. □

Remark 6.3. The class of η in $H^1(P, M)$ is nonzero. If $\eta = dZ$, then $\eta(x) = 0$ implies $Z \in k[N]$, so Z is upper triangular. But $\Delta_y Z$ then has zero diagonal, contrary to the diagonal of $\eta(y)$. Nonvanishing of this finite-group class alone does not establish a Galois lifting obstruction; the arithmetic pairing below provides that obstruction.

6.2. **An actual lifting cocycle and its local contractions.** Let F be a field of characteristic different from p containing μ_p, fix an identification $k \simeq \mu_p$, and choose a compatible primitive p^2th root of unity. Write the cyclotomic character modulo p^2 as

$$\chi = 1 + p\kappa : G_F \longrightarrow (\mathbf{Z}/p^2\mathbf{Z})^\times, \qquad \kappa \in H^1(F, k).$$

Let $\rho : G_F \to P$ be a continuous representation, and use a, b, η for their pullbacks along ρ. The action on M is $g \cdot T = \mathrm{Ad}(\rho(g))T$. All trace contractions use the cochain cup product with this action; in particular,

$$\mathrm{tr}(e \cup Q)(g, h) = \mathrm{tr}\big(e(g, h)\,(gh) \cdot Q\big), \tag{6.8}$$

$$\mathrm{tr}(X \cup \eta)(g, h) = \mathrm{tr}\big(X(g)\, g \cdot \eta(h)\big). \tag{6.9}$$

Kummer theory gives lifts A, B of a, b to continuous cocycles with values in $(\mathbf{Z}/p^2\mathbf{Z})(\chi)$, satisfying

$$A(gh) = A(g) + \chi(g)A(h), \qquad B(gh) = B(g) + \chi(g)B(h). \tag{6.10}$$

This is Lemma 2.6.

Choose continuous lifts $C_{ij}(g)$ of $\rho(g)_{ij}$ to $\mathbf{Z}/p^2\mathbf{Z}$ for $i < j$, taking $C_{23} = B$ and $C_{i,i+1} = A$ for $i \neq 2$. Define

$$R_{ii}(g) = \chi(g)^{p-i}, \qquad R_{ij}(g) = \chi(g)^{p-j}C_{ij}(g) \quad (i < j), \tag{6.11}$$

with zero entries below the diagonal. This cochain reduces to ρ. Its diagonal and first superdiagonal satisfy the multiplicative identities exactly: for $j = i + 1$, the corresponding entry of $R(g)R(h)$ is

$$\chi(g)^{p-j}\chi(h)^{p-j}(C_{ij}(g) + \chi(g)C_{ij}(h)),$$

which equals $R_{ij}(gh)$ by (6.10). Writing J for the strictly upper triangular algebra in M, the actual factor set consequently has the form

$$R(g)R(h)R(gh)^{-1} = I + pe(g, h), \qquad e(g, h) \in J^2. \tag{6.12}$$

Multiplication by p identifies the kernel of reduction with the additive group M. Associativity gives $e \in Z^2(F, M)$, and its class is the obstruction to lifting ρ. In particular, the inverse factor in (6.12) is part of the definition. Since η and its conjugates by P are upper triangular,

$$\operatorname{tr}(e \cup \eta) = 0 \tag{6.13}$$

as a cochain.

Lemma 6.4. *Let H be a closed subgroup of G_F whose image under ρ is contained in V or is cyclic. There is a matrix $Q_H \in M$ with $\eta|_H = dQ_H$. The scalar cochain $L_H = \operatorname{tr}(e|_H \cup Q_H)$ is a 2-cocycle, and the primitive can be chosen so that*

$$[L_H] = \begin{cases} 0, & \rho(H) \subset V, \\ 0, & \rho(H) = \langle g \rangle \text{ with } a(g)b(g) \neq 0, \\ [\kappa|_H \cup a|_H], & \rho(H) = \langle g \rangle \text{ with } a(g) \neq 0,\ b(g) = 0. \end{cases} \tag{6.14}$$

Proof. The primitive exists by Lemma 6.2. The Leibniz rule and (6.13) give $dL_H = 0$. Let $\mathcal{B} \subset M$ be the space of matrices zero below the first subdiagonal. Upper triangular conjugation preserves $\mathcal{B}$: in an entry $(UQU^{-1})_{ij}$ with $i \geq j + 2$, a possibly nonzero summand involves Q_{rs} with $r \geq i$ and $s \leq j$, so $r \geq s + 2$. Moreover, $\operatorname{tr}(J^2\mathcal{B}) = 0$. If $\rho(H) \subset V$, then the common primitive (6.5) belongs to $\mathcal{B}$, so $L_H = 0$ pointwise.

Suppose $\rho(H) = \langle g \rangle$ with $a(g)b(g) \neq 0$. Any primitive satisfies

$$[g - I, Q_H] = \eta(g)g,$$

whose right side is upper triangular. The regular strictly upper triangular matrix $g - I$ is conjugate to N by an invertible upper triangular matrix: its kernels are the standard complete flag, and an adapted Jordan basis gives the conjugation. It suffices to show that $[N, Q']$ upper triangular forces $Q' \in \mathcal{B}$. Its entries are

$$[N, Q']_{ij} = Q'_{i+1,j} - Q'_{i,j-1},$$

with out-of-range entries zero. These expressions vanish for $i > j$; starting with column 1 and proceeding right gives $Q'_{ij} = 0$ for $i \geq j + 2$. Thus $Q_H \in \mathcal{B}$, and again $L_H = 0$ pointwise.

It remains to consider $\rho(H) = \langle g \rangle$ with $a(g) \neq 0$ and $b(g) = 0$. By the conjugacy calculation above, there is $v \in V$ such that $g = vg_0^{a(g)}v^{-1}$. Since a identifies this cyclic group with $\mathbf{F}_p$, it follows that

$$\rho(h) = vg_0^{a(h)}v^{-1} \qquad (h \in H).$$

We construct a lift on the full group H. Regard H_0 in (6.7) also as an integer matrix. Its only chain of length $p - 2$ runs from e_p to $-e_1$; consequently

$$H_0^{p-2} = -E_{1p}, \qquad H_0^{p-1} = 0$$

hold over $\mathbf{Z}$. Over k set

$$T_0 = \log(I + H_0) = \sum_{r=1}^{p-2} \frac{(-1)^{r+1}}{r} H_0^r.$$

All denominators are invertible, and the truncated logarithm and exponential are inverse, so $g_0^a = \exp(aT_0)$ for $a \in \mathbf{F}_p$. Define an upper unipotent matrix U by

$$Ue_1 = e_1, \qquad Ue_2 = e_2, \qquad Ue_j = T_0^{p-j} e_p \quad (3 \le j \le p). \tag{6.15}$$

For $j \ge 3$, one has $H_0^{p-j} e_p = e_j$, and higher powers have strictly smaller row indices; hence these columns have diagonal entry one and zero entries below it. Also $T_0^{p-2} e_p = -e_1$. Checking columns $1, 2, 3$ separately and using successive powers for the remaining columns gives $T_0 U = U H_0$. Thus, with $W = vU$,

$$\rho(h) = W \exp(a(h) H_0) W^{-1}.$$

Lift W to an upper unipotent matrix $\widetilde{W}$ over $\mathbf{Z}/p^2\mathbf{Z}$, and retain the same integer matrix H_0 over this ring. Define weights

$$w_1 = p - 2, \qquad w_2 = p - 3, \qquad w_i = p - i \quad (3 \le i \le p),$$

and put $D_\chi(h) = \operatorname{diag}(\chi(h)^{w_i})_{i=1}^p$. Each nonzero entry of H_0 has weight difference one, including both E_{12} and $-E_{13}$. Therefore $D_\chi(h) H_0 D_\chi(h)^{-1} = \chi(h) H_0$ exactly. Set

$$S(h) = \widetilde{W} \exp(A(h) H_0) D_\chi(h) \widetilde{W}^{-1}, \qquad \exp(tH_0) = \sum_{r=0}^{p-2} \frac{t^r}{r!} H_0^r. \tag{6.16}$$

Every factorial is a unit modulo p^2, and $H_0^{p-1} = 0$ holds in this ring. The binomial identity in degrees at most $p-2$ therefore proves $\exp(sH_0)\exp(tH_0) = \exp((s+t)H_0)$ exactly. The weight identity and (6.10) now show that $S(h_1)S(h_2) = S(h_1 h_2)$. Its reduction is $\rho|_H$. The maps A and χ are continuous with finite image, so S is a continuous lift on H.

Write $R(h) = (I + pX(h))S(h)$. Since R and S are upper triangular, so is X. Conjugation by $\widetilde{W}$ preserves diagonals, and the weights $p - i$ in (6.11) differ from w_i by $(1, 1, 0, \dots, 0)$. It follows that

$$e|_H = dX, \qquad \operatorname{diag} X(h) = \kappa(h)(1, 1, 0, \dots, 0). \tag{6.17}$$

For any primitive Q_H, the full cochain Leibniz identity gives

$$d \operatorname{tr}(X \cup Q_H) = \operatorname{tr}(e|_H \cup Q_H) - \operatorname{tr}(X \cup \eta|_H).$$

On H, equation (6.4) reads $\operatorname{diag} \eta(h) = (0, a(h), -a(h), 0, \dots, 0)$. Upper triangular conjugation preserves this diagonal. The trace of a product of upper triangular matrices is the pairing of their diagonals, so (6.17) gives

$$\operatorname{tr}(X \cup \eta|_H)(h_1, h_2) = \kappa(h_1) a(h_2).$$

This proves the last line of (6.14), including its positive sign with the factor-set convention (6.12). □

6.3. **Arithmetic realization and nonliftability.**

Proof of Theorem 6.1. Apply Proposition 5.3 to the module identification above. It gives a surjection $\rho : G_K \to P$ with quotient a. On the complement generated by $y^\lambda x$ one has $b = (1 + \lambda)a$. At the three distinguished primes the local b-characters are therefore $0, a, a/2$, respectively, up to V-conjugacy, which preserves a and b.

Use the section (6.11) and its actual factor e. Lemma 6.4 provides common local primitives Q_v at every prime. At v_1, v_2, v_3, the invariants of $L_v = \operatorname{tr}(e|_v \cup Q_v)$ are $1/p, 0, 0$. Every other cyclic image contributes either zero or the invariant of $\kappa \cup a$, which is zero by (5.4). An image contained in V contributes zero, including at the new ramification prime. Thus

$$\sum_v \operatorname{inv}_v[L_v] = 1/p. \tag{6.18}$$

Together with (6.13), Corollary 2.11 excludes every lift to $\mathrm{GL}_p(\mathbf{Z}/p^2\mathbf{Z})$, with arbitrary additional ramification. □

Corollary 6.5. *If $p \geq 5$, K is a global field of characteristic different from p with $\mu_p \subset K$ and $\mu_{p^2} \not\subset K$, k is a finite field of characteristic p, and $n \geq p$, then there is a continuous representation $G_K \to \mathrm{GL}_n(k)$ that does not lift to $\mathrm{GL}_n(W_2(k))$.*

Proof. Let $\rho : G_K \to \mathrm{GL}_p(\mathbf{F}_p)$ be the representation constructed above. Its scalar extension $\rho \otimes_{\mathbf{F}_p} k$ cannot lift, by Lemma 2.2. If $(\rho \otimes_{\mathbf{F}_p} k) \oplus 1^{\oplus(n-p)}$ lifted, then Lemma 2.4 would give a lift of its first summand, a contradiction. For $p = 5$, Theorem 1.2 shows that this dimension bound is sharp. For larger p, the construction supplies dimension p without determining the least possible dimension of a nonliftable representation of G_K in coefficient characteristic p. □

7. A six-dimensional counterexample in characteristic three

The construction in Section 6 uses a regular unipotent matrix of size p. In characteristic three a different finite image gives a counterexample in dimension six. The essential additional property is that the dual cocycle vanishes on an entire normal subgroup, which accommodates the new ramification introduced by approximation.

Theorem 7.1. *Let K be a global field of characteristic different from three such that $\mu_3 \subset K$ and $\mu_9 \not\subset K$. Put*

$$N = E_{16} + E_{23} + E_{24} - E_{45}, \qquad x = I + N + \tfrac{1}{2}N^2, \qquad y = I + E_{14} + E_{36},$$

and let $P = \langle x, y\rangle \subset \mathrm{GL}_6(\mathbf{F}_3)$. Then there is a continuous surjection $\rho : G_K \twoheadrightarrow P$ that has no continuous lift to $\mathrm{GL}_6(\mathbf{Z}/9\mathbf{Z})$.

7.1. **The finite image and the dual cocycle.** All matrices in this subsection are over $\mathbf{F}_3$. We use blocks of sizes $(2, 2, 2)$ and put $M = M_6(\mathbf{F}_3)$, with its conjugation action. Since $N^2 = -E_{25}$ and $N^3 = 0$, the generators and their commutator are

$$x = I + E_{16} + E_{23} + E_{24} + E_{25} - E_{45}, \qquad z = [x, y] = I + E_{15} + E_{26}.$$

Here $[x, y] = xyx^{-1}y^{-1}$. The matrix z is central, and x, y, z have order three. Their adjacent blocks satisfy

$$A_x = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \quad B_x = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \qquad A_y = B_y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

so $A_xB_y - A_yB_x = I_2$, which also verifies the displayed commutator. The first adjacent blocks of x and y are linearly independent, whereas $z - I$ has nonzero block 13 and zero adjacent blocks. Thus the 27 matrices

$$y^b z^c x^a, \qquad a, b, c \in \mathbf{F}_3,$$

are distinct and comprise P. Every element has order dividing three, since the cube of a strictly three-block upper triangular matrix is zero. In particular,

$$V = \langle y, z\rangle \simeq C_3^2, \qquad P = V \rtimes \langle x\rangle, \qquad xyx^{-1} = zy. \tag{7.1}$$

Write

$$s(y^b z^c x^a) = a, \qquad t(y^b z^c x^a) = b$$

for the two additive characters of P.

Lemma 7.2. *There is a cocycle $\eta \in Z^1(P, M)$ with block upper triangular values such that $\eta|_V = 0$ and η is inner on every cyclic subgroup. If $g \in P$ and*

$$W_g = (\log g)_{13}, \qquad \log g = (g - I) - \tfrac{1}{2}(g - I)^2,$$

where W_g is extended by zero outside block 13, then one can choose a cyclic primitive Q_g satisfying

$$dQ_g(g) = \eta(g), \qquad \mathrm{tr}(W_gQ_g) = s(g)(1 - t(g)^2). \tag{7.2}$$

On the whole subgroup V, one may take the common primitive $Q = 0$.

Proof. Set

$$H = \mathrm{diag}(1,2,2,1,1,2) + 2E_{34}, \qquad B = E_{16} + E_{24} + 2E_{35}.$$

The displayed generators give

$$\mathrm{Ad}(v)H = H \quad (v \in V), \qquad \mathrm{Ad}(x)H = H + B, \qquad \mathrm{Ad}(g)B = B \quad (g \in P).$$

Thus $\mathrm{Ad}(g)H = H + s(g)B$. Since $-(a+b)^2 = -a^2 + ab - b^2$ in $\mathbf{F}_3$, the formula

$$\eta(g) = s(g)H - s(g)^2 B \tag{7.3}$$

satisfies $\eta(gh) = \eta(g) + \mathrm{Ad}(g)\eta(h)$ and vanishes on V.

Put $g_\lambda = y^\lambda x$ for $\lambda \in \mathbf{F}_3$. At $g_0 = x$, the single matrix

$$Q_0 = E_{32} + E_{42} - E_{53} + E_{54} + E_{61}$$

satisfies

$$\mathrm{Ad}(x)Q_0 - Q_0 = H - B = \eta(x).$$

Since $W_x = E_{16}$, its contraction is $\mathrm{tr}(W_x Q_0) = 1$.

For $\lambda \neq 0$, so that $\lambda^2 = 1$, an explicit primitive is

$$Q_\lambda = \begin{pmatrix} 2 & 0 & \lambda & 0 & 0 & 0 \\ \lambda & 2 & 2 & 1 & 0 & 0 \\ 2\lambda & 2 & 2 & 2 & 0 & 0 \\ \lambda & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda & 0 & 0 & 0 \end{pmatrix}.$$

Direct multiplication, using $\lambda^2 = 1$, gives

$$g_\lambda Q_\lambda - Q_\lambda g_\lambda = (H - B)g_\lambda,$$

and hence $\Delta_{g_\lambda} Q_\lambda = H - B = \eta(g_\lambda)$. The block $(Q_\lambda)_{31}$ is zero, so its contraction with every matrix supported in block 13 is zero. Together with the calculation for Q_0, this proves

$$\mathrm{tr}(W_{g_\lambda} Q_\lambda) = 1 - \lambda^2 \qquad (\lambda \in \mathbf{F}_3). \tag{7.4}$$

Every element outside V is V-conjugate to a nonzero power of exactly one g_λ: its characters determine $a = s(g) \neq 0$ and $\lambda = t(g)/a$, and conjugation by powers of y changes the remaining central exponent arbitrarily. Conjugate the corresponding primitive by the same element of V. The cocycle identity and $\eta|_V = 0$ make it a primitive on the conjugate cyclic subgroup.

To check its contraction, write $(\log g)_{12} = A$, $(\log g)_{23} = B_0$, and $F = (Q_g)_{31}$. Block upper triangularity of $dQ_g(g)$ gives $B_0 F = FA = 0$. Conjugation by $v \in V$ leaves F unchanged and replaces W_g by $W_g + A_v B_0 - AB_v$, whose additional terms have zero trace contraction with F. Finally $\log(g^a) = a \log g$ for $a \in \mathbf{F}_3$. Formula (7.4) therefore gives $a(1-\lambda^2) = s(g)(1 - t(g)^2)$, since $a^2 = 1$ for $a \neq 0$. For $g \in V$, take $Q_g = 0$. □

7.2. **The actual factor cocycle and its local contraction.** Let F be a field of characteristic different from three containing μ_3. Fix an identification $\mathbf{F}_3 \simeq \mu_3$, and write $\chi = \chi_9 = 1 + 3\kappa$ for the cyclotomic character. For a representation $\rho : G_F \to P$, write

$$\rho(g) = \begin{pmatrix} I_2 & A_g & C_g \\ 0 & I_2 & B_g \\ 0 & 0 & I_2 \end{pmatrix}.$$

The entries of A and B are additive characters. Lemma 2.6 gives entrywise χ-twisted lifts $\widehat{A}, \widehat{B}$ to $\mathbf{Z}/9\mathbf{Z}$. Choose any continuous lift $\widehat{C}$ of C, and define

$$R(g) = \begin{pmatrix} \chi(g)^2 I_2 & \chi(g)\widehat{A}_g & \widehat{C}_g \\ 0 & \chi(g) I_2 & \widehat{B}_g \\ 0 & 0 & I_2 \end{pmatrix}. \tag{7.5}$$

The diagonal and adjacent blocks are multiplicative. Consequently its actual factor cocycle satisfies

$$R(g)R(h)R(gh)^{-1} = I + 3e(g,h), \qquad e(g,h) \text{ is supported in block 13.} \tag{7.6}$$

By Proposition 2.1, $[e]$ is the obstruction to lifting ρ. We continue to write η, s, t for their pullbacks from P. Since η is block upper triangular,

$$\operatorname{tr}(e \cup \eta) = 0 \tag{7.7}$$

as a cochain. Cup products throughout this section include the residual adjoint action, as in (6.8) and (6.9).

Lemma 7.3. *Let $D \subset G_F$ be a closed subgroup whose residual image is cyclic or contained in V. If $\rho(D) \subset V$, then $Q_D = 0$ is a common primitive and $\operatorname{tr}(e|_D \cup Q_D) = 0$. Otherwise choose a generator g_0 of $\rho(D)$ with $s(g_0) = 1$ and a primitive Q_D from Lemma 7.2. Then*

$$[\operatorname{tr}(e|_D \cup Q_D)] = \big(1 - t(g_0)^2\big)[\kappa|_D \cup s|_D] \quad \text{in } H^2(D, \mathbf{F}_3). \tag{7.8}$$

Proof. The assertion for V follows from $\eta|_V = 0$. For a cyclic image, write $\rho(h) = g_0^{u(h)}$, so $u = s|_D$. Set $L_0 = \log g_0$, and lift its strict blocks arbitrarily to a matrix $\widehat{L}$ over $\mathbf{Z}/9\mathbf{Z}$. Then $\widehat{L}^3 = 0$. Write $\widehat{W}$ for its block 13, extended by zero; both $\widehat{L}\widehat{W}$ and $\widehat{W}\widehat{L}$ are zero. Restrict a Kummer lift of s to obtain a χ-twisted lift U of u on D, and put

$$D_\chi(h) = \operatorname{diag}(\chi(h)^2 I_2, \chi(h) I_2, I_2), \qquad R'(h) = \exp(U(h)\widehat{L}) D_\chi(h),$$

where $\exp(T\widehat{L}) = I + T\widehat{L} + (T^2/2)\widehat{L}^2$. Only the invertible denominator 2 occurs. Conjugation gives

$$D_\chi(h)\widehat{L}D_\chi(h)^{-1} = \chi(h)\widehat{L} + 3\kappa(h)\widehat{W}.$$

The two terms commute, and the second has zero product with $\widehat{L}$. Using $U(hh') = U(h) + \chi(h)U(h')$ in the exponential law therefore gives

$$R'(h)R'(h')R'(hh')^{-1} = I + 3\kappa(h)u(h')W_{g_0}. \tag{7.9}$$

In particular its normalized factor is $e' = \kappa \cup u\, W_{g_0}$, with the displayed positive sign.

The sections R and R' have the same block diagonal. Thus $R = (I + 3X)R'$ for a strictly block upper triangular cochain X, and $e = e' + dX$. Since $dQ_D = \eta|_D$, the cochain Leibniz identity is

$$d\operatorname{tr}(X \cup Q_D) = \operatorname{tr}((e - e') \cup Q_D) - \operatorname{tr}(X \cup \eta).$$

The last term vanishes identically: the adjoint action preserves the block upper triangular matrices, and X is strictly block upper triangular. Finally W_{g_0} commutes with g_0, so the full cup action in $e' \cup Q_D$ contributes the constant $\operatorname{tr}(W_{g_0} Q_D)$. Equations (7.2) and (7.9) prove (7.8). □

This calculation applies over every completion, including the primes above three. In particular it does not require the local Kummer lifts in the calculation to agree with those chosen in (7.5).

7.3. **Arithmetic realization and nonliftability.**

Proof of Theorem 7.1. The identities $xyx^{-1} = zy$ and $xzx^{-1} = z$ identify V with the module $\mathbf{F}_3[C_3]/\mathbf{F}_3\nu$ of Section 5, with cyclic vector y. Apply Proposition 5.3, writing s for its quotient character and $\alpha = \kappa \cup s$. In characteristic three its parameters are

$$(r_1, r_2, r_3) = (1, 1, 1), \qquad (\lambda_1, \lambda_2, \lambda_3) = (2, 0, 1).$$

For the resulting surjection $\rho : G_K \to P$, the local images are cyclic or contained in V at every prime.

Form the actual section (7.5) and choose common local primitives Q_v as in Lemma 7.3. At the three distinguished primes the weights $1 - \lambda_i^2$ are $0, 1, 0$, so $L_v = \operatorname{tr}(e|_v \cup Q_v)$ has invariants $0, 1/3, 0$. Every other cyclic image contributes a multiple of $\operatorname{inv}_v(\alpha) = 0$; every image in V

contributes zero with $Q_v = 0$. This includes the full decomposition group at the approximation prime. Consequently

$$\sum_v \mathrm{inv}_v[L_v] = 1/3. \tag{7.10}$$

Equation (7.7) and Corollary 2.11 now exclude an unrestricted lift. The local formula applies regardless of any additional ramification of the Kummer lifts defining the section. □

7.4. **An explicit cyclic subextension and coefficient fields.** For $K = \mathbf{Q}(\zeta)$, where $\zeta^2 + \zeta + 1 = 0$, the cyclic quotient in Proposition 5.3 may be specified explicitly. Take

$$\pi_1 = 3 + \zeta, \qquad \pi_2 = 4 + \zeta, \qquad \pi_3 = 6 + \zeta, \qquad v_i = (\pi_i),$$

and

$$A = \pi_1^2 \pi_2 \pi_3 = 139 + 142\zeta, \qquad L = K\big((139 + 142\zeta)^{1/3}\big).$$

The norms are $7, 13, 31$, so the arithmetic Frobenius values of κ are $2, 1, 1$, while the valuations of A are $2, 1, 1$. Formula (5.3) gives invariant $1/3$ for $\kappa \cup \mathrm{Kum}(A)$ at each v_i. At every other prime away from three both characters are unramified. The unique prime above three has invariant zero by reciprocity. Thus (5.4) holds, and Proposition 5.3 retains this prescribed quotient while constructing the required P-extension.

Corollary 7.4. *If K is a global field of characteristic different from three with $\mu_3 \subset K$ and $\mu_9 \not\subset K$, k is a finite field of characteristic three, and $n \geq 6$, then there is a continuous representation $G_K \to \mathrm{GL}_n(k)$ with no lift to $\mathrm{GL}_n(W_2(k))$.*

Proof. Extend the coefficients of the representation in Theorem 7.1 from $\mathbf{F}_3$ to k and add $n - 6$ trivial summands. Lemmas 2.2 and 2.4 show that a lift would give a lift of the original representation, a contradiction. □

7.5. **Consequences for global fields and for Q.**

Proof of Corollary 1.5. The positive statements follow from Theorems 1.1 and 1.2. For each odd p, Dirichlet's theorem gives a prime $q \equiv 1 + p \pmod{p^2}$. Apply Theorem 1.3 or 1.4 to $\mathbf{F}_q(t)$, and use its coefficient-extension assertion. For the sharp uniform dimension bound, take $p = 5$ and $K = \mathbf{F}_{11}(t)$. □

Proof of Corollary 1.6. For $p = 2$, apply Theorem 1.1. For odd p, put $m_3 = 6$ and $m_p = p$ for $p \geq 5$. More generally, let K be a global field of characteristic different from p, let $L = K(\mu_p)$, and suppose $\mu_{p^2} \not\subset L$. By Theorem 1.3 or 1.4, there is a nonliftable k-representation V of G_L of dimension m_p. The induction argument of [22, §2.1, proofs of Lemmas 2.2–2.3 in the arXiv version] applies to the open subgroup G_L. The restriction of $\mathrm{Ind}_{G_L}^{G_K} V$ contains V as the direct summand indexed by the identity coset. Thus the induced representation of dimension $[L : K]m_p$ is nonliftable by Lemma 2.4. The same lemma allows us to add trivial summands in every higher dimension.

For $K = \mathbf{Q}$ we have $[\mathbf{Q}(\mu_p) : \mathbf{Q}] = p - 1$ and $[\mathbf{Q}(\mu_{p^2}) : \mathbf{Q}(\mu_p)] = p$. The stated bounds follow. □

8. Semisimple representations over arbitrary finite fields

Theorem 8.1. *Let K be a global field and let k be a finite field of characteristic p. Every continuous semisimple representation $\bar{\rho} : G_K \to \mathrm{GL}_n(k)$ admits a continuous lift to $\mathrm{GL}_n(W_2(k))$.*

Semisimplicity is over k, and neither the determinant nor the allowed ramification is prescribed. We apply Böckle's cohomological lifting criterion [2, Theorem 1.5] using the trace rigidity of Mazur and Carayol. Passage to the endomorphism field of each irreducible constituent and restriction of scalars treat general finite coefficient fields. Restriction and corestriction along $K(\mu_p)/K$ then descend the vanishing of the obstruction.

8.1. **Trace detection.** We use the cyclic restriction kernel defined in Section 2, including cyclic subgroups of order divisible by p.

Lemma 8.2. *Let k be any field, let Γ be a finite group, and let $\rho : \Gamma \to \mathrm{GL}(V)$ be an absolutely irreducible finite-dimensional representation. With $M = \mathrm{End}_k(V)$, the map*

$$H^1(\Gamma, M) \longrightarrow k^\Gamma, \qquad [c] \longmapsto (g \longmapsto \mathrm{tr}(c(g)\rho(g))) \tag{8.1}$$

is well-defined and injective. Consequently $H^1_{\mathrm{cyc}}(\Gamma, M) = 0$.

Proof. The function in (8.1) is the ε-coefficient of the trace of $\rho_c(g) = (1 + \varepsilon c(g))\rho(g)$ over $k[\varepsilon]/(\varepsilon^2)$. Changing c by a coboundary conjugates ρ_c and hence preserves its trace. If this coefficient vanishes for every g, then the Mazur–Carayol trace theorem [5, Theorem 1] identifies ρ_c with the constant deformation. The conjugating matrix reduces to a scalar by absolute irreducibility, so rescaling it gives a strict conjugacy and $[c] = 0$. For a cyclically trivial class, each $\rho_c(g)$ is conjugate to $\rho(g)$ on $\langle g \rangle$, so the coefficient vanishes for every g. □

Remark 8.3. The lemma uses ordinary k-valued traces on all elements of Γ and the full coefficient module $\mathrm{End}_k(V)$, including when $p \mid \dim V$.

8.2. **The cyclotomic case.** Böckle's local lifting theorem gives a $W_2(k)$-lift at every nonarchimedean prime [2, Theorem 1.1]. At archimedean primes the following stronger conclusion holds.

Lemma 8.4 ([2, Lemma 2.3])**.** *Let k be a finite field. Every representation of $G_{\mathbf{R}}$ or $G_{\mathbf{C}}$ in $\mathrm{GL}_n(k)$ lifts to $\mathrm{GL}_n(W(k))$.*

For the remainder of this subsection suppose $\mathrm{char}\, K \neq p$. Let L be the field cut out by $\bar\rho$, and put $H = \mathrm{Gal}(L(\mu_p)/K)$ and $M = \mathrm{End}_k(k^n)$. The invariant perfect pairing $(A, B) \mapsto \mathrm{Tr}_{k/\mathbf{F}_p}\mathrm{tr}(AB)$ identifies M^D with $M(1)$. In Böckle's definition of global unobstructedness, we may take all cyclic subgroups of H as the auxiliary subgroups. His lifting criterion then gives the following corollary [2, Theorem 1.5].

Corollary 8.5. *If $H^1_{\mathrm{cyc}}(H, M(1)) = 0$, then $\bar\rho$ lifts continuously to $\mathrm{GL}_n(W_2(k))$.*

Remark 8.6. The field trace is nondegenerate because $k/\mathbf{F}_p$ is separable, without division by $[k : \mathbf{F}_p]$. At real primes the modified groups in degrees 1 and 2 agree with ordinary cohomology, so no oddness or imaginarity hypothesis is imposed.

Proposition 8.7. *Suppose that $\bar\rho|_{G_{K(\mu_p)}}$ is absolutely irreducible. Then $\bar\rho$ admits a continuous lift to $\mathrm{GL}_n(W_2(k))$. In particular, this applies to every absolutely irreducible representation when $\mu_p \subset K$.*

Proof. Let $\Delta = \ker(\bar\chi_p : H \to \mathbf{F}_p^\times)$. The restriction of $M(1)$ to Δ is M, and the residual representation of Δ is absolutely irreducible. Lemma 8.2 therefore kills the restriction of every class in $H^1_{\mathrm{cyc}}(H, M(1))$. Restriction on H^1 is injective because $[H : \Delta]$ is prime to p, by restriction–corestriction. Corollary 8.5 applies. If $\mu_p \subset K$, then $\Delta = H$. □

Remark 8.8. For $p = 2$, $K(\mu_2) = K$, so Proposition 8.7 already includes every absolutely irreducible representation. The argument below treats all semisimple representations uniformly.

8.3. **Restriction of scalars and cyclotomic descent.**

Lemma 8.9. *Let k be a finite field, let Γ be a finite group, and let V be an irreducible finite-dimensional $k[\Gamma]$-module. Then*

$$D = \mathrm{End}_{k[\Gamma]}(V)$$

is a finite field extension of k. The action of Γ on V is D-linear, and V, viewed as a $D[\Gamma]$-module, is absolutely irreducible.

Proof. By Schur's lemma and Wedderburn's little theorem, D is a finite field. Over D, the module V is still simple, and its endomorphism ring is D since every element of D commutes with the scalar D-action. Absolute irreducibility follows from [37, Proposition 9.2.5]. □

Lemma 8.10. *Let D/k be a finite extension of finite fields, of degree d. The ring $W_2(D)$ is free of rank d over $W_2(k)$. If a continuous representation of a profinite group on a D-vector space V of dimension m has a $W_2(D)$-lift, then its underlying k-representation has a $W_2(k)$-lift of dimension $md = \dim_k V$.*

Proof. The Teichmüller lifts of a k-basis of D form a $W_2(k)$-basis of $W_2(D)$ by [8, (2.33) and its proof]. Restricting scalars thus gives a free rank-md lift with reduction V. Lifting a change of basis realizes the prescribed residual matrices. □

Proof of Theorem 8.1. If $\operatorname{char} K = p$, then [2, Theorem 1.7] gives a $W(k)$-lift. Henceforth $\operatorname{char} K \neq p$. First suppose $\mu_p \subset K$. For each irreducible constituent V_i, Lemma 8.9 makes V_i absolutely irreducible over its finite endomorphism field D_i. Proposition 8.7 lifts it over $W_2(D_i)$, and Lemma 8.10 gives a $W_2(k)$-lift of the same k-dimension. Taking direct sums proves the theorem in this case.

For general K, put $F = K(\mu_p)$. By Clifford's theorem [37, Theorem 5.3.1], restriction to the normal subgroup $\bar{\rho}(G_F)$ of the residual image preserves semisimplicity. The case just proved kills the restriction of the lifting obstruction $\omega \in H^2(K, \operatorname{End}_k(k^n))$. Since $\operatorname{cor}_{F/K} \operatorname{res}_{F/K}(\omega) = [F : K]\omega$ and $[F : K] \mid p - 1$, we have $\omega = 0$. This descends obstruction vanishing, without requiring descent data on a chosen lift over F. □

In particular, Theorem 8.1 applies to every irreducible representation over k. Trace detection enters the proof only over the endomorphism field of each irreducible constituent.

Remark 8.11 (Scope of the semisimple theorem). The proof concerns the full adjoint module and unrestricted lifts. It does not give a prescribed determinant, preserve a prescribed set of ramified primes, or extend a chosen $W_2(k)$-lift to $W_3(k)$. Sections 9 and 10 show that the freedom to enlarge the ramification set is necessary over both number fields and function fields, even in dimension two. For a nonsplit extension, separately lifting the subrepresentation and quotient does not lift the extension class; semisimplification alone therefore gives no assertion about the original representation.

9. Prescribed ramification over a number field

We first construct an absolutely irreducible counterexample over $\mathbf{Q}$. Throughout this section, $q = 1951$ and $S = \{5, q, \infty\}$. The following theorem and Corollary 9.8 prove Theorem 1.8.

Theorem 9.1. *Put $q = 1951$, $p = 5$, $k = \mathbf{F}_5$, and $S = \{5, q, \infty\}$. There exists a continuous surjection*

$$\bar{\rho} : G_{\mathbf{Q},S} \longrightarrow \mathrm{SL}_2(\mathbf{F}_5) \subset \mathrm{GL}_2(\mathbf{F}_5)$$

which is absolutely irreducible, is even, and is unramified at every finite prime other than q, with the following properties.

(i) *There is no continuous lift $G_{\mathbf{Q},S} \to \mathrm{GL}_2(\mathbf{Z}/25\mathbf{Z})$ of $\bar{\rho}$, with any determinant. Consequently there is no lift $G_{\mathbf{Q},S} \to \mathrm{GL}_2(\mathbf{Z}_5)$.*

(ii) *There is a finite-image lift on the same $G_{\mathbf{Q},S}$ over $\mathcal{O} = \mathbf{Z}_5[\sqrt{5}]$, whose residue field is still $\mathbf{F}_5$.*

The computational input to part (i) is the exact relation in Proposition 9.3. The remaining argument is given explicitly below and uses neither a computation of a universal deformation ring nor a calculation of a full ray class module. The arithmetic certificate is part of the proof; its verification uses exact number-field and ideal arithmetic and does not assume the completeness of a computed class group or unit group.

9.1. The residual representation and its projective field.

Set

$$f(X) = X^5 - X^4 - 780X^3 - 1795X^2 + 3106X + 344, \tag{9.1}$$

and let L be its splitting field over $\mathbf{Q}$. The example in Doud–Moore [14, §2, §4, and Table 1] gives an even complex Artin representation ρ of conductor q with projective field L. Thus $\mathrm{Gal}(L/\mathbf{Q}) \simeq A_5$; L is totally real and ramified only at q, with inertia of order 5 there.

Here is a construction with the required original residue field. Write $\chi = \det \rho$. Conductor exponent one means that the inertia representation at q is tame with a one-dimensional fixed subspace. Its eigenvalues are 1 and a primitive fifth root, since its projective image has order 5. Hence χ^5 is unramified at all finite primes of $\mathbf{Q}$, so it is trivial. Therefore

$$\rho_0 = \rho \otimes \chi^2, \qquad \det \rho_0 = \chi^5 = 1.$$

The finite image of ρ_0 is the binary icosahedral group, isomorphic to $\mathrm{SL}_2(\mathbf{F}_5)$. Compose its quotient map with such an isomorphism and the natural two-dimensional representation over $\mathbf{F}_5$ to obtain $\bar\rho$. This construction uses an abstract finite quotient, not a choice of reduction of complex matrix entries.

The natural representation of $\mathrm{SL}_2(\mathbf{F}_5)$ is absolutely irreducible: the upper and lower nontrivial unipotent subgroups have different unique fixed lines even over $\overline{\mathbf{F}}_5$. Complex conjugation has image $\pm I$, since its projective image is trivial. Thus $\bar\rho$ is even. Its projective kernel cuts out precisely L. The binary field itself need not be totally real.

Write $\Gamma = \mathrm{Gal}(L/\mathbf{Q})$, acting on the five ordered roots of (9.1). At a prime $w \mid q$, let D_w and C_w be the decomposition and inertia subgroups in Γ. Tame Frobenius acts on C_w by the q-power map. Since $q \equiv 1 \pmod 5$, D_w centralizes C_w. The centralizer of a 5-cycle in A_5 is its cyclic subgroup: a permutation commuting with a fixed 5-cycle is determined by the image of one letter, and is a power of that cycle. Thus

$$D_w = C_w \simeq C_5. \tag{9.2}$$

In particular the residue field at w is $\mathbf{F}_q$.

9.2. **The adjoint module and a projection.** Let $M = \mathfrak{sl}_2(\mathbf{F}_5)$ with the adjoint action of $\mathrm{SL}_2(\mathbf{F}_5)$. Its center acts trivially, so M is a $\Gamma \simeq A_5$-module. We use the following concrete realization. Let

$$V = \{(a_1, \dots, a_5) \in \mathbf{F}_5^5 : \textstyle\sum a_i = 0\}, \qquad M_{\mathrm{perm}} = V/\mathbf{F}_5(1,1,1,1,1).$$

If b_i is the image of $e_i - e_5$, then (b_1, b_2, b_3) is a basis and $b_4 = -b_1 - b_2 - b_3$. Indeed $e_i - e_5$, $1 \le i \le 4$, form a basis of V, and their sum is $(1,1,1,1,1)$ in characteristic five. The first three images are independent: in a linear combination of $e_1 - e_5, e_2 - e_5, e_3 - e_5$, the fourth coordinate is zero, so this combination can be constant only if it is zero. Put

$$u_1 = (1,0,0)^t, \quad u_2 = (0,1,0)^t, \quad u_3 = (0,0,1)^t, \quad u_4 = (-1,-1,-1)^t, \quad u_5 = 0.$$

For a permutation σ of the roots, its action matrix R_σ has jth column $u_{\sigma(j)} - u_{\sigma(5)}$, $1 \le j \le 3$. All matrices in this subsection are over $\mathbf{F}_5$ and act on columns.

For completeness, take $s = (12)(34)$ and $t = (135)$. The matrices

$$a = \begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}, \qquad b = \begin{pmatrix} 0 & 4 \\ 1 & 1 \end{pmatrix}$$

generate $\mathrm{SL}_2(\mathbf{F}_5)$, and their projective images correspond to s and t, respectively. Relative to the basis $E_{12}, \mathrm{diag}(1,-1), E_{21}$ of M, the matrix

$$C = \begin{pmatrix} 4 & 0 & 3 \\ 3 & 2 & 2 \\ 1 & 2 & 1 \end{pmatrix}, \qquad \det C = 4,$$

satisfies $C\,\mathrm{Ad}(a) = R_s C$ and $C\,\mathrm{Ad}(b) = R_t C$. To make these checks explicit, the four matrices are

$$\mathrm{Ad}(a) = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 0 \end{pmatrix}, \quad R_s = \begin{pmatrix} 0 & 1 & 4 \\ 1 & 0 & 4 \\ 0 & 0 & 4 \end{pmatrix},$$

$$\mathrm{Ad}(b) = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 4 & 4 \\ 4 & 2 & 1 \end{pmatrix}, \quad R_t = \begin{pmatrix} 4 & 4 & 4 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Multiplication by the displayed C proves both intertwining identities. Moreover $a^{-1}b = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$; its powers give all upper elementary matrices, and conjugating by a gives all lower elementary

matrices. Gaussian elimination shows that these generate $\mathrm{SL}_2(\mathbf{F}_5)$. The kernel of the adjoint action on $\mathfrak{sl}_2(\mathbf{F}_5)$ is its scalar subgroup $\{\pm I\}$.

The permutation action on M_{perm} is faithful. If a permutation σ acts trivially, then $\sigma(e_i - e_j) - (e_i - e_j)$ is a constant vector for all $i \neq j$. Its support has size at most four, so this constant vector is zero. Since $1 \neq -1$ in $\mathbf{F}_5$, this forces $\sigma(i) = i$ and $\sigma(j) = j$. Thus $\sigma = 1$. The intertwining identities therefore identify the group generated by s, t with the adjoint image of $\mathrm{SL}_2(\mathbf{F}_5)$, of order 60. Both permutations are even, so this group is A_5, and $M \simeq M_{\mathrm{perm}}$.

Every automorphism of A_5 is induced by conjugation by a permutation of the five letters [12, §8.2, Exercise 8.2.2]. Since V and its constant line are S_5-stable, the isomorphism class of M_{perm} is independent of the chosen identification $\Gamma \simeq \mathrm{PSL}_2(\mathbf{F}_5)$.

Fix

$$H = \langle t \rangle, \qquad t = (135), \qquad F = L^H, \qquad [F : \mathbf{Q}] = 20.$$

The averaging idempotent $e = \frac{1}{3}(1 + t + t^2)$ acts as

$$e = \begin{pmatrix} 0 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 0 \end{pmatrix}, \qquad e(x, y, z)^t = y(3, 1, 3)^t. \tag{9.3}$$

Indeed $1/3 = 2$ in $\mathbf{F}_5$, so the displayed matrix is $2(I + R_t + R_t^2)$. It satisfies $e^2 = e$ and $R_t e = e R_t = e$, and its image is exactly $\mathbf{F}_5(3, 1, 3)^t$. Thus M^H is one-dimensional and $\ker e = \{(x, 0, z)^t : x, z \in \mathbf{F}_5\}$.

Lemma 9.2. *For every subgroup $C_5 \subset \Gamma$ of order 5, the restriction $e : M^{C_5} \to M^H$ is nonzero.*

Proof. Write $\mathbf{1} = (1, 1, 1, 1, 1)$. For the standard dot product on $\mathbf{F}_5^5$, one has $V^\perp = \mathbf{F}_5\mathbf{1} \subset V$. Its restriction to V therefore has radical exactly $\mathbf{F}_5\mathbf{1}$, and descends to a nondegenerate invariant symmetric form on M_{perm}. Since $(e_i - e_5) \cdot (e_j - e_5) = \delta_{ij} + 1$, its Gram matrix is

$$G = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

A generator of C_5 has a lift to $\mathrm{SL}_2(\mathbf{F}_5)$ whose fifth power is I: multiply either lift by $-I$ if its fifth power is $-I$. This lift is a nontrivial unipotent $I + N$ with $N^2 = 0$. After conjugation in $\mathrm{GL}_2(\mathbf{F}_5)$ we may take $N = E_{12}$. For $X = \left(\begin{smallmatrix} r & s \\ t & -r \end{smallmatrix}\right)$, the equation $[E_{12}, X] = 0$ says $t = 0$ and $2r = 0$. Thus M^{C_5} is precisely the line spanned by N, and this line is isotropic for $X \mapsto \mathrm{tr}(X^2)$ because $N^2 = 0$. Direct multiplication gives

$$C^t G C = 2G_{\mathrm{tr}}, \qquad G_{\mathrm{tr}} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Thus every C_5-fixed line in the permutation model is isotropic. On the other hand, the quadratic form on $\ker e = \{y = 0\}$ is

$$(x, 0, z) G (x, 0, z)^t = 2(x^2 + xz + z^2).$$

It is anisotropic over $\mathbf{F}_5$: if $z \neq 0$, then a zero would give a root of $T^2 + T + 1$, whose discriminant $-3 = 2$ is nonsquare; if $z = 0$, then $x = 0$. Hence no fixed line is contained in $\ker e$. □

Two particular automorphisms in the certified ordering will be used:

	$(\sigma(1), \ldots, \sigma(5))$	R_σ
g_2	$(1, 3, 2, 5, 4)$	$\begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$
g_7	$(5, 1, 4, 2, 3)$	$\begin{pmatrix} 0 & 1 & 4 \\ 0 & 0 & 4 \\ 4 & 4 & 3 \end{pmatrix}$

Here the subscripts are labels, not powers. Put $h = (3,1,3)^t$, so that $e = h(0,1,0)$. The middle coordinates of $R_{g_2}h$ and $R_{g_7}h$ are respectively $3+1+2\cdot 3 = 0$ and $4\cdot 3 = 2$ in $\mathbf{F}_5$. Consequently $eR_{g_2}e = 0$ and $eR_{g_7}e = 2e$, and hence

$$3eR_{g_2}e = 0, \qquad 3eR_{g_7}e = e. \tag{9.4}$$

The factor 3 is important: these are the scalars for norm composed with conjugation and extension, without division by $[L:F]$.

9.3. **The exact arithmetic certificate.** For $E = F, L$, let

$$\mathfrak{m}_E = (25q)\mathcal{O}_E, \qquad X_E = \mathrm{Cl}_{\mathfrak{m}_E}(E)/5\,\mathrm{Cl}_{\mathfrak{m}_E}(E).$$

There is no infinite component in these moduli. We write ray class groups additively, but fractional ideals multiplicatively. The class $[(a)]$ of a principal ideal need not vanish in a ray class group: it vanishes when $a \equiv 1 \pmod{\mathfrak{m}_E}$, with the congruence interpreted locally at the primes dividing $\mathfrak{m}_E$.

Ideal extension and ideal norm induce maps

$$j : X_F \longrightarrow X_L, \qquad n : X_L \longrightarrow X_F.$$

The modulus of L is extended from F, and norms of ray-principal elements are ray-principal. Consequently these maps are well defined and satisfy

$$nj = 3, \qquad jn = 1 + t + t^2 = 3e. \tag{9.5}$$

For $g \in \Gamma$, define the unnormalized operator

$$T_g = ngj : X_F \longrightarrow X_F. \tag{9.6}$$

On an ideal J prime to the modulus this is represented by $\mathrm{N}_{L/F}(g(J\mathcal{O}_L))$.

There are four primes of F above q, each with ramification index 5 and residue degree 1. The extension L/F splits completely at each of them: its decomposition subgroup is $H \cap D_w = 1$, by (9.2). Label by $\mathfrak{p}_1$ the first such prime in the certificate. Choose the explicitly supplied integral element a_1 with congruences

$$\begin{cases} a_1 \equiv 3 \pmod{\mathfrak{p}_1^5}, \\ a_1 \equiv 1 \pmod{\mathfrak{p}^5} & \text{for } \mathfrak{p} \mid q,\ \mathfrak{p} \neq \mathfrak{p}_1, \\ a_1 \equiv 1 \pmod{\mathfrak{p}^2} & \text{for } \mathfrak{p} \mid 5. \end{cases} \tag{9.7}$$

These are the full powers in $\mathfrak{m}_F$. Put $A_1 = [(a_1)] \in X_F$. The residue 3 is not a fifth power in $\mathbf{F}_q^\times$; in fact its order is 1950.

Proposition 9.3 (Exact arithmetic certificate)**.** *The accompanying data specify fractional ideals $I_2, I_6, G_1, \ldots, G_6$ of F, prime to $\mathfrak{m}_F$, and an element $\alpha \in F^\times$ such that*

$$\begin{aligned} D &:= \frac{\mathrm{N}_{L/F}(g_2(I_2\mathcal{O}_L))\, I_6}{\mathrm{N}_{L/F}(g_7(I_6\mathcal{O}_L))\,(a_1)^2} \\ &= G_1^{30570} G_2^{20} G_4^5(\alpha), \qquad \alpha \equiv 1 \pmod{\mathfrak{m}_F}. \end{aligned} \tag{9.8}$$

In particular, in the true ray class quotient X_F,

$$T_{g_2}([I_2]) - T_{g_7}([I_6]) + [I_6] = 2A_1. \tag{9.9}$$

Verification and deduction. The exact objects and checks are described in Appendix A. Both field models are certified with their maximal orders. The norm components in the numerator and denominator of (9.8) are independently recomputed from the degree-60 field, its automorphisms, and the embedding $F \hookrightarrow L$. The ideal identity is verified using exact ideal arithmetic. The full ray congruence for the factored element α is checked separately; it is not inferred from a putative class group relation. All these checks have passed.

Set $B = G_1^{6114} G_2^4 G_4$. Equation (9.8) is exactly $D = B^5(\alpha)$, so $[D] = 0$ in X_F. Taking ray classes in the definition of D proves (9.9). There is no assumption that I_2, I_6 or the G_i generate a class group. □

Remark 9.4. A class group computation was used to find this witness. Its completeness is not used in the proof: an explicit equality of fractional ideals with a verified ray-principal factor remains true even if a search computation missed class group or unit group relations. In particular, no GRH assumption is imported from that search into Proposition 9.3.

9.4. **Consequences of a hypothetical lift.** Suppose there is a lift $\rho_{25} : G_{\mathbf{Q},S} \to \mathrm{GL}_2(\mathbf{Z}/25\mathbf{Z})$. Its determinant δ has values in $1 + 5\mathbf{Z}/25\mathbf{Z}$, a group of exponent 5, since $(1 + 5r)^5 = 1$ modulo 25. The scalar twist

$$\rho_{25}^{\circ}(g) = \delta(g)^2 \rho_{25}(g)$$

has determinant $\delta(g)^5 = 1$ and unchanged residual representation. It is still defined on the same $G_{\mathbf{Q},S}$. It therefore suffices to exclude determinant-one lifts.

Projectivize ρ_{25}° to obtain a lift of the projective residual representation to $\mathrm{PSL}_2(\mathbf{Z}/25\mathbf{Z})$. The reduction kernel is canonically the additive group M:

$$M \longrightarrow \ker\big(\mathrm{PSL}_2(\mathbf{Z}/25\mathbf{Z}) \to \mathrm{PSL}_2(\mathbf{F}_5)\big), \qquad X \longmapsto [I + 5X].$$

Indeed $\det(I + 5X) = 1 + 5\,\mathrm{tr}(X)$ modulo 25, so the determinant-one condition is exactly $\mathrm{tr}(X) = 0$ in $\mathbf{F}_5$. A projective class in the reduction kernel has a representative reducing to I after a unique choice of sign; two representatives of the form $I+5X$ cannot differ by $-I$, because their reductions are both I. The product law is $(I+5X)(I+5Y) = I+5(X+Y)$, and conjugation induces the usual residual adjoint action. This proves both the asserted identification and its equivariance. Let $N = \ker(G_{\mathbf{Q},S} \to \Gamma) = G_{L,S_L}$. Restriction of this projective lift to N is an additive homomorphism $N \to M$, equivariant for the conjugation action of Γ.

Lemma 9.5. *This restriction factors through a Γ-equivariant linear map $\varphi : X_L \to M$.*

Proof. It kills commutators and fifth powers. By class field theory it therefore suffices to bound the conductors of exponent-5 abelian characters of G_{L,S_L}. The field L is unramified at 5. At each prime $v \mid 5$, the 5-adic logarithm gives $(1 + 5\mathcal{O}_{L_v})^5 = 1 + 25\mathcal{O}_{L_v}$; hence conductor exponent at most 2 suffices. At $w \mid q$, the principal units are pro-q, so every exponent-5 character is trivial on them and has conductor exponent at most 1. The larger exponent 5 in $q\mathcal{O}_L$ is harmless. At real primes an order-2 inertia group has no nontrivial map to an exponent-5 group. These bounds give the finite modulus $\mathfrak{m}_L$ and prove the factorization. Equivariance follows from functoriality of Artin reciprocity, or directly from conjugation in the lifted group. □

Define

$$\psi = \varphi \circ j : X_F \longrightarrow M^H.$$

The target is M^H because extended ideals are H-invariant. Using (9.5) and equivariance, for $y \in X_F$ we get

$$\psi(T_g y) = \varphi(jngjy) = 3eR_g\varphi(jy) = 3eR_g e\psi(y).$$

Thus (9.4) implies

$$\psi T_{g_2} = 0, \qquad \psi T_{g_7} = \psi. \tag{9.10}$$

Applying ψ to (9.9) now gives

$$2\psi(A_1) = 0, \qquad \text{hence } \psi(A_1) = 0. \tag{9.11}$$

We show that any lift instead forces this value to be nonzero.

9.5. **Tame inertia forces a nonzero value.**

Lemma 9.6. *Let $u = I + E_{12} \in \mathrm{SL}_2(\mathbf{F}_5)$. Every matrix $A \in \mathrm{GL}_2(\mathbf{Z}/25\mathbf{Z})$ reducing to u satisfies*

$$A^5 = I + 5E_{12} \neq I.$$

Its projective fifth power is nontrivial as well.

The identity is Serre's power formula for $p = 5$ and $N = E_{12}$ [33, Chap. IV, §3.4, proof of Lemma 3, pp. IV-23–IV-24]. Since $I+5E_{12}$ is nonscalar, the fifth power is nontrivial projectively. Lifting a change of basis gives the same projective conclusion for every nontrivial residual unipotent.

Fix $w \mid \mathfrak{p}_1$ in L. Define $x_w \in X_L$ using a CRT element which is 3 modulo w^5 and 1 at the full modulus powers of every other prime dividing $\mathfrak{m}_L$. Such an element is a unit at these primes, and its principal ideal determines the class. Its class is unchanged by changing the CRT representative. The group C_w fixes x_w, because its residue degree is one and the prescribed residues are rational integers. Thus $\varphi(x_w) \in M^{C_w}$.

Lemma 9.7. *The map induced by a hypothetical lift satisfies $\varphi(x_w) \neq 0$.*

Proof. Choose a generator τ of the pro-5 part of tame inertia over $\mathbf{Q}_q$. It maps onto the projective inertia group C_w. Its binary residual image is a nontrivial unipotent of order 5; the pro-5 choice excludes a scalar factor of order 2. The element τ^5 belongs to N, and is a generator of the pro-5 tame inertia subgroup over L_w before passage to its abelian quotient. By Lemma 9.6, the projective lift has nontrivial value on τ^5.

Under local reciprocity the exponent-5 quotient of tame inertia over L_w is generated by the image of any non-fifth-power residue unit in $\mathbf{F}_q^\times$. The class of 3 is such a generator. The ray class x_w represents this local unit image in the global abelian quotient, up to the harmless inverse determined by the reciprocity convention. Its image under φ is consequently a nonzero scalar multiple of the image of τ^5, and is nonzero. □

Since L/F splits at $\mathfrak{p}_1$, the three primes above $\mathfrak{p}_1$ are the H-orbit of w. The CRT conditions (9.7) give the exact identity in the ray class quotient

$$j(A_1) = \sum_{h \in H} h x_w.$$

Indeed the product of the three corresponding CRT elements has the same residue 3 at each of these three primes, and residue 1 at all other modulus primes, as a_1 does after extension to L. It follows that

$$\psi(A_1) = 3e\varphi(x_w).$$

By Lemma 9.7, $\varphi(x_w)$ is a nonzero vector in the C_w-fixed line. Lemma 9.2 shows that its projection under e is nonzero. Thus $\psi(A_1) \neq 0$, contradicting (9.11). This proves Theorem 9.1(i).

9.6. **A finite-image lift over a ramified quadratic ring.** Put $\mathcal{O} = \mathbf{Z}_5[\sqrt{5}]$, whose residue field is $\mathbf{F}_5$. Dorobisz constructs a lift $s : \mathrm{SL}_2(\mathbf{F}_5) \to \mathrm{GL}_2(\mathcal{O})$ of the natural representation. Its determinant is trivial because $\mathrm{SL}_2(\mathbf{F}_5)$ is perfect [13, Lemma 9 and Proposition 20(i) in the arXiv version]. Thus $s \circ \bar{\rho}$ is a continuous finite-image lift on $G_{\mathbf{Q},S}$, proving Theorem 9.1(ii).

Corollary 9.8. *For every finite extension $k'/\mathbf{F}_5$, the representation $\bar{\rho} \otimes_{\mathbf{F}_5} k'$ has no lift on $G_{\mathbf{Q},S}$ over $W_2(k') = W(k')/25W(k')$. In particular it has no lift over $W(k')$.*

Proof. By Lemma 2.2, a lift over $W_2(k')$ on $G_{\mathbf{Q},S}$ would give a lift over $\mathbf{Z}/25\mathbf{Z}$ on the same group, contradicting Theorem 9.1(i). A lift over $W(k')$ would give a lift over $W_2(k')$ by reduction. □

Remark 9.9. The example separates the demand for a $W(k)$-valued lift from the existence of a characteristic-zero point after coefficient extension. The obstruction concerns lifts over the original Witt ring and its unramified extensions. It does not rule out characteristic-zero lifting after a finite ramified extension of coefficients, as the lift in Theorem 9.1(ii) demonstrates.

10. Prescribed ramification over function fields

The freedom to introduce new ramification in Theorem 1.7 is also necessary over function fields. Here the obstruction is obtained from the monodromy of a family of four-point covers. The pth power of a separating Dehn twist fixes a residual representation but has no fixed point on the set of its lifts; specialization realizes this permutation as arithmetic Frobenius.

Theorem 10.1. *For every prime $p \geq 5$, there is a finite set Σ_p of rational primes, containing the prime divisors of $p(p^2-1)$, with the following property. For every prime $\ell \notin \Sigma_p$, there are a power $q = \ell^a$, an element $\lambda \in \mathbf{F}_q \setminus \{0,1,2\}$, and a continuous representation*

$$\bar{\rho} : G_{K,S} \longrightarrow \mathrm{SL}_2(\mathbf{F}_p), \qquad K = \mathbf{F}_q(x), \quad S = \{0, \lambda, 1, \infty\},$$

whose geometric image is $\mathrm{SL}_2(\mathbf{F}_p)$ and which has no lift $G_{K,S} \to \mathrm{GL}_2(\mathbf{Z}/p^2\mathbf{Z})$. Its tame inertia images at $0, \lambda, 1, \infty$ have orders $4, 3, 4, 6$, respectively. For every finite extension $k/\mathbf{F}_p$, its scalar extension has no $W_2(k)$-lift on the same group.

The conclusion concerns all lifts, with arbitrary determinant and arbitrary inertia orders at the points of S. The representation is geometrically absolutely irreducible. Its inflation to G_K does lift by Theorem 1.2; consequently every such lift is ramified at a prime outside S.

10.1. **A separating twist on the set of lifts.** Fix $p \geq 5$, and put $R = \mathbf{Z}/p^2\mathbf{Z}$ and $G = \mathrm{SL}_2(\mathbf{F}_p)$. Set

$$\begin{aligned} N = E_{12}, \qquad u = I + N, \qquad A = C = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \\ B = A^{-1}u = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}, \qquad D = A^{-1}u^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}. \end{aligned} \tag{10.1}$$

These matrices satisfy $AB = u$, $CD = u^{-1}$, and $ABCD = I$. Their orders are $4, 3, 4, 6$, respectively. Each of the pairs (A, B) and (C, D) generates G: it generates A and $u^{\pm 1}$, hence all upper and lower elementary matrices. The common centralizer of either pair in $M_2(\mathbf{F}_p)$ is the scalar subspace. Indeed, a matrix commuting with u has the form $\left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right)$, and commuting also with A forces $b = 0$. The natural representation of G is absolutely irreducible, since the unique u-invariant line over $\overline{\mathbf{F}}_p$ is $\langle e_1 \rangle$, which A does not preserve.

Lemma 10.2. *Every $V \in \mathrm{GL}_2(R)$ reducing to u satisfies $V^p = I + pN$.*

This is the $n = 1$ power formula in [33, Chap. IV, §3.4, proof of Lemma 3, p. IV-24], applied to $V = I + N + pX$ with $N^2 = 0$.

Define

$$H = \mathrm{red}^{-1}(G) \subset \mathrm{GL}_2(R).$$

Its order is $p^4|G| = p^5(p^2-1)$. Let $\mathcal{F}$ be the set of H-conjugacy classes of tuples $(a, b, c, d) \in H^4$ with $abcd = 1$ whose reductions are G-conjugate to (A, B, C, D). The tuples need not generate H. The set $\mathcal{F}$ is nonempty, since the integral matrices displayed in (10.1) have product I over R. Surjectivity of $H \to G$ allows each class to be represented by a tuple reducing exactly to (A, B, C, D).

For any tuple of product one, put

$$T(a, b, c, d) = (gag^{-1}, gbg^{-1}, c, d), \qquad g = ab. \tag{10.2}$$

This is the square of the Hurwitz move $(a, b, c, d) \mapsto (aba^{-1}, a, c, d)$, and represents the Dehn twist separating the first two punctures from the last two. The product ab is unchanged, so T^m conjugates the first two entries by $(ab)^m$ and fixes the last two. In particular, T^p fixes the residual tuple and acts on $\mathcal{F}$.

Lemma 10.3. *The permutation T^p of $\mathcal{F}$ has order p and no fixed point.*

Proof. Choose a representative (a, b, c, d) with the specified reduction. Lemma 10.2 gives $(ab)^p = I + pN$. If its class were fixed, then some $Q \in H$ would conjugate the tuple to its T^p-image. Modulo p, the matrix Q centralizes G, so its reduction is I or $-I$. Multiplying Q by a central sign, write $Q = I + pX$. Since c and d are unchanged, we have

$$[X, C] = [X, D] = 0 \quad \text{in } M_2(\mathbf{F}_p).$$

The centralizer calculation above makes X scalar. Hence Q is scalar over R, whereas the first entry changes by

$$(I + pN)a(I - pN) - a = p[N, A] = p \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0.$$

This contradiction proves the absence of fixed points. Moreover, $(ab)^{p^2} = I$ for every representative, so T^{p^2} is the identity on $\mathcal{F}$. Since $\mathcal{F}$ is nonempty, T^p has order p. □

10.2. **Hurwitz spaces for all possible lift images.** Put

$$B_0 = \mathbf{P}^1_{\mathbf{Z}[1/|H|]} \setminus \{0, 1, 2, \infty\}, \qquad \mathcal{U} = \mathbf{P}^1_{B_0} \setminus \{x = 0, x = \lambda, x = 1, x = \infty\}.$$

Write U_λ for the fiber of $\mathcal{U} \to B_0$ at λ. The four excluded sections of B_0 are disjoint because 2 is invertible on the base. The point $x = 2$ gives a section of $\mathcal{U} \to B_0$.

We use Hurwitz spaces with an identified deck group, so that their geometric points correspond to inner conjugacy classes of surjections. For a finite group J of order invertible on the base, the space of connected J-covers of $\mathbf{P}^1$ with four branch points is finite étale over the configuration space. Its geometric fibers classify tame covers, compatibly with field automorphisms, and its complex monodromy is the Hurwitz braid action [31, Theorem 4.11 and Remark 4.15(ii)]. Pull these spaces back to B_0 along $\lambda \mapsto \{0, \lambda, 1, \infty\}$.

Let $\mathcal{R} \to B_0$ be the resulting space for G. The following construction includes representations whose image is a proper subgroup of H.

Proposition 10.4. *There is a finite étale B_0-scheme $\mathcal{L}$ and a finite étale morphism*

$$\mathcal{L} \longrightarrow \mathcal{R} \tag{10.3}$$

with the following geometric fibers. If Ω is algebraically closed, $\lambda \in B_0(\Omega)$, and $r : \Pi_\lambda \twoheadrightarrow G$ represents a point of $\mathcal{R}_\lambda(\Omega)$, where $\Pi_\lambda = \pi_1(U_\lambda)$, then

$$\mathcal{L}_{[r]}(\Omega) \simeq \{\psi \in \mathrm{Hom}_{\mathrm{cont}}(\Pi_\lambda, H) : [\mathrm{red} \circ \psi]_G = [r]_G\}/H. \tag{10.4}$$

The bijection respects field automorphisms and reduction of representations. In particular, no surjectivity to H is required.

Proof. For each representative $J \leq H$ of an H-conjugacy class with $\mathrm{red}(J) = G$, let $\mathcal{C}_J$ be the pulled-back Hurwitz space of connected J-covers. Let $\mathcal{D}_J$ be its locus where each of the four inertia groups has nontrivial image in G. This locus is open and closed. Indeed, the tame local description along a branch section makes the inertia subgroup, up to conjugacy in J, locally constant in the étale topology; equivalently, one may use the uniqueness of tame deformations with prescribed branch sections [31, Proposition 4.5]. The condition is also invariant under $N_H(J)$.

We construct the reduction morphism directly on these pulled-back spaces. By [31, Propositions 4.10 and 4.13], $\mathcal{D}_J$ has an étale covering by charts carrying actual J-cover families. On each chart, take the quotient by $K_J = \ker(J \to G)$. This gives a tame G-cover with the same four branch sections. Formation of the quotient commutes with base change: on invariant affine charts this follows from the averaging idempotent, since $|K_J|$ is invertible. The tame local description also shows that the quotient is a smooth family of curves. The coarse classifying transformation [31, Theorem 2.1] gives a morphism from each chart to $\mathcal{R}$.

On an overlap, the two J-families have isomorphic geometric fibers, hence their quotient families have the same geometric G-classes. The two morphisms to $\mathcal{R}$ therefore agree: the diagonal of $\mathcal{R} \to B_0$ is open and closed, so their equalizer, which contains every geometric point of the overlap, is the whole overlap. Étale descent gives $\mathcal{D}_J \to \mathcal{R}$. This argument uses actual families on charts and does not require a base-change theorem for the universal property of coarse moduli spaces.

Changing the identification of the deck group by $N_H(J)$ acts on $\mathcal{D}_J$; inner conjugation by J acts trivially. On the quotient G this change is inner conjugation, so the reduction morphism is invariant under $N_H(J)/J$. Define

$$\mathcal{L} = \coprod_{\substack{[J]_H \\ \mathrm{red}(J)=G}} \mathcal{D}_J/(N_H(J)/J).$$

These orbit quotients exist as finite étale B_0-schemes: after an étale base change trivializing the covers, they are quotients of finite constant sets. The reduction morphisms descend to (10.3). A morphism between finite étale B_0-schemes is itself finite étale, by the same local description.

For completeness, let $\mathrm{Surj}_*(\Pi_\lambda, J)$ denote the continuous surjections for which all four residual inertia images are nontrivial. The geometric fiber of $\mathcal{L}$ over λ is

$$\coprod_{\substack{[J]_H \\ \mathrm{red}(J)=G}} \mathrm{Surj}_*(\Pi_\lambda, J)/N_H(J).$$

Every H-valued homomorphism with the stated residual conditions occurs, by taking J to be its image and conjugating J to a chosen representative. Any H-conjugacy between homomorphisms with image J is given by an element of $N_H(J)$. This proves the bijection with all such H-classes. Taking the fiber over $[r]_G$ gives (10.4); the nontrivial residual inertia condition is automatic in this fiber. All constructions commute with field automorphisms. □

At the complex parameter $\lambda = 1/2$, the tuple (10.1) defines a point $r \in \mathcal{R}(\overline{\mathbf{Q}})$. Here we use the presentation of the geometric fundamental group by four peripheral generators of product one. Let Z be the connected component of $\mathcal{R}_{\overline{\mathbf{Q}}}$ containing r, and put

$$L = \mathcal{L} \times_{\mathcal{R}} Z.$$

Then Z is a smooth connected curve and $L \to Z$ is finite étale, with fiber $\mathcal{F}$ at r. Along each of the four ordered sections, the order of the tame inertia group is locally constant in the étale topology, by the local description used in the proof of Proposition 10.4. Thus the locus $\mathcal{R}_{(4,3,4,6)} \subset \mathcal{R}$ with these ordered inertia orders is open and closed over $\mathbf{Z}[1/|H|]$. By (10.1), the map from Z to $\mathcal{R}$ factors through this locus.

A loop moving λ around 0 acts by T, up to the inverse convention for the braid action. Its pth power fixes r and lifts to a closed loop on Z. Its action on $\mathcal{F}$ is T^p, or its inverse.

Let

$$\Gamma = \mathrm{im}\big(\pi_1(Z, r) \longrightarrow \mathrm{Sym}(\mathcal{F})\big).$$

The kernel defines a connected finite étale right Γ-torsor $Y \to Z$. We regard $\mathcal{F}$ as the given left Γ-set and use the convention $(y\delta, f) \sim (y, \delta f)$ for the contracted product. There is an isomorphism

$$L \simeq Y \times^{\Gamma} \mathcal{F}. \tag{10.5}$$

This uses the action on the entire fiber, whether or not L is connected. Lemma 10.3 supplies an element $\gamma \in \Gamma$ of order p with no fixed point on $\mathcal{F}$. With our convention, if arithmetic Frobenius on a torsor fiber satisfies $F(y) = y\delta$, then its action on the associated fiber is $f \mapsto \delta f$, since $F[y, f] = [y\delta, f] = [y, \delta f]$. Changing y conjugates δ and hence preserves the absence of fixed points.

10.3. **Specialization and arithmetic Frobenius.** We use the following consequence of the finite-field Chebotarev theorem [26, Theorem 1.1].

Lemma 10.5. *Let $Y_0 \to Z_0$ be a finite étale right torsor under a finite constant group Γ over a finite field $\mathbf{F}_{q_0}$. Suppose that Y_0 and Z_0 are smooth geometrically connected curves. For every $\gamma \in \Gamma$, some finite extension $\mathbf{F}_q/\mathbf{F}_{q_0}$ admits a point $z \in Z_0(\mathbf{F}_q)$ whose arithmetic Frobenius conjugacy class in the torsor is the class of γ.*

Indeed, for the conjugacy class C of γ, the cited theorem gives $\frac{|C|}{|\Gamma|}q + O(\sqrt{q})$ such points over $\mathbf{F}_q$. This is positive for all sufficiently large powers q of q_0.

Lemma 10.6. *Let U be a smooth geometrically connected curve over $\mathbf{F}_q$ with an $\mathbf{F}_q$-point, let G be a finite group, and put $\Pi = \pi_1(U_{\overline{\mathbf{F}}_q})$. A continuous representation $r : \Pi \to G$ extends to a continuous representation $\pi_1(U) \to G$ if and only if its G-conjugacy class is fixed by arithmetic Frobenius.*

Apply Drinfeld–Lau descent [11, Theorem II(A)] to the classifying stack BG of the finite constant group G. A Frobenius-fixed conjugacy class gives a geometric G-torsor together with a choice of isomorphism to its Frobenius pullback. The cited equivalence identifies such pairs with G-torsors on U. Choosing a point in the geometric fiber realizes the specified representative r.

Proof of Theorem 10.1. We first specify the data retained under specialization. Descend Z, L, Y, their maps (including $Z \to \mathcal{R}_{(4,3,4,6)} \to \mathcal{R} \to B_0$), every right deck transformation $y \mapsto y\delta$ ($\delta \in \Gamma$), and the identifications

$$\begin{aligned} L &\simeq \mathcal{L} \times_{\mathcal{R}} Z, \qquad L \simeq Y \times^{\Gamma} \mathcal{F}, \\ Y \times \Gamma &\xrightarrow{\sim} Y \times_Z Y, \qquad (y, \delta) \longmapsto (y, y\delta) \end{aligned} \tag{10.6}$$

to a number field E. Include the inverse isomorphisms and the finitely many identities expressing the group action. The finite group Γ and its finite left set $\mathcal{F}$ are kept constant. These finite-presentation objects, maps, and identities descend simultaneously by [35, Tag 01ZM].

Spread the data over $A = \mathcal{O}_E[1/M]$ for a positive integer M, enlarging M as necessary. We may arrange that $Z_A \to B_{0,A}$ is finite étale of its original positive degree, that $Y_A \to Z_A$ is finite étale of degree $|\Gamma|$, and that all identities in (10.6) hold over A. Finite étale morphisms and isomorphisms descend after shrinking the base by the limit results in [35, Section 32.8]. The schemes Z_A and Y_A are smooth relative curves with nonempty fibers, by these positive-degree maps. Their generic fibers are geometrically connected; after further shrinking A, all their fibers are geometrically connected by [35, Tag 055G]. Also invert $|H|$ in A.

Let Σ_p be the finite set of rational primes dividing M. All the preceding data have been chosen from p before choosing ℓ. For any $\ell \notin \Sigma_p$, choose a prime v of E over ℓ, with residue field $\mathbf{F}_{q_0}$. We obtain nonempty smooth geometrically connected curves Y_v and Z_v and a right torsor $Y_v \to Z_v$ under the same constant group Γ. Moreover,

$$L_v \simeq \mathcal{L} \times_{\mathcal{R}} Z_v, \qquad L_v \simeq Y_v \times^{\Gamma} \mathcal{F}.$$

The first identity and Proposition 10.4 identify each fiber with the entire set of geometric lift classes. The second retains the same permutation action of Γ on $\mathcal{F}$, including the element γ with no fixed point. No specialization assertion about full fundamental groups is needed. In particular, no assumption that $\ell \nmid |\Gamma|$ is used. The covers represented by the Hurwitz fibers are tame because their groups are subgroups of H and $\ell \nmid |H|$; no assertion about tame boundary ramification of $Y_v \to Z_v$ is required.

Apply Lemma 10.5 to the element γ constructed above. It gives a finite extension $\mathbf{F}_q/\mathbf{F}_{q_0}$ and a point $z \in Z_v(\mathbf{F}_q)$ for which Frobenius acts without fixed points on the geometric fiber $(L_v)_z$. Let $\lambda \in B_0(\mathbf{F}_q)$ be its image. The point z determines a Frobenius-invariant G-conjugacy class of surjections

$$r_\lambda : \Pi \longrightarrow G, \qquad \Pi = \pi_1\big((U_\lambda)_{\overline{\mathbf{F}}_q}\big).$$

Since $x = 2$ is an $\mathbf{F}_q$-point of U_λ, Lemma 10.6 extends r_λ to an actual continuous G-valued representation $\bar{\rho}$ of $\pi_1(U_\lambda) = G_{K,S}$. This step concerns a representation, not the existence of a universal cover on a coarse moduli space. Its geometric image is G. Since z maps to $\mathcal{R}_{(4,3,4,6)}$, its inertia images at $0, \lambda, 1, \infty$ have orders $4, 3, 4, 6$.

If a lift $\rho : G_{K,S} \to \mathrm{GL}_2(R)$ existed, then its image would lie in H. Its geometric image J would satisfy $\mathrm{red}(J) = G$, and its residual inertia at each point of S would be nontrivial. Thus its geometric restriction defines a point of $(L_v)_z(\overline{\mathbf{F}}_q)$ by the complete fiber formula (10.4), whether or not $J = H$. For the selected Frobenius lift $F_q \in \pi_1(U_\lambda)$ we have

$$\rho(F_q \alpha F_q^{-1}) = \rho(F_q)\rho(\alpha)\rho(F_q)^{-1} \qquad (\alpha \in \Pi),$$

and $\rho(F_q) \in H$. Hence this geometric H-conjugacy class is Frobenius-invariant, contradicting the choice of z. This proves nonliftability over R. The assertion for finite coefficient extensions follows from Lemma 2.2, applied to the same profinite group $G_{K,S}$. □

Remark 10.7. The construction proves existence for every ℓ outside a finite set depending on p. It does not specify that set, the power q, or a numerical value of λ. These choices come from good reduction of a finite Hurwitz cover and the finite-field realization step. In particular, the theorem does not classify liftability for an arbitrary prescribed pair (K, S). Its dimension is optimal: every one-dimensional representation over a finite field has a Teichmüller lift on the same profinite group.

10.4. **An everywhere-unramified example.**

Proof of Corollary 1.10. Fix $p \geq 5$ and $\ell \notin \Sigma_p$ as in Theorem 10.1. Take q, λ and $\bar{\rho}$ from that theorem, and put $U = \mathbf{P}^1_{\mathbf{F}_q} \setminus \{0, \lambda, 1, \infty\}$. Let $C \to \mathbf{P}^1_{\mathbf{F}_q}$ be the normalization in the extension

$$y^{12} = \frac{x(x-\lambda)}{x-1}.$$

We have $\ell \nmid 12$. The valuations of the right-hand side at $0, \lambda, 1, \infty$ are $1, 1, -1, -1$, respectively. The polynomial is Eisenstein at the x-adic valuation over $\overline{\mathbf{F}}_q$, so the cover has degree 12 and is geometrically connected. It is tame and totally ramified at these four points and étale elsewhere. The normalization is a smooth projective curve, and Riemann–Hurwitz gives

$$2g(C) - 2 = -2 \cdot 12 + 4(12 - 1) = 20.$$

Write $V = C \times_{\mathbf{P}^1} U$. The connected finite étale cover $V \to U$ identifies $\pi_1(V)$ with a subgroup of index 12 in $\pi_1(U)$. Each residual inertia order divides 12, so total ramification of index 12 kills the residual inertia at every point of $C \setminus V$ by Abhyankar's lemma [35, Tag 0BRM]. Thus $\bar{\rho}|_{\pi_1(V)}$ factors through $\pi_1(C) = G_{\mathbf{F}_q(C),\varnothing}$. Over $\overline{\mathbf{F}}_q$ the cover $V \to U$ is cyclic. Hence the image of $\pi_1(V_{\overline{\mathbf{F}}_q})$ is normal in $\mathrm{SL}_2(\mathbf{F}_p)$ with cyclic quotient. Since $\mathrm{SL}_2(\mathbf{F}_p)$ is perfect for $p \geq 5$, this image, and therefore the geometric image on C, is all of $\mathrm{SL}_2(\mathbf{F}_p)$.

Let $M = M_2(\mathbf{F}_p)$ with the adjoint action through $\bar{\rho}$. By Theorem 10.1 and Proposition 2.1, the lifting obstruction $\beta \in H^2(\pi_1(U), M)$ is nonzero. By functoriality, the obstruction on $\pi_1(V)$ is $\mathrm{res}(\beta)$. Since $\mathrm{cor}\,\mathrm{res}(\beta) = 12\beta$ and $p \nmid 12$, this restriction is nonzero. A lift on $\pi_1(C)$ would pull back to a lift on $\pi_1(V)$. Lemma 2.2 applied to $\pi_1(C)$ gives the same conclusion after every finite extension of $\mathbf{F}_p$. □

Remark 10.8 (Characteristic-zero coefficients). Let X denote either U or C in the preceding proof. Its residual representation is geometrically absolutely irreducible of rank two and has determinant one. For this representation, de Jong's theorem [10], in the form of [4, Corollary 1.4 and Remark 1.5], gives a finite flat $\mathbf{Z}_p$-deformation ring for $\pi_1(X)$ with determinant fixed to one. The ring has residue field $\mathbf{F}_p$, so its generic fiber is nonzero and has a point over a finite extension $E/\mathbf{Q}_p$. The deformation ring maps into $\mathcal{O}_E$ by integrality, giving a characteristic-zero lift on $\pi_1(X)$. Any finite coefficient field affording an integral lift on $\pi_1(X)$ must be ramified over $\mathbf{Q}_p$. Indeed, if $E/\mathbf{Q}_p$ were unramified with residue field k, then reduction modulo p^2 would give a $W_2(k)$-lift, contrary to Theorem 10.1 or Corollary 1.10. If additional ramification is allowed, then Böckle and Khare give an $\mathrm{SL}_2(\mathbf{Z}_p)$-lift on $\pi_1(X \setminus T)$ for some finite set T of closed points of X [3, Theorem 1.1]. Their hypotheses hold for both curves since the image is full, $p \geq 5$, and the boundary ramification is tame.

Remark 10.9 (Three tame punctures on the rational curve). Let $U = \mathbf{P}^1_{\mathbf{F}_q} \setminus S$, where S has at most three geometric points. Let k be a finite field of odd characteristic $p \nmid q$. Every tamely ramified, geometrically absolutely irreducible representation $\sigma : \pi_1(U) \to \mathrm{GL}_2(k)$ lifts to $W_2(k)$ on $\pi_1(U)$, with determinant the Teichmüller lift of $\det \sigma$.

To see this, put $r = |S(\overline{\mathbf{F}}_q)|$. For $r = 0$ the geometric fundamental group is trivial. Suppose $1 \leq r \leq 3$, and put $\bar{U} = U_{\overline{\mathbf{F}}_q}$ and $M = \mathrm{ad}^0(\sigma)$. The trace pairing on M is perfect since p is odd. By Schur's lemma M has no geometric invariants. Poincaré duality gives $H^2_c(\bar{U}, M) = 0$. The Grothendieck–Ogg–Shafarevich formula in the tame case gives $\chi_c(\bar{U}, M^\vee(1)) = 3(2-r)$ [30, Theorem 1]. The compactly supported groups in degrees zero and two for $M^\vee(1)$ vanish as well, so duality gives $\dim_k H^1(\bar{U}, M) = 3(r-2)$. The localization sequence now gives a surjection

$$H^1(\bar{U}, M) \longrightarrow \bigoplus_{s \in S(\overline{\mathbf{F}}_q)} H^1(I_s, M),$$

where I_s is geometric inertia at s [17, Lemmas 1.6 and 1.8]. Each summand has dimension $\dim_k M^{I_s} \geq 1$. Indeed, wild inertia acts trivially and has p-cohomological dimension zero. For a generator of tame inertia, the dimensions of the coinvariants and invariants agree, and the centralizer of a single 2-by-2 matrix contains a nonzero trace-zero matrix. Thus $3(r-2) \geq r$,

forcing $r = 3$, and the surjection between spaces of dimension three is a Frobenius-equivariant isomorphism.

The group $\pi_1(\bar{U})$ has p-cohomological dimension at most one by affine vanishing [17, Lemma 2.1]. The same bound holds for I_s by its wild and tame inertia sequence. Since the finite constant field has p-cohomological dimension one, the Hochschild–Serre sequences for U and for the local fields, together with Shapiro's lemma on each Frobenius orbit of punctures, identify the map

$$H^2(\pi_1(U), M) \longrightarrow \bigoplus_{v \in S} H^2(K_v, M), \qquad K = \mathbf{F}_q(x),$$

with $H^1(G_{\mathbf{F}_q}, -)$ of the preceding isomorphism. It is thus injective. The obstruction with the chosen determinant takes values in $H^2(\pi_1(U), M)$ since the fixed-determinant congruence kernel is the trace-zero module. Since $p \nmid 2$, Böckle's fixed-determinant local lifting result [2, Corollary 1.2] gives a lift with the chosen determinant at each v. Injectivity therefore kills the global obstruction.

11. Scope and questions not settled here

The preceding results leave the following questions.

(1) *A fixed cross-characteristic global field.* Fix an odd prime p and a global field K of characteristic different from p. The induction in the proof of Corollary 1.6 gives a counterexample over K whenever $\mu_{p^2} \not\subset K(\mu_p)$. In particular it settles the all-dimensions question over $\mathbf{Q}$ for every finite coefficient field. For a given K with $\mu_{p^2} \subset K(\mu_p)$, the existence of an unrestricted counterexample over K is not decided by that argument.

For example, if K is a function field with complete constant field $\mathbf{F}_q$ and $d = \operatorname{ord}_{\mathbf{F}_p^\times}(q)$, then $K(\mu_p) = K\mathbf{F}_{q^d}$ has complete constant field $\mathbf{F}_{q^d}$. The induction applies exactly when $v_p(q^d - 1) = 1$. For a fixed ground characteristic $\ell \neq p$, if $p^2 \mid \ell^{\operatorname{ord}_{\mathbf{F}_p^\times}(\ell)} - 1$, then every finite field of characteristic ℓ containing μ_p already contains μ_{p^2}.

(2) *Minimal dimensions for absolute Galois groups.* As the global field varies, the least possible dimension of a counterexample in characteristic three is either five or six. For $p \geq 7$, the interval $5 \leq n < p$ is not fully classified. For the fixed field $\mathbf{Q}$, Corollary 1.6 gives upper bounds on the least counterexample dimension, and Theorem 1.2 gives the lower bound five. The exact minimum over $\mathbf{Q}$ remains undetermined here for odd p.

(3) *Prescribed function fields and ramification sets.* For $p \geq 5$, Theorem 10.1 gives four rational branch points on $\mathbf{P}^1$, and Corollary 1.10 gives $S = \varnothing$ on a curve of genus 11. On $\mathbf{P}^1$, Remark 10.9 proves liftability in the tame, geometrically absolutely irreducible rank-two case with at most three geometric punctures. Thus four is the minimum possible number of geometric branch points in this setting for the good characteristics of Theorem 10.1. The lifting problem for a prescribed curve and a prescribed ramification set remains undetermined in general.

In rank two and characteristic three, the reduction $\mathrm{GL}_2(\mathbf{Z}/9\mathbf{Z}) \to \mathrm{GL}_2(\mathbf{F}_3)$ admits a group-theoretic section [25, Theorem 7.1]. Composing with the section lifts every $\mathbf{F}_3$-valued representation on its original profinite group. This lift extends to $W_2(k)$ for every finite $k/\mathbf{F}_3$. For a given finite field $k \supsetneq \mathbf{F}_3$, can a function field K of characteristic different from three and a finite set S of its primes be chosen so that $G_{K,S}$ has a geometrically absolutely irreducible two-dimensional k-representation, with no model over $\mathbf{F}_3$, that fails to lift to $W_2(k)$ on that group?

(4) *Everywhere-unramified equal-characteristic lifting.* For $\operatorname{char} K = p$ and $S = \varnothing$, neither universal $W_2(k)$-liftability nor universal $W(k)$-liftability is asserted. The projective curve C falls outside the affine hypothesis of [2, Theorem 1.7]. For instance the smooth elliptic curve

$$C/\mathbf{F}_5 : \quad y^2 = x^3 + 3x + 2$$

has $\#C(\mathbf{F}_5) = 5$ and Frobenius trace one. It is ordinary, $H^1(C_{\overline{\mathbf{F}}_5}, \mathbf{F}_5)$ has dimension one, and Frobenius acts on it trivially: the unit root of $X^2 - X + 5$ reduces to 1 modulo 5. The

geometric degree-two group vanishes by the characteristic-p curve bound [34, Expose X, Corollary 5.2] and the low-degree comparison. The Hochschild–Serre sequence therefore gives

$$H^2(\pi_1(C), \mathbf{F}_5) \simeq H^1(G_{\mathbf{F}_5}, H^1(C_{\overline{\mathbf{F}}_5}, \mathbf{F}_5)) \simeq \mathbf{F}_5.$$

This only exhibits a nonzero obstruction *space*. It does not exhibit a nonzero matrix lifting obstruction or a nonliftable representation.

(5) *Higher Witt length and extra conditions.* The positive cross-characteristic lifting theorems do not assert compatible $W_r(k)$-lifts for all r, extension of a prescribed partial lift, a prescribed determinant, or a prescribed local deformation condition. In equal characteristic, Böckle's theorem gives a lift to $W(k)$ on G_K, and on $G_{K,S}$ for nonempty finite S [2, Theorem 1.7].

The matrix identities used in the proofs are established for their stated parameters. The supplementary finite computations provide independent checks of the formulas; the arguments do not extrapolate from these finite samples.

Appendix A. Exact data and reproducibility

The accompanying directory `certificate/` is part of the computer-assisted proof of Theorem 1.8. The data are plain text. No binary BNF or BNR object, conditional ray class basis, or file from the exploratory search is required.

A.1. **The degree twenty model and data schema.** The polynomial defining F in the certificate is

$$\begin{aligned} Q(Y) = {} & Y^{20} - 10Y^{19} - 189Y^{18} + 1986Y^{17} + 10973Y^{16} - 134092Y^{15} \\ & - 222499Y^{14} + 4068119Y^{13} - 96247Y^{12} - 59019653Y^{11} \\ & + 54341353Y^{10} + 383922159Y^9 - 584577823Y^8 - 812465877Y^7 \\ & + 1810126828Y^6 - 116786163Y^5 - 1091570098Y^4 + 232224745Y^3 \\ & + 187325384Y^2 - 7148897Y - 5573516. \end{aligned}$$

The file `certificate_data.gp` specifies a five-entry vector

$$\texttt{certificate_data} = [Q, \mathcal{B}, c, \mathcal{G}, (v, \mathcal{A})].$$

Here $\mathcal{B}$ is an integral basis of $\mathcal{O}_F$, $\mathcal{G}$ is the list $(G_1, \ldots, G_6)$, and

$$v = (30570, 20, 0, 5, 0, 0)^t.$$

The six-entry list c has the following interpretation:

`c[1]`: the HNF matrix of the ideal I_2 in the basis $\mathcal{B}$;
`c[2]`: the basis-coordinate column of an element b_6 with $I_6 = (b_6)$;
`c[3]`: the HNF matrix of $\mathrm{N}_{L/F}(g_2(I_2\mathcal{O}_L))$;
`c[4]`: the element $\mathrm{N}_{L/F}(g_7(b_6))$;
`c[5]`: the basis-coordinate column of the CRT marker a_1;
`c[6]`: the HNF matrix of D.

The matrix $\mathcal{A}$ has 61 rows. If its rth row is (a_r, m_r), then $\mathcal{A}$ represents the exact element $\alpha = \prod_{r=1}^{61} a_r^{m_r}$. Negative exponents are allowed; the ray congruence is a local congruence of units. This compact representation avoids expanding very large algebraic numbers.

The file `provenance_data.gp` gives $Q, \mathcal{B}$, a polynomial P defining L, a supplied integral basis of L, the embedding $F \hookrightarrow L$, a relative cubic model, the four needed automorphisms, and all five roots of (9.1) in L. It also includes the component list c, which the provenance checker compares exactly with the list in `certificate_data.gp`. Thus the ideal equality cannot accidentally refer to a different number field model or differently ordered root action.

A.2. **Field and norm provenance.** The script `verify_provenance.gp` checks all of the following.

(1) Both defining polynomials are irreducible. Their supplied maximal orders are certified with `nfcertify`. Their signatures are $(20, 0)$ and $(60, 0)$ and their discriminants are q^{16} and q^{48}, respectively.
(2) The quintic has Galois group A_5 and the five supplied distinct elements of L are its roots. The supplied automorphisms satisfy P and act on these roots by their specified permutations.
(3) The order-three automorphism acts as (135). The supplied embedding of F is fixed by it. The relative cubic reconstruction has the same absolute polynomial and base embedding, and relative discriminant of norm one.
(4) Extending I_2, applying g_2, and taking the relative ideal norm reproduces `c[3]`. Applying g_7 to b_6 and taking the relative element norm reproduces `c[4]`.
(5) All CRT congruences for a_1 and the exact formula for D are satisfied. The primes above q have the asserted decomposition and ramification data, and 3 has order 1950 modulo q.

These checks establish the arithmetic provenance of the two operators appearing in the proof, not just an equality among unattached ideals.

A.3. **The ideal identity and ray congruence.** The script `verify_certificate_text.gp`, using `verify_gcd_functions.gp`, reconstructs F from Q and $\mathcal{B}$, checks the CRT conditions, verifies that all relevant ideals are prime to $\mathfrak{m}_F$, and verifies that every entry of v is divisible by five. It checks the ideal product

$$D\prod_{i=1}^{6} G_i^{-v_i}\prod_{r=1}^{61}(a_r)^{-m_r} = \mathcal{O}_F$$

by exact ideal-gcd cancellation, which takes 639 cancellation steps for the supplied data.

To explain this verification mathematically, a fractional ideal is first written as a quotient of integral ideals. For two integral ideal factors J^a, K^b of opposite exponent signs, put $d = J + K$, the ideal gcd. Replace them by

$$(J/d)^a(K/d)^b d^{a+b}.$$

This preserves the product exactly. Canonical HNF ideal matrices identify and combine equal factors. Successful removal of every factor proves the displayed product identity, without factoring large norms or assuming that a group of classes has been completely computed.

A fresh residue-unit computation modulo $(25q)\mathcal{O}_F$ verifies $\alpha \equiv 1 \pmod{\mathfrak{m}_F}$. The independent script `verify_alpha_direct.gp` checks the same congruence by exact modular multiplication and cross-multiplication of the compact factors. Thus the congruence is independently checked without using class group coordinates.

A.4. **Commands and interpretation.** From the accompanying `certificate/` directory, run:

```
gp -f -q -s 256M verify_provenance.gp
gp -f -q -s 256M verify_certificate_text.gp
gp -f -q -s 256M verify_alpha_direct.gp
python3 verify_representation.py
```

The first two arithmetic checkers produce, respectively, the success markers

```
PROVENANCE_VERIFIED_WITHOUT_BNF
PORTABLE_TEXT_CERTIFICATE_VERIFIED_WITHOUT_BNF
```

The third script provides an independent check of the ray congruence; the Python script checks the finite-field representation identities. The original certificate used PARI/GP 2.17.2 for exact number-field and ideal operations [40]. The combined output of the four checkers is in `certificate/verify_all.log`. The directory also contains their individual logs, the exact input data, and the scripts. The file `certificate/SHA256SUMS.txt` records their checksums. The certificate checks are exact computations, not formal proofs in a proof assistant.

The proof needs only the identities and congruences stated above. It does not use a numerical dimension for a Hom space, completeness of a ray class presentation, a Morita equivalence, a computed global cohomology group, or an unverified conditional class group output.

Appendix B. Matrix identities and supplementary exact checks

The proofs in Sections 3, 4, 6, 7, and 10 establish the matrix identities for their stated parameters. The directory `audit/` also contains independent exact checks. These serve to detect transcription and sign errors; finite samples are not used to justify an assertion for all ranks, all primes, or all finite coefficient fields.

The characteristic-two checker expands the Boolean-polynomial identities for the quadratic form, root automorphisms, squares, and root relations in odd ranks $3, 5, 7, 9$. The low-dimensional checker verifies polynomial matrix identities over $\mathbf{Z}$ and modulo 3, including the lower entries of conjugation, the three-layer blocks, the upper-unitriangular cube, and the Frobenius appearing in the Witt carry. The counterexample checker verifies all 27 elements and all 729 cocycle equations in characteristic three, and checks the higher-prime formulas at $p = 5, 7, 11, 13$. It separately checks normalized factors and local contraction signs at $p = 3, 5, 7$.

The fixed-ramification checker verifies the adjoint/permutation intertwiner, averaging projection, all six order-five fixed lines, the two norm-conjugation scalars, and all 625 lifts modulo 25 of the standard residual unipotent. Its additional exact checks concern the projective congruence kernel and an independent matrix realization of the lift over $\mathbf{Z}_5[\sqrt{5}]$ in Theorem 9.1(ii). The function-field checker additionally verifies the tuple in (10.1), the two generated groups and their centralizers for $p = 5$, all 625 possible first-order conjugators, and the power identity of Lemma 10.2 for every lift of $I + E_{12}$ at $p = 5, 7, 11$. Section 10 combines the Hurwitz-space construction with the finite-field Chebotarev theorem. Run the independent matrix checkers from the project root:

```
python3 audit/check_p2_polynomials.py
python3 audit/check_lowdim.py
python3 audit/counterexample_audit.py
python3 audit/check_fixedS_matrices.py
python3 audit/check_function_field_fixedS.py
```

The logs record the checks actually performed. These calculations and the mathematical audit do not constitute proof-assistant verification.

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